
Software Requirements Specification

for

PetCare

Version 1.4 approved

Prepared by Jasmine Tye Jia Wen

Keng Jia Chi

Sih Jia Qi

Lim Chun Wen

Hng Cherng Khai

Nanyang Technological University

<08/11/2024>

Table of Contents

1. Introduction.....	1
1.1 Purpose.....	1
1.2 Document Conventions.....	1
1.3 Intended Audience and Reading Suggestions.....	1
1.4 Product Scope.....	1
1.5 References.....	2
2. Overall Description.....	3
2.1 Product Perspective.....	3
2.2 Product Functions.....	3
2.3 User Classes and Characteristics.....	4
2.4 Operating Environment.....	5
2.5 Design and Implementation Constraints.....	8
2.6 Assumptions and Dependencies.....	9
3. External Interface Requirements.....	10
3.1 User Interfaces.....	10
3.2 Hardware Interfaces.....	10
3.3 Software Interfaces.....	10
3.4 Communications Interfaces.....	11
4. System Features.....	12
4.1 Register.....	12
4.2 Login.....	14
4.3 Reset Password.....	15
4.4 Logout.....	18
4.5 View Home Page.....	19
4.6 View Personal Account.....	23
4.7 Edit Personal Information.....	24
4.8 View All Articles.....	26
4.9 Filter Articles.....	28
4.10 Search Articles.....	30
4.11 View Article Details.....	32
4.12 Add New Article.....	34
4.13 View Posted Articles.....	36
4.14 Edit Article.....	38
4.15 Delete Article.....	41
4.16 View Pet Information.....	44
4.17 Edit Pet Information.....	45

4.18 Add Pet.....	47
4.19 Delete Pet.....	49
4.20 Display Active Missing Pets.....	52
4.21 Report Pet as Missing.....	54
4.22 Report Sighting.....	57
4.23 Mark Missing Pet as Found.....	59
4.24 Explore PetCare Amenities.....	62
4.25 Filter PetCare Amenities.....	65
4.26 Search Petcare Amenities.....	66
4.27 Add Adoption Pet.....	68
4.28 Edit Adoption Pet.....	70
4.29 Delete Pet Adoption.....	72
4.30 Display adoption Pet (User).....	74
4.31 Display Adoption Pet (Adoption Center).....	76
4.32 Interact with Chatbot.....	77
4.33 Respond to User’s PetCare App Guidance Queries.....	80
4.34 Respond to User’s Pet Care Related Queries.....	82
5. Other Nonfunctional Requirements.....	84
5.1 Performance Requirements.....	84
5.2 Safety Requirements.....	85
5.3 Security Requirements.....	86
5.4 Software Quality Attributes.....	86
Appendix A: Data Dictionary.....	88
B.1 Use Case Model.....	92
B.2 Entity Class Diagram.....	93
B.3 Control and Boundary Class Diagram.....	94
B.4 Sequence Diagram.....	94
B.5 State Machine Diagram.....	123
B.6 System Architecture Model.....	124
C.1 Postman API Documentation.....	125

Revision History

Name	Date	Reason For Changes	Version
Jasmine	26/10/2024	Initial write-up.	1.0
Cherng Khai	27/10/2024	Initial write-up on section 2.5, 2.6, 2.7 and 5.	1.1

Chun Wen	05/11/2024	Modified System Architecture Diagram	1.2
Jia Chi	06/11/2024	Included section 3.	1.3
Jia Qi	08/11/2024	Included section 5, finalizing documentation.	1.4

1. Introduction

1.1 Purpose

PetCare is a mobile application that aims to create a seamless and user-friendly platform for pet owners, adoption centers and animal lovers. The primary goal of *PetCare* is to strengthen the bond between pet owners and their furry companions by offering a comprehensive suite of tools, providing them with a reliable resource for things related to pet care and adoption. Whether it is for adopting new pets or managing pet's records, the application will provide an effective solution for all the stakeholders involved in the process.

1.2 Document Conventions

This section describes the conventional standards used throughout this document. It is imperative that all readers pay attention to the standards listed in this section.

Font:	Times New Roman
Heading:	Bold, Size 18
Sub-heading 1:	Bold, Size 14
Sub-heading 2:	Bold, Italic 12
Content:	Normal, Size 12
Emphasis:	Italic, Size 12
Technical Standards:	IEEE 830-1998

Refer *Appendix A: Data Dictionary* for the definitions of special terms used throughout this documentation.

1.3 Intended Audience and Reading Suggestions

The three main roles that utilize the *PetCare* application are pet owners, and adoption centers. Pet owner is a sub role of app users, have access to all the functionalities that an app user has and additionally, will utilize the features for people with pets such as the pet information page.

The stakeholders include the adoption centers and pet shelters, mobile app developers and third party service providers such as Google Maps API, Nominatim API, Tavily API and Gemini API.

1.4 Product Scope

PetCare helps connect pet seekers with adoption centers while also offering a range of supportive features for pet owners. By leveraging GPS technology, users can easily browse through available

pets at nearby adoption centers, ensuring they can find their ideal companion. The app also includes an AI-powered chatbot that provides personalized pet care advice, making it easier for adopters to care for their pets. In addition, the app offers tools to record and manage pet details, buy or sell second-hand pet items, and search for missing pets, making it a comprehensive resource for pet lovers.

1.5 References

- i) *Firebase Documentation* (n.d.) Firebase.. Retrieved October 25, 2024, from <https://firebase.google.com/docs>
- ii) *ReactJS Documentation* (n.d.) React. Retrieved October 25, 2024, from <https://reactjs.org/docs/getting-started.html>
- iii) *Expo documentation.* (n.d.). Expo Documentation. Retrieved October 25, 2024, from <https://docs.expo.dev/>
- iv) *Nearby search.* (n.d.). Google for Developers. Retrieved October 25, 2024, from <https://developers.google.com/maps/documentation/places/web-service/search-nearby>
- v) *Welcome to Apache Maven.* (n,d). Maven. Retrieved October 25, 2024, from <https://maven.apache.org/>
- vi) *Langchain | Tavily AI.* (n.d.). Retrieved October 25, 2024, from <https://docs.tavily.com/docs/integrations/langchain>
- vii) *How to load PDFs |  LangChain.* (n.d.). Retrieved October 25, 2024, from https://python.langchain.com/docs/how_to/document_loader_pdf/
- viii) *Get started with the Gemini API.* (n.d.). Google AI for Developers. Retrieved October 25, 2024, <https://ai.google.dev/gemini-api/docs>

2. Overall Description

2.1 Product Perspective

The *PetCare* application is a new, self-contained product designed to streamline pet management, pet adoption, and the processes involved in searching for missing pets. This application is not merely a follow-on product in an existing family but a standalone platform that integrates essential features such as user authentication, personal information management, pet profiles, informative articles, and the ability to adopt pets or report missing ones. Additionally, the app provides information on nearby *PetCare* amenities and includes a chatbot for user support. The *PetCare* app will serve as a critical component of a larger pet welfare ecosystem, collaborating with local shelters and, in the future, with veterinary services and pet supply vendors to create a comprehensive user experience. The app's functionality includes interfaces with external databases for real-time information and a user-friendly interface that allows easy navigation and interaction.

2.2 Product Functions

2.2.1 Registration: Allows users to create new accounts by providing the necessary details.

2.2.2 Login: Allow users to sign in to their own respective account and access the features based on their roles.

2.2.3 Nearby *Petcare* Amenities: Allow users to explore nearby pet related amenities based on their current location.

2.2.4 Pet Information: Allow users to create, edit and delete information about their own pet.

2.2.5 Adoption Center: Provides a platform for users to explore any pet that is available for adoption and allows adoption centers to manage their adoption listing that includes adding, editing and removing adoption.

2.2.6 Chatbot: An interactive virtual assistant designed to enhance user experience by providing instant support and information. It allows users to engage in conversational interactions, enabling them to inquire about various aspects of pet care. The chatbot can also guide users in navigating the application.

2.2.7 Articles: Provides a platform for users to view articles that are created by other users. A normal user can add , edit and delete their own article.

2.2.8 Missing Pet: Provides a platform for users to report their missing pet and allow others to report their sighting.

2.2.9 Personal Account: Allows users to view and manage their account details.

2.3 User Classes and Characteristics

2.3.1 Pet Owners & Pet Enthusiasts

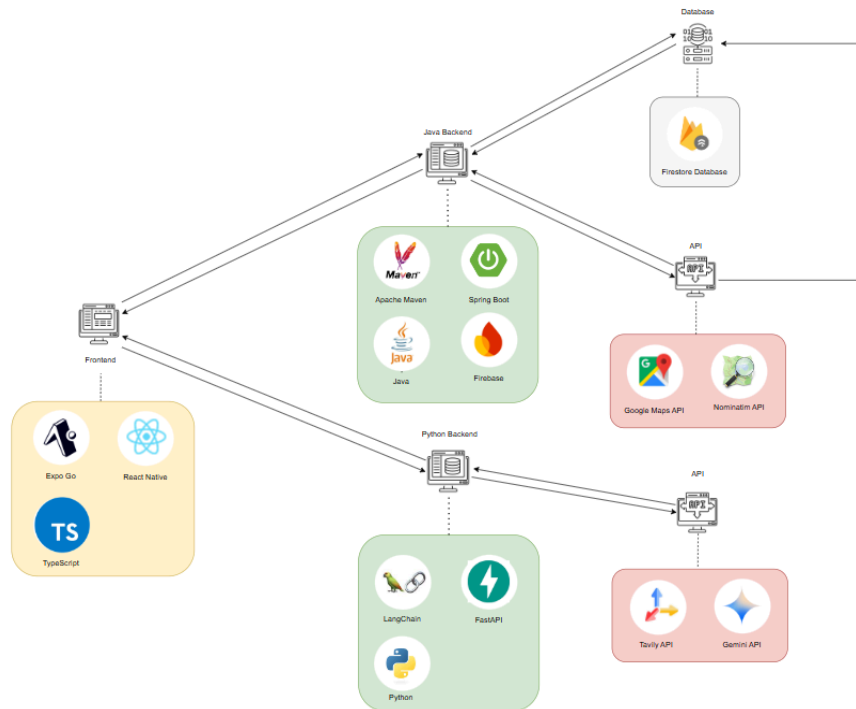
Attributes	Description
Frequency of Use	High
Subset of Functions used	All except add, delete & edit of adoption
Technical Expertise	Low
Characteristics	The <i>PetCare</i> development team predicts that this demographic of users will use the mobile app to enhance their pet-related experience, such as seeking help from other users to locate a missing pet, finding nearby pet-care amenities, reading and posting articles about their pets, and searching for pets to adopt from shelters.

2.3.2 Adoption Center

Attributes	Description
Frequency of Use	High
Subset of Functions used	All
Technical Expertise	Low
Characteristics	The <i>PetCare</i> development team predicts that this demographic of Adoption Center will use the mobile app to get more pet enthusiastic to adopt a pet and streamline the appointment process by having a generic view of pets from different adoption centers.

2.4 Operating Environment

This section of the report outlines the conditions and requirements under which the *PetCare* application will function effectively.



Tech Stack Diagram

2.4.1 FrontEnd Development

Utilizing the following for building user-interface:

- React.js
- TypeScript
- Expo Go

2.4.2 Front-End Dependencies

The following dependencies are utilized in the *PetCare* application along with their respective versions, as of time of development:

Dependency	Version
@expo/vector-icons	^14.0.2
@firebase/database	^1.0.8
@firebase/storage	^0.13.2

@react-leaflet/core	^2.1.0
@react-native-async-storage	^1.24.0
@react-native-firebase/app	^21.3.0
@react-native-firebase/auth	^21.3.0
@react-native-firebase/firestore	^21.3.0
@react-navigation/native	^6.0.2
@react-navigation/stack	^6.4.1
axios	^1.7.7
axios-retry	^4.5.0
expo	~51.0.28
expo-constants	~16.0.2
expo-file-system	~17.0.1
expo-font	~12.0.9
expo-image-picker	~15.0.7
expo-linear-gradient	^13.0.2
expo-location	~17.0.1
expo-router	~3.5.23
expo-splash-screen	~0.27.5
expo-status-bar	~1.12.1
expo-system-ui	~3.0.7
expo-web-browser	~13.0.3
firebase	^10.14.1
geolib	^3.3.4
leaflet	^1.9.4
react	^18.2.0

react-dom	18.2.0
react-leaflet	^4.2.1
react-native	0.74.5
react-native-date-picker	^5.0.7
react-native-fast-image	^8.6.3
react-native-geolocation-service	^5.3.1
react-native-gesture-handler	~2.16.1
react-native-image-picker	^5.7.0
react-native-linear-gradient	^2.8.3
react-native-maps	1.14.0
react-native-modal-datetime-picker	^18.0.0
react-native-permissions	^4.1.5
react-native-picker-select	^9.3.1
react-native-reanimated	~3.10.1
react-native-safe-area-context	4.10.5
react-native-screens	3.31.1
react-native-snap-carousel	^3.9.1
react-native-vector-icons	^10.2.0
react-native-web	^0.19.12
typescript	~5.3.3

2.4.3 Backend Development

Utilizing the following for server-side logic and data storage

- Java
- Python
- Firebase

2.4.4 External Third-Party API

Utilizing the following for added functionality:

- GoogleMaps API
- Nominatim API
- Tavily API
- Gemini API

2.4.5 Operating Platform

Development conducted through VS code 1.87.2 (universal) and managed through GitHub for version control.

2.5 Design and Implementation Constraints

2.5.1 External API Limitations

The following sub-section covers all external API usage limitations that may or may not affect certain functionalities of the mobile application.

2.5.1.1 Google Maps API

Rate limits and usage quotas must be managed to prevent service disruption, especially in high-traffic scenarios.

2.5.1.2 Nominatim API

It has usage restrictions like rate limits and requires accreditation. Consider fallbacks in case of outages.

2.5.1.3 Tavily API & Gemini API

API documentation must be thoroughly reviewed for potential constraints, such as request limitations or the need for specific authorization protocols.

2.5.2 Design Standards

The following sub-section covers all design standards adopted by *PetCare* application.

Design Standards	Details
------------------	---------

User Interface Design	The UI design of PetCare application follows the wireframes previously designed and agreed upon by the development team and stakeholders of <i>PetCare</i> .
Responsive Design	The application will be designed to provide an optimal viewing experience across a variety of devices and screen sizes, utilizing flexible layouts and scalable graphics.
Color Palette	A defined color palette is used throughout the design of <i>PetCare</i> application to help maintain <i>PetCare</i> 's brand recognition and identity.

2.6 Assumptions and Dependencies

1. *PetCare* depends on external APIs to obtain related information.
 - a. GOOGLE_MAP_API
 - b. TAVILY_API
 - c. GEMINI_API
2. The data obtained from the external APIs are accurate and updated.
3. *PetCare* assumes that users have access to a smartphone or tablet with an internet connection capable of supporting the app.
4. *PetCare* assumes that users have a basic understanding on how to navigate mobile applications and will be able to use the app's features without extensive technical support.
5. *PetCare* must have a stable Internet connection to access all of the app's functionalities due to its dependence on real-time data and location-based services.
6. *PetCare* uses third-party APIs such as Google Map APIs for location-based services. The reliance on these third party service providers means that changes or deprecation in third-party APIs could negatively affect the app's functionality.

3. External Interface Requirements

3.1 User Interfaces

Refer to PetCare_UI.pdf.

3.2 Hardware Interfaces

3.2.1 Client-side Requirements

The *PetCare* application is compatible with all mobile devices. The device must support a mobile browser that meets the technical specifications outlined in Section 2.4.1 *Production Environment of PetCare*.

3.2.2 Server-side Requirements

The *PetCare* application's back-end must be hosted on a dedicated server. The server will be configured using Spring Boot in Java and must be operational prior to initiating the back-end processes. This setup will enable the application to perform Create, Read, Update, and Delete (CRUD) operations on the Firestore database, which is hosted on the cloud.

3.3 Software Interfaces

3.3.1 Software Components and Version

The *PetCare* application uses Firebase services such as authentication, Firestore and cloud storage as the database to handle all the feature implementation.

3.3.2 Software Architecture

3.3.2.1 Expo Framework and React Native: These outline the structure of your frontend, focusing on how you build, deploy, and render the app across iOS and Android. These sections explain the design decisions that influence the app's cross-platform functionality, performance, and user interface behavior.

3.3.2.2 Maven and Spring Framework: The use of Maven and Spring on the backend highlights architectural choices for dependency management, build automation, and a well-organized project structure, which are crucial aspects of software architecture. The choice to use Spring Framework also implies a specific approach to handling backend services and RESTful API structure, which defines how your application logic is layered and connected.

3.4 Communications Interfaces

The *PetCare* application communication architecture must follow a client-server model. Each communication must go through a REST-styled Application Programming Interface (API) provided by the back-end server. Each request must also be served over HyperText Transfer Protocol Secure (HTTPS) which is sufficient for local development

Communication from client to server must invoke GET and POST requests. Communication from server to client must serve data standardized in JavaScript Object Notation (JSON) format.

All stakeholders are advised to begin by reading section go through *Appendix C.1: Postman API Documentation* for detailed information regarding *PetCare*'s API endpoints,

4. System Features

4.1 Register

4.1.1 Description and Priority

Allow new users to register an account in the *PetCare* application.

Overall Priority	High
------------------	------

4.1.2 Stimulus/ Response Sequences

Use Case ID:	UC_REC_1		
Use Case Name:	Register Account		
Created By:	Keng Jia Chi	Last Updated By:	Keng Jia Chi
Date Created:	25th August 2024	Date Last Updated:	25th August 2024

Actor:	User
Description:	Allows first time users to create an account in <i>PetCare</i> application.
Preconditions:	1. The email address given is not registered in the system.
Postconditions:	An account is successfully created for the user.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The system allows the user to register with Email and password. 2. The user chooses to register with email and password. 3. The system requests the user to input the following information fields: a. Name b. Email address c. Password d. Confirm Password 4. The user selects the “Register” button. 5. The system validates the required fields. 6. If the user’s details are valid, a <i>PetCare</i> account will be

	successfully created. 7. The system redirects the user to the <i>PetCare</i> application login page.
Alternative Flows:	AF-S6: If email address is already in database 1. The system will display “The email is already in use. Please use a different email.”. 2. System returns to Step 3. AF-S6: If the password does not meet the complexity requirements. 1. The system will display “Passwords must contain at least one uppercase letter, one lowercase letter, one numeric character, and one special character.”. 2. System returns to Step 3. AF-S6: If the password does not meet the length requirements. 1. The system will display “Passwords must be at least 6 characters long.”. 2. System returns to Step 3. AF-S6: If password is not the same as confirm password. 1. The system will display Passwords and confirm passwords do not match.”. 2. System returns to Step 3. AF-S6: If the user did not complete all the required fill. 1. The system will display "Please fill in all the fields.”. 2. System returns to Step 3.
Exceptions:	None
Includes:	None
Special Requirements:	The system needs to validate user input data.
Assumptions:	None
Notes and Issues:	None

4.1.3 Functional Requirement

1. *PetCare* shall allow users to register for an account.

1.1 *PetCare* shall allow users to create an account using email address and password.

1.1.1 *PetCare* must display text fields for the user to enter their information.

1.1.1.1 The text field must consist of username.

1.1.1.2 The text field must consist of an email address.

1.1.1.3 The text field must consist of a password.

1.1.1.4 The text field must consist of a confirmation password.

1.1.2 *PetCare* shall verify the field filled in by the user before creating an account.

1.1.2.1 The email address has never registered in the system.

1.1.2.2 The password given must contain at least 1 uppercase character, 1 lowercase character and 1 special character and 1 number.

1.1.2.3 The password given must contain at least 6 characters.

1.1.2.4 The password given must match with the confirmed password.

1.1.3 *PetCare* shall create an account for the user upon verification and navigate the user to the login page.

4.2 Login

4.2.1 Description and Priority

Allow users to login into their existing *PetCare* account.

Overall Priority	High
------------------	------

4.2.2. Stimulus/ Response Sequences

Use Case ID:	UC_LOGIN_1		
Use Case Name:	Login		
Created By:	Keng Jia Chi	Last Updated By:	Keng Jia Chi
Date Created:	25th August 2024	Date Last Updated:	25th August 2024

Actor:	User
Description:	Allows users to login into their <i>PetCare</i> account using their email and password.
Preconditions:	User has registered for an account.
Postconditions:	User is logged into the <i>PetCare</i> application and is navigated to the home screen of the application.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The system allows the user to login with Email and password. 2. The user inputs his/her email and password. 3. The user selects the “Login” button. 4. If the user’s login credentials are valid, the system will direct the user to PetCare’s home page.

Alternative Flows:	AF-S3: If the user's login credentials are invalid 1. The system shall display "Incorrect email/password. Please try again." to the user. 2. The system returns to Step 1. AF-S3: If the user's did not provide either email or password or both 1. The system shall display "Please fill in both email and password." to the user. 2. The system returns to Step 1.
Exceptions:	None
Includes:	None
Special Requirements:	None
Assumptions:	The user has an existing <i>PetCare</i> account
Notes and Issues:	None

4.2.3. Functional Requirement

2. *PetCare* shall allow users to log in to the account they have created previously.

2.1 *PetCare* shall allow users to log in to their account with their email address and password.

2.1.1 *PetCare* must display two text fields for the user to enter their email address and password.

2.1.1.1 *PetCare* must display one text field for Email Address.

2.1.1.2 *PetCare* must display one text field for Password.

2.1.2 *PetCare* must display a "Login" button.

2.1.3 When the user clicks "Login", *PetCare* must verify the information obtained from the text fields.

2.1.4 If email or password fields or both are empty, *PetCare* shall display "Please fill in both email and password." to the user.

2.1.4 If email and password entered are incorrect, *PetCare* shall display "Incorrect email/password. Please try again" to the user.

2.1.5 If email and password matches, *PetCare* shall log the user in and navigate the user to the homepage of the application.

4.3 Reset Password

4.3.1 Description and Priority

Allow users to reset their existing *PetCare* account password.

Overall Priority	High
------------------	------

4.3.2. Stimulus/ Response Sequences

Use Case ID:	UC_RP_1		
Use Case Name:	Reset Password		
Created By:	Keng Jia Chi	Last Updated By:	Keng Jia Chi
Date Created:	25th August 2024	Date Last Updated:	25th August 2024

Actor:	User
Description:	Allows users to reset their password when they have forgotten it or need to change it for security reasons.
Preconditions:	1. The user has an existing <i>PetCare</i> account. 2. The user must have access to the email address associated with their <i>PetCare</i> account. 3. The user must have access to their email account.
Postconditions:	1. The user password is successfully reset. 2. The user can log in with the new password
Priority:	High
Frequency of Use:	Moderate
Flow of Events:	1. The user clicks on the “Forgot Password” button. 2. The system displays the password reset page. 3. The user enters their registered email address. 4. If the email address is valid, the system sends a password reset link to the user's email. 5. The user clicks on the reset link in their email. 6. The system displays a new password creation page. 7. The user enters a new password and confirms it. 8. If the password meets the complexity requirements, the system updates the user's password in the database. 9. The system shall display “Password reset successfully” to the user. 10. The user is redirected to the login page.
Alternative Flows:	AF-S4: The user’s email address is invalid 1. The system will display “Invalid email address” to the user. 2. The system will prompt the users to re-enter their email address.

	3. The system returns to Step 3. AF-S8: The user's password does not meet the complexity requirements 1. The system will display "Invalid password" with password guidelines to the user. 2. The system will prompt the users to enter a new password. 3. The system returns to Step 7.
Exceptions:	1. The password reset link expires before the user clicks on it.
Includes:	None
Special Requirements:	1. The password reset link should expire after 24 hours. 2. The new password must meet the system's complexity requirements.
Assumptions:	1. The user has access to their registered email account.
Notes and Issues:	None

4.3.3. Functional Requirement

3. *PetCare* shall allow users to reset their password.

3.1 *PetCare* shall display a "Forgot Password" button on the login page.

3.2 When the user taps on the "Forgot Password" button, *PetCare* shall redirect the user to the reset page.

3.3 *PetCare* shall display one text field for the user's registered email address.

3.4 *PetCare* shall verify the email address entered by the user.

3.4.1 If the email address is valid, *PetCare* shall send a password reset link to the user's email.

3.4.2 If the email address is invalid, *PetCare* shall display an "Invalid email address" message and prompt the user to re-enter their email.

3.5 When the user clicks on the reset link, *PetCare* shall display a new password creation page.

3.6 *PetCare* shall display two text fields.

3.6.1 *PetCare* must display one text field for a new password.

3.6.2 *PetCare* must display one text field to confirm the password.

3.7 *PetCare* shall verify that the new password meets the strong password format requirements.

3.7.1 The password given must contain at least 1 uppercase character, 1 lowercase character and 1 special character and 1 number.

3.7.2 The password given must contain at least 6 characters.

3.7.3 The password given must match with the confirmed password.

3.8. If the password meets the requirements, *PetCare* shall update the user's password in the database.

3.8.1 If the password does not meet the password format requirements, *PetCare* shall display an "Invalid password" message with password guidelines and prompt the user to re-enter a new password.

3.9 After successful password reset, *PetCare* shall display a "Password reset successfully" message on the screen.

3.10 *PetCare* shall redirect the user to the login page after successful password reset.

4.4 Logout

4.4.1 Description and Priority

Allow users to log out from *PetCare* accounts.

Overall Priority	High
------------------	------

4.4.2. Stimulus/ Response Sequences

Use Case ID:	UC_LOGOUT_1		
Use Case Name:	Log Out		
Created By:	Keng Jia Chi	Last Updated By:	Keng Jia Chi
Date Created:	25th August 2024	Date Last Updated:	25th August 2024

Actor:	User
Description	Allows users to securely log out of their account.
Preconditions:	1. The user has navigated to the Personal Account Information page.
Postconditions:	The user is logged out of their account
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user clicks the "Log Out" button. 2. The system displays a confirmation dialog.

Actor:	User
Description	Allows users to securely log out of their account.
Preconditions:	1. The user has navigated to the Personal Account Information page.
	3. The user confirms the logout action. 4. The system terminates the user's session. 5. The system clears any local session data. 6. The system redirects the user to the login page..
Alternative Flows:	AF-S3: The user cancels log out 1. The user clicks “Cancel” on the confirmation dialog. 2. The system closes the dialog.
Exceptions:	None
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

4.4.3. Functional Requirement

4. *PetCare* shall display a “Logout” button in the Personal Account page for users to log out from their account.

4.1 When the user clicks the "Logout" button, *PetCare* shall display a confirmation dialog.

4.1.1 The confirmation dialog shall ask the user to confirm their intention to log out.

4.1.2 The confirmation dialog shall provide “Yes” and “No” options.

4.2. If the user confirms the logout action:

4.2.1 *PetCare* shall terminate the user’s session.

4.2.2 *PetCare* shall clear any local session data.

4.2.3 *PetCare* shall redirect the user to the login page.

4.3 If the user cancels the logout action:

4.3.1 *PetCare* shall close the confirmation dialog.

4.3.2 *PetCare* shall return the user to the Personal Account page.

4.5 View Home Page

4.5.1 Description and Priority

Overall Priority	High
------------------	------

4.5.2 Stimulus/ Response Sequences

Use Case ID:	UC_HOME_1		
Use Case Name:	View Home Page		
Created By:	Jasmine Tye	Last Updated By:	Jasmine Tye
Date Created:	25th August 2024	Date Last Updated:	19th October 2024

Actor:	User
Description:	The Home page provides a comprehensive view of relevant information to the user. It displays details about the user's pets, shows new pets available for adoption in shelters, provides information about nearby amenities related to pet care, and includes a news article carousel featuring top pet-related articles.
Preconditions:	1. The user must have <i>PetCare</i> installed 2. The user must be logged in and authenticated. 3. The user has granted location permissions to the app.
Postconditions:	1. The Home page displays a personalised view including: a. Bottom navigation bar that allows users to redirect to Missing Pet, Adoption Center, Nearby, or Account page b. The user's pets with details like name and photos. c. Pet related articles in the form of a carousel view
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user opens <i>PetCare</i> and navigates to the Home page. 2. <i>PetCare</i> retrieves and displays the user's registered pets, showing each pet's name and image under the "My Pets" section. 3. <i>PetCare</i> displays a "+" icon to allow the user to add a new pet profile: - If there are registered pets, <i>PetCare</i> displays each pet's image with a "+" icon as the last item. - If there are no registered pets, <i>PetCare</i> displays only the "+" icon. 4. The user can tap on a pet's image or name to view, edit, or delete the pet's details. 5. <i>PetCare</i> retrieves and displays a news article carousel to showcase top pet-related articles.

	6. <i>PetCare</i> automatically rotates the carousel every 5 seconds. 7. The user manually swipes through the articles in the carousel to change the article being shown in the carousel. 8. The user taps on an article in the carousel to view its full content. 9. <i>PetCare</i> displays a “See All” button above the news carousel, allowing the user to navigate to the “Browse Articles” page to view all articles. 10. <i>PetCare</i> displays a bottom navigation bar with icons that redirect the user to different sections on tap: a. Missing: Redirects to the “Missing Pets” page. b. Adoption: Redirects to the “Adoption Center” page. c. Nearby: Redirects to the “<i>PetCare</i> Amenities” page. d. Account: Redirects to the “Account” page. 11. <i>PetCare</i> displays a fixed chatbot icon at the bottom right corner, above the navigation bar, which the user can tap to access the ‘Chatbot’ page.
	AF-S2: No pets registered under user 1. System displays a message stating “You have no registered pets” AF-S3: No article added by user 1. System displays a message stating “There are currently no article”
Exceptions:	EX-1 Data Retrieval Failure 1. If any data retrieval fails, the system displays an error message prompting the user to try again later.
Includes:	UC_PI_01 - View Pet Information UC_ARC_1 - View All Articles
Special Requirements:	None
Assumptions:	1. Users have Internet access
Notes and Issues:	None

4.5.3 Functional Requirement

5. *PetCare* shall retrieve and display the user’s registered pets on the home page.

5.1 *PetCare* shall display each pet’s name and image as an icon under the “My Pets” section.

5.1.1 If there are registered pets, *PetCare* shall display each pet’s image with a “+” icon to add a new pet profile.

5.1.2 If there are no registered pets under “My Pets”, *PetCare* shall display only the “+” icon to prompt the user to add a pet profile.

5.1.3 *PetCare* shall allow the user to tap on a pet's image or name to access the selected pet's details.

5.1.4 *PetCare* shall allow the user to view, edit, or delete the details of each registered pet.

5.2 *PetCare* shall display a news article carousel on the Home page to showcase the latest 5 pet-related articles.

5.2.1 *PetCare* shall display a thumbnail image on the carousel.

5.2.2 *PetCare* shall display the title of the article on the carousel.

5.2.3 *PetCare* shall display the name of the poster below the title of the article on the carousel.

5.2.4 *PetCare* shall display the profile picture of the article's poster to the left of the poster's name.

5.2.5 *PetCare* shall display the date the article was published.

5.2.6 *PetCare* shall automatically rotate through the articles in the carousel every 10 seconds.

5.2.7 *PetCare* shall allow the user to manually swipe through the articles in the carousel.

5.2.8 *PetCare* shall allow the user to tap on an article in the carousel to view its full content.

5.3 *PetCare* shall display a "See All" button beside the "Browse Articles" text above the news carousel.

5.3.1 When the user taps on the "See All" button, *PetCare* redirects the user to the Browse Articles page, which displays all articles posted by all users of the app.

5.4 *PetCare* shall display a bottom navigation bar on the Home page that provides navigation to different sections of the app.

5.4.1 *PetCare* shall redirect the user to the "Missing Pets" page when the "Missing" icon is tapped.

5.4.2 *PetCare* shall redirect the user to the "Adoption Center" page when the "Adoption" icon is tapped.

5.4.3 *PetCare* shall redirect the user to the "Explore Pet-Care Amenities" page when the "Nearby" icon is tapped.

5.4.4 *PetCare* shall redirect the user to the "Personal Account" page when the "Account" icon is tapped.

5.5 *PetCare* shall display a fixed chatbot icon on the bottom right corner of the Home page, above the bottom navigation bar.

5.5.1 When the user taps on the chatbot icon, *PetCare* shall redirect the user to the 'Chatbot' page.

4.6 View Personal Account

4.6.1 Description and Priority

Allow users to view their personal account information in the *PetCare* application..

Overall Priority	Medium
------------------	--------

4.6.2. Stimulus/ Response Sequences

Use Case ID:	UC_ACC_1		
Use Case Name:	View Personal Account		
Created By:	Keng Jia Chi	Last Updated By:	Keng Jia Chi
Date Created:	25th August 2024	Date Last Updated:	25th August 2024

Actor:	User
Description:	Allow users to view their personal account information.
Preconditions:	1. The user is logged into their account. 2. The user has navigated to the Account Information page.
Postconditions:	The user's account information is displayed.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. The user selects the "Personal Account" page. 2. The system retrieves the user's information from the database. 3. The system displays the following information: a. Name b. Email address c. Phone number d. Address 4. The user can edit their information using the included use case Edit Information. 5. The user logs out from their existing account using the included use case Log Out.
Alternative Flows:	None
Exceptions:	None

Includes:	1. Edit Information 2. Log Out
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

4.6.3. Functional Requirement

6. *PetCare* shall allow users to view their personal account information.

6.1 When the user selects the “Personal Account” page, *PetCare* shall retrieve the user’s information from the database.

6.2 *PetCare* shall display the following user information on the Personal Account page:

6.2.1 *PetCare* shall display the user’s name.

6.2.2 *PetCare* shall display the user’s email address.

6.2.3 *PetCare* shall display the user’s phone number.

6.2.4 *PetCare* shall display the user’s address.

6.2.5 *PetCare* shall display the user’s profile picture.

4.7 Edit Personal Information

4.7.1 Description and Priority

Allow users to edit their personal account information in the *PetCare* application.

Overall Priority	Medium
------------------	--------

4.7.2. Stimulus/ Response Sequences

Use Case ID:	UC_ACC_2		
Use Case Name:	Edit Personal Information		
Created By:	Keng Jia Chi	Last Updated By:	Keng Jia Chi
Date Created:	25th August 2024	Date Last Updated:	25th August 2024

--	--	--	--

Actor:	User
Description:	Allow users to edit their personal account information.
Preconditions:	1. The user has navigated to the Account Information page.
Postconditions:	The user's account information is updated with the new details.
Priority:	Medium
Frequency of Use:	Medium
Flow of events:	1. The user clicks the "Edit Information" button. 2. The system makes the field editable. 3. The user enters the new information. 4. The user clicks the "Save" button. 5. The system validates the new information. 6. If valid, the system updates the database with the new information. 7. The system displays an "Information update successfully" message to the user and shows the updated information.
Alternative Flows:	AF-S6: Information is invalid 1. The system displays an error message. 2. The user is prompted to correct the information. 3. Return to step 3.
Exceptions:	None
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

4.7.3. Functional Requirement

7. PetCare shall display an "Edit Information" button in the Personal Account page for users to edit their personal account information.

7.1 *PetCare* shall display an "Edit Information" button on the Personal Account page.

7.2 When the user taps on the "Edit Information" button, *PetCare* shall make the information fields editable.

7.2.1 *PetCare* shall allow editing of the user's name.

7.2.2 *PetCare* shall allow editing of the user's email address.

7.2.3 *PetCare* shall allow editing of the user's phone number.

7.2.4 *PetCare* shall allow editing of the user's address.

7.2.5 *PetCare* shall allow editing of the user's profile picture.

7.3 *PetCare* shall provide a "Save" button for users to submit their changes.

7.4 When the user clicks the "Save" button, *PetCare* shall validate the new information.

7.4.1 If the information is valid, *PetCare* shall update the user's information in the database.

7.4.2 If the information is invalid, *PetCare* shall display an error message and prompt the user to enter the correct information.

7.5 After successful update, *PetCare* shall display an "Information updated successfully" message to the user.

7.6 *PetCare* shall display the updated information on the Personal Account page after a successful update.

4.8 View All Articles

4.8.1 Description and Priority

Existing users of *PetCare* are able to browse all pet-related articles. Users are able to access articles by tapping "See All" on the Home page. *PetCare* displays articles in a card format with a thumbnail, title, published date, poster's name and profile picture.

Overall Priority	Medium
------------------	--------

4.8.2 Stimulus/ Response Sequences

Use Case ID:	UC_ARC_1		
Use Case Name:	Browse All Articles		
Created By:	Jasmine Tye	Last Updated By:	Jasmine Tye
Date Created:	31st August 2024	Date Last Updated:	19th October 2024

Actor:	User
Description:	Allows a user to browse, filter and search all pet-related articles.

Preconditions:	1. The user has tapped on the “See All” button in the Home page. 2. The user is logged in and authenticated. 3. The user is navigated to the Home Page.
Postconditions:	1. The articles are displayed with specific details, including a thumbnail image, title, poster's name, profile picture, and the relative time since the article was published. 2. The user can tap on any article to view its full content.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. The system displays the Browse Articles section in the Home page. 2. The user taps on the “See All” button under the Browse Articles section. 3. The system redirects the user to the “Browse Articles” page. 4. The system displays a bottom navigation bar that has a “All” and “Posted” section. 5. The system highlights the “All” section in the bottom navigation bar of the “Browse Articles” page. 6. The system will highlight the “ALL” category and display all articles under all categories by default. 7. For each article, the system displays: a. A thumbnail image of the article b. Title of the article c. The name of the poster who posted the article d. The profile picture of the poster e. The date the article was published. 8. The user scrolls through the list of articles. 9. The user taps on an article to view its full content. 10. Use Case continue on UC_ARC_4
Alternative Flows:	AF-S5: No Articles Available 1. If no articles are available in a selected category, PetCare displays a message saying “No articles available”
Exceptions:	None
Includes:	UC_HOME_1 - View Home Page UC_ARC_4 - View Article Details
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

4.8.3 Functional Requirement

8. *PetCare* shall manage and display all pet-related articles.

8.1 *PetCare* shall display a bottom navigation bar with two sections: "ALL" and "POSTED" (which only displays the articles posted by the user).

8.1.1 The "ALL" section of the bottom navigation is selected by default, *PetCare* shall display all articles.

8.1.2 Under the "ALL" bottom navigation tab, *PetCare* shall display all articles categorized into: Lifestyle, Grooming, Community, Health and Wellness, and Others.

8.2 *PetCare* shall display article cards in a specific format.

8.2.1 *PetCare* shall display a thumbnail image of the article.

8.2.2 *PetCare* shall display the title of the article.

8.2.2.1 If the title of the article exceeds the width of the article card, the rest of the title is hidden and an ellipsis(...) is shown.

8.2.3 *PetCare* shall display the name of the poster who posted the article.

8.2.4 *PetCare* shall display the profile picture of the poster who posted the article.

8.2.5 *PetCare* shall display the date the article was published.

4.9 Filter Articles

4.9.1 Description and Priority

Existing users of *PetCare* can filter articles by category by selecting a different category tab. *PetCare* will dynamically display only articles associated with the selected category.

Overall Priority	Medium
------------------	--------

4.9.2 Stimulus/ Response Sequences

Use Case ID:	UC_ARC_2		
Use Case Name:	Filter Articles		
Created By:	Jasmine Tye	Last Updated By:	Jasmine Tye
Date Created:	5 September 2024	Date Last Updated:	20th October 2024

Actor:	User
Description:	Allows a user to filter articles by category, such as Lifestyle, Grooming, Community, Health and Wellness, and Others.

Preconditions:	1. The user is on the “Browse Articles” page.
Postconditions:	1. Search results for articles related to the input keyword(s) are displayed to the user. 2. If no results are found, a message is displayed indicating no articles are available for the searched keyword.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. The system categorizes and displays articles into the following sections: Lifestyle, Grooming, Community, Health and Wellness and Others. 2. The user taps a category from the available options. 3. Upon category selection, the system filters the content and displays only the articles associated with the selected category. 4. The user can switch between categories by tapping on another category. Each time a new category is selected, the system dynamically updates the displayed content to match the newly selected category. 5. The user scrolls through the list of articles within the selected category. 6. The user taps on an article to view its full content. 7. Use Case continue on UC_ARC_4
Alternative Flows:	AF-S3: No Articles Available in Category 1. If no articles are available in a selected category, PetCare displays a message saying “No articles available”
Exceptions:	None
Includes:	None
Special Requirements:	The system must filter the results without refreshing or reloading the page.
Assumptions:	The Articles page must be fully functionable and display accurate results.
Notes and Issues:	None

4.9.3 Functional Requirement

9. *PetCare* shall categorize all articles into the following sections: Lifestyle, Grooming, Community, Health and Wellness, and Others.

9.1 *PetCare* shall display category buttons on the "Browse Articles" page that allow users to filter articles based on selected categories.

9.1.1 The "ALL" category shall be selected by default, displaying articles across all categories.

9.1.2 The selected category button shall be highlighted in purple, while unselected category buttons shall be gray.

9.2 When a user taps a category button, *PetCare* shall filter the articles to display only those associated with the selected category.

9.2.1 The system shall dynamically update the displayed articles in real-time without refreshing or reloading the page.

9.2.2: Users can tap on another category at any time to switch the filtered view to the newly selected category.

9.3 *PetCare* shall allow users to scroll through the list of articles within the selected category.

9.4 *PetCare* shall allow users to tap on any article within the filtered list to view its full content (continuing with UC_ARC_4 - View Article Details).

4.10 Search Articles

4.10.1 Description and Priority

Existing users of *PetCare* can search and further filter the pet-related articles by entering keywords in a search bar on the "Browse Posted Articles" or "Browse Articles" page. As the user is typing in the search bar, *PetCare* displays relevant articles that match the keywords.

Overall Priority	Medium
------------------	--------

4.10.2 Stimulus/ Response Sequences

Use Case ID:	UC_ARC_3		
Use Case Name:	Search Articles		
Created By:	Jasmine Tye	Last Updated By:	Jasmine Tye
Date Created:	5 September 2024	Date Last Updated:	19th October 2024

Actor:	User
Description:	Allows a user to search for pet-related articles using keywords.

Preconditions:	1. The user is on the “Browse Articles” page.
Postconditions:	1. The user can view articles filtered by the selected category. 2. If no articles are available in the selected category, a message is displayed.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. The user inputs a keyword into the search bar. 2. As the user types, the system switches to the "ALL" category view and displays articles that match the input keyword. 3. The system displays a “X” icon at the right end of the search bar to clear the search. 4. If the user taps on the "X" icon, the system clears the search and returns to the default “ALL” category view. 5. User will select the article to view its full content. 6. Use Case continue on UC_ARC_4
Alternative Flows:	AF-S2: No Result Found 1. When a user is searching a keyword in the search bar, if there are no results, PetCare displays a message saying “No articles with the {user input} keyword(s) available. Please use a different keyword.”
Exceptions:	None
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

4.10.3 Functional Requirement

10. *PetCare* shall display a search bar on the "Browse Articles" page with placeholder text “Search anything here” when no keyword is entered.

10.1 *PetCare* shall allow the user to type keywords into the search bar to search for articles.

10.1.1 Upon entering a keyword, the system shall automatically switch to the "ALL" category, displaying all relevant articles matching the keyword across all categories.

10.2 *PetCare* shall filter the articles dynamically as the user types in the search bar, displaying results that match the entered keyword.

10.2.1 The system shall dynamically update the displayed articles in real-time without refreshing or reloading the page.

10.3 *PetCare* shall display an “X” icon at the right end of the search bar when there is text in the search field.

10.3.1 When the user taps on the “X” icon, the system shall clear the search bar text and revert to displaying all articles in the "ALL" category by default.

10.5 *PetCare* shall allow users to scroll through the filtered list of articles based on the search results.

10.5 *PetCare* shall allow users to tap on any article within the filtered list to view its full content (continuing with UC_ARC_4 - View Article Details).

4.11 View Article Details

4.11.1 Description and Priority

Existing users of *PetCare* can view the full details of a selected article. This can be done by selecting an article from the "Browse All Articles" page, the "Posted" section, or the article carousel on the Home page. Upon selection, the system displays a new page containing the article's thumbnail image, category, title, full text, poster's name, and publication date.

Overall Priority	High
------------------	------

4.11.2 Stimulus/ Response Sequences

Use Case ID:	UC_ARC_4		
Use Case Name:	View Article Details		
Created By:	Jasmine Tye	Last Updated By:	Jasmine Tye
Date Created:	31st August 2024	Date Last Updated:	19th October 2024

Actor:	User
Description:	Allows the user to view the full details of the selected article.
Preconditions:	1. The user has selected an article either from the article carousel in Home page, browse all articles page or posted articles page.
Postconditions:	1. The user can view the full details of the selected article,

	including the article's content and all related information.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user opens the "Browse All" articles page and taps on an article. 2. The system opens a new page displaying the full article details: a. The thumbnail image of the article b. The category of the article c. The title of the article d. The full text of the article e. The name of the poster f. The profile picture of the poster g. The relative time since the article was published 3. The user scrolls through the article content.
Alternative Flows:	AF-S6: Accessing an Article from Home page 1. The user taps on an article from the article carousel in the Home page. 2. The user continues with Step 2. AF-S6: Accessing an Article from Posted article page 1. The user taps on an article from the "Posted Article" page. 2. The user continues with Step 2. AF-S3: Back button 1. The user taps on the 'Back' button. 2. The system redirects the user back to the previous page.
Exceptions:	EX3: Article Not Found 1. If the selected article has been deleted or is unavailable, <i>PetCare</i> displays a popup message saying "Article not found" 2. The system redirects the user back to the page they were previously at.
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

4.11.3 Functional Requirement

11. *PetCare* shall allow users to tap on an article to view its full details from any of the following locations: The "Browse All Articles" page, The "Posted" section in the bottom navigation bar and the article carousel on the Home page

11.1 Upon selection of an article, *PetCare* shall open a new page displaying the following article details:

11.1.1 *PetCare* shall display the thumbnail image of the article.

11.1.2 *PetCare* shall display the category label of the article.

11.1.3 *PetCare* shall display the title of the article.

11.1.4 *PetCare* shall display the full content of the article text.

11.1.5 *PetCare* shall display the name of the poster.

11.1.6 *PetCare* shall display the date the article was published.

11.2 *PetCare* shall allow users to scroll through the full content of the article.

11.3 *PetCare* shall display a back button that allows users to navigate back to the previous page.

4.12 Add New Article

4.12.1 Description and Priority

Existing users of *PetCare* can create and submit new pet-related articles. *PetCare* will facilitate the creation of a new article by allowing users to fill out a form that includes allowing users to fill out a form that includes fields for the article title, body text, category selection, and a thumbnail image. Once submitted, the article will be stored in the database and appear in both the "All Articles" section and the "Posted Articles" section, with visibility in the latter restricted to the original poster.

Overall Priority	Medium
------------------	--------

4.12.2 Stimulus/ Response Sequences

Use Case ID:	UC_ARC_5		
Use Case Name:	Add New Article		
Created By:	Jasmine Tye	Last Updated By:	Jasmine Tye
Date Created:	31st August 2024	Date Last Updated:	19th October 2024

Actor:	User
--------	------

Description:	Allows a user to create and submit a new pet-related article to the <i>PetCare</i> application.
Preconditions:	1. The user is on the “Posted Articles” page of the application.
Postconditions:	1. A new article is created and stored in the system. 2. The system assigns the article to the poster in the database. 3. The new article appears in the “All Articles” section 4. The new article appears in the “Posted Articles” section only to the original poster.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. The user taps on the file icon located at the bottom right of the bottom navigation bar in the “Posted Articles” page. 2. The system displays the article creation form. 3. The user fills in the following information: a. Title of the article b. Body text of the article c. Category of the article from the given options (Lifestyle, Grooming, Community, Health and Wellness, Others) d. Thumbnail image for the article 4. The user taps on the “Confirm” button. 5. The system saves the new article to the database. 6. The system displays a success popup message. 7. The system redirects the user to the “Posted” section where the new article is now visible and posted.
Alternative Flows:	AF-S3: User Cancels Article Creation 1. During step 3, the user can tap on the “Back” button to stop the article creation. 2. The system displays a popup message to confirm the action. 3. If the user taps on the “Ok” button, the system closes the form and returns the user back to the “Posted Articles” page without creating a new article. 4. If the user taps on the “No” button, the system closes the popup message and the user returns to the article creation form.
Exceptions:	EX3: Image Upload Failure 1. If there is an issue with uploading the image, <i>PetCare</i> displays an error message and asks the user to re-upload the image.
Includes:	None
Special Requirements:	None
Assumptions:	- The user publishes articles that are related to pets. - The user selects the correct category for the article.

Notes and Issues:	None
-------------------	------

4.12.3 Functional Requirement

12. *PetCare* shall allow all users to add a new article to the database and display it in “Browse Posted Articles” and “Browse All Articles”.

12.1 *PetCare* shall provide a file icon located at the bottom right of the “Posted Articles” page that allows users to initiate the creation of a new article.

12.2 When the user taps on the file icon, *PetCare* shall display an article creation form with the following fields that are compulsory to add an article to the database:

12.2.1 A text field to input the title of the article.

12.2.2 A text field to input the body text of the article.

12.2.3 A selection field to choose the article category from the given options.

12.2.4 An image picker to upload a thumbnail image from the user’s device.

12.3 *PetCare* must ensure that the user fills in all the fields before allowing article creation.

12.3.1 *PetCare* shall display an alert if all fields are not filled.

12.4 *PetCare* must verify all input information.

12.4.1 *PetCare* must verify that the article title does not exceed 20 words.

12.4.2 *PetCare* must verify that the article body text has a minimum of 500 characters.

12.4.3 *PetCare* must provide an alert to the user to show feedback regarding the validation issues.

12.5 *PetCare* must create an article with the user input together with the user details as the poster of the article to the database.

12.5.1 The newly posted article shall be displayed under both the "All Articles" and "Posted" sections, with visibility in the "Posted" section restricted to the poster.

12.6 *PetCare* must redirect the user to the “Browse Posted Article” page, where the new article card shall be visible to the user who posted it.

4.13 View Posted Articles

4.13.1 Description and Priority

Existing users of *PetCare* can view all articles that they have posted. The system shall retrieve and display a list of the user’s articles under the “Browse Posted Articles” page.

Overall Priority	Medium
------------------	--------

4.13.2 Stimulus/ Response Sequences

Use Case ID:	UC_ARC_6		
Use Case Name:	View Posted Articles		
Created By:	Jasmine Tye	Last Updated By:	Jasmine Tye
Date Created:	31st August 2024	Date Last Updated:	31st August 2024

Actor:	User
Description:	Allows users to view the articles that each user has posted onto the <i>PetCare</i> app.
Preconditions:	1. The user has navigated to the “Browse Articles” page and has tapped on the “Posted” section in the bottom navigation bar.
Postconditions:	1. The user can see a list of all the articles that they have posted under the “Posted” section. 2. The user is able to interact with their posted articles.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. The user navigates to the “Browse Articles” page. 2. The user taps on the “Posted” section on the bottom navigation bar. 3. The system displays the “Posted Article” page. 4. The system retrieves and displays all articles posted by the user. 5. The user can scroll through the list of their posted articles. 6. The user can tap on article(s) to view the full details of the article.
Alternative Flows:	AF-S3: No articles posted 1. If the user has not posted any articles, the system displays a message stating “No articles posted. Share your articles.”
Exceptions:	EX1: Network Error 1. If there is a network or database issue while fetching the articles, <i>PetCare</i> displays an error message indicating that articles could not be loaded and suggests the user to try again later.
Includes:	UC_ARC_4 - View Article Details
Special Requirements:	None
Assumptions:	None

Notes and Issues:	None
-------------------	------

4.13.3 Functional Requirement

13. *PetCare* shall retrieve and display all articles that the logged in user has posted.
- 13.1 *PetCare* shall switch the view and highlight the “POSTED” section in the bottom navigation bar.
 - 13.2 *PetCare* shall display article cards in a specific format.
 - 13.2.1 *PetCare* shall display a thumbnail image of the article.
 - 13.2.2 *PetCare* shall display the title of the article.
 - 13.2.2.1 If the title of the article exceeds the width of the article card, the rest of the title is hidden and an ellipsis(...) is shown.
 - 13.2.3 *PetCare* shall display the name of the poster who posted the article.
 - 13.2.4 *PetCare* shall display the profile picture of the poster who posted the article.
 - 13.2.5 *PetCare* shall display the date the article was published.
 - 13.3 *PetCare* shall allow searching of articles that the logged in user has posted (UC_ARC_3).

4.14 Edit Article

4.14.1 Description and Priority

Existing users of *PetCare* can make changes to their previously posted articles. Users are required to navigate to the “Posted” section of the app and select the vertical ellipsis icon next to the article. This action reveals a "Edit" option, which opens a form pre-filled with the article's current details. Users can modify any fields which will then be reflected in the database.

Overall Priority	Medium
------------------	--------

4.14.2 Stimulus/ Response Sequences

Use Case ID:	UC_ARC_7		
Use Case Name:	Edit Article		
Created By:	Jasmine Tye	Last Updated By:	Jasmine Tye

Date Created:	31st August 2024	Date Last Updated:	31st August 2024
---------------	------------------	--------------------	------------------

Actor:	User
Description:	Allows a user to edit an article that they have previously posted on the <i>PetCare</i> app.
Preconditions:	1. The user has posted at least one article. 2. The user has selected an article from the “Posted Article” page.
Postconditions:	1. The user’s changes to the article are saved and updated in the <i>PetCare</i> database. 2. The edited article is updated in both the “Posted Article” and “All Article” sections.
Priority:	Medium
Frequency of Use:	Low
Flow of Events:	1. The user navigates to the “Posted” section from the bottom navigation bar in the Browse Articles page. 2. The user selects an article that they previously posted. 3. The user taps on the vertical ellipsis icon on the top right corner of the article detail page which is only visible to the original poster of the article. 4. <i>PetCare</i> displays the “Edit” and “Delete” options. 5. The user taps on the “Edit” options. 6. <i>PetCare</i> redirects the user to the Edit Article form, pre-filled with the existing details of the article: a. Title of the article. b. Body text of the article. c. Selected category of the article. d. Existing thumbnail image of the article. 7. The user can modify any of the article details: a. Changing the title b. Updating the body text c. Selecting a different category d. Uploads a new thumbnail image 8. The user taps on the “Save” button to submit the changes. 9. The system displays a success message. 10. The system updates the edited article details in both the “Posted” and “All” sections. 11. The system redirects the user back to the article view to show the updated content.
Alternative Flows:	AF-S3: User Cancels Editing 1. During steps 6-7, the user can tap on the “Back” button to stop the editing process. 2. The system displays a popup message to confirm the action. 3. If the user taps on the “Ok” button, the system closes the form and returns the user back to the article details without making any changes.

	4. If the user taps on the “No” button, the system closes the popup message and the user returns to the edit process.
Exceptions:	EX1: Network Error 1. If there is a network or database issue while updating the article details, <i>PetCare</i> displays an error message indicating that articles could not be updated and suggests the user to try again later.
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

4.14.3 Functional Requirement

14. *PetCare* must allow only the poster of the article to edit the details of the article.

14.1 *PetCare* shall display a vertical ellipsis icon on the top right corner of the article detail page, visible only to the original poster of the article.

14.2 *PetCare* must present the “Edit” option when the user taps on the ellipsis.

14.3 *PetCare* must display the “Edit Article Form”, pre-filled with the article’s existing details, including:

14.3.1 *PetCare* must pre-fill the article thumbnail image with the article’s existing thumbnail image.

14.3.2 *PetCare* must pre-fill the article title with the article’s existing article title.

14.3.3 *PetCare* must pre-fill the article body text with the article’s existing body text.

14.3.4 *PetCare* must select the article category option with the article’s existing article category.

14.4 *PetCare* must allow the user to modify any of the article details, including:

14.4.1 *PetCare* must allow the user to pick a different article thumbnail image.

14.4.2 *PetCare* must allow the user to modify the article title.

14.4.3 *PetCare* must allow the user to modify the article body text.

14.4.4 *PetCare* must allow the user to select a different article category.

14.5 *PetCare* must ensure that the user fills in all the fields before allowing article details modification.

14.5.1 *PetCare* shall display an alert if all fields are not filled.

14.6 *PetCare* must verify all input information.

14.6.1 *PetCare* must verify that the article title does not exceed 20 words.

14.6.2 *PetCare* must verify that the article body text has a minimum of 500 characters.

14.6.3 *PetCare* must provide an alert to the user to show feedback regarding the validation issues.

14.7 *PetCare* must display loading feedback when edit requests are being processed.

14.8 *PetCare* must update the article details with the modified data in the database.

14.8.1 *PetCare* must display a success feedback to inform the user about a successful article update.

14.9 *PetCare* must redirect the user to the article page with the modified data displayed.

4.15 Delete Article

4.15.1 Description and Priority

Existing users of *PetCare* can delete their previously posted articles. Users are required to navigate to the “Posted” section of the app and select the vertical ellipsis icon next to the article. This action reveals a "Delete" option, which prompts a confirmation dialog. Once confirmed, the article is permanently removed from the *PetCare* database and is no longer visible in either the "Posted" or "All" sections.

Overall Priority	Medium
------------------	--------

4.15.2 Stimulus/ Response Sequences

Use Case ID:	UC_ARC_8		
Use Case Name:	Delete Article		
Created By:	Jasmine Tye	Last Updated By:	Jasmine Tye
Date Created:	31st August 2024	Date Last Updated:	31st August 2024

Actor:	User
Description:	Allows the user to delete an article that they have previously posted on the <i>PetCare</i> app.
Preconditions:	1. The user has posted at least one article. 2. The user has selected an article from the “Posted Article” page.
Postconditions:	1. The selected article is permanently removed from the <i>PetCare</i> database. 2. The deleted article is no longer visible in both the “Posted” and “All” sections in the “Browse Articles” page.

Priority:	Medium
Frequency of Use:	Low
Flow of Events:	1. The user navigates to the “Posted” section from the bottom navigation bar in the Browse Articles page. 2. The user selects an article that they previously posted. 3. The user taps on the vertical ellipsis icon on the top right corner of the article detail page which is only visible to the original poster of the article. 4. <i>PetCare</i> displays the “Edit” and “Delete” options. 5. The user taps on the “Delete” option. 6. <i>PetCare</i> displays a popup box asking the user to confirm the deletion, “Are you sure you want to delete this article? This action cannot be undone.” 7. The user taps on the “Yes” button to confirm the deletion. 8. <i>PetCare</i> permanently deletes the article from the database. 9. <i>PetCare</i> updates the “Posted” and “All” sections to remove the deleted article from view. 10. <i>PetCare</i> displays a confirmation message indicating that the article has been deleted successfully. 11. The system redirects the user back to the article view to show the updated content.
Alternative Flows:	AF-S3: User Cancels Deletion 1. In Step 7, if the user taps on the “No” button in the confirmation popup, <i>PetCare</i> closes the pop-up box without deleting the article. 2. <i>PetCare</i> redirects the user back to the article details without making any changes.
Exceptions:	EX1: Network Error 1. If there is a network or database issue while updating the article details, <i>PetCare</i> displays an error message indicating that articles could not be deleted and suggests the user to try again later.
Includes:	UC_ARC_4 - View Article Details
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

4.15.3 Functional Requirement

15. *PetCare* must allow only the poster of the article to delete the article.

15.1 *PetCare* shall display a vertical ellipsis icon on the top right corner of the article detail page, visible only to the original poster of the article.

15.2 *PetCare* must present the “Delete” option when the user taps on the ellipsis.

15.3 *PetCare* must display a delete confirmation modal to confirm the deletion.

15.3.1 *PetCare* must display the “No” option to close the delete confirmation modal.

15.4 *PetCare* must permanently delete the article from the database.

15.4.1 *PetCare* must update both the “Posted” and “All” sections to remove the deleted article from view.

15.5 *PetCare* must redirect the user to the “Browse Posted Articles” page to show the user that the article has been deleted.

4.16 View Pet Information

4.16.1 Description and Priority

PetCare users that have added at least one pet will be able to view their pet details including name, breed, sex, age and so on. It will display the information in a modal when users click on a pet image on the Home Screen.

Overall Priority	High
------------------	------

4.16.2 Stimulus/ Response Sequences

Use Case ID:	UC_PI_01		
Use Case Name:	View Pet Information		
Created By:	Sih Jia Qi	Last Updated By:	Sih Jia Qi
Date Created:	26th August 2024	Date Last Updated:	26th August 2024

Actor:	User
Description:	Allow users to view their pet information such as name, date of birth, breed, sex, weight, coat color, special markings and medic conditions.
Preconditions:	1. The user has a device with Petcare installed. 2. The user is logged in and authenticated. 3. The user has added at least a pet.
Postconditions:	Display the pet(s) information on the Pet Information modal.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user clicks on the pet image located in the My Pets section on the Home page to view the details of their registered pet. 2. The system displays the following details on the Pet Information modal: - Pet name - Date of birth - Breed - Sex

	- Weight - Coat color - Special markings - Medical conditions 3. On the Pet Information modal, the user can click on the “Edit” icon to edit the pet details. 4. On the Pet Information modal, the user can click on the “Delete” icon to delete the pet and its details from the database.
Alternative Flows:	None
Exceptions:	EX1: Data retrieval error 1. If the system is unable to connect to the database, the system returns a message prompting the user to try again later.
Includes:	UC_PI_02 - Edit Pet Information UC_PI_04 - Delete Pet
Special Requirements:	None
Assumptions:	Database is connected successfully.
Notes and Issues:	None

4.16.3 Functional Requirement

16. *PetCare* must allow users to view their pets information.

16.1 *PetCare* must display each pet’s details on a Pet Information modal including

16.2.1 Pet Name

16.2.2 Date of Birth

16.2.3 Breed

16.2.4 Sex

16.2.5 Weight

16.2.6 Coat Color

16.2.7 Special Markings

16.2.8 Medic Conditions

4.17 Edit Pet Information

4.17.1 Description and Priority

Existing users can edit their pet information on the edit pet page. An edit pet form will be populated on the edit pet screen when the user clicks on the edit icon on the pet information modal.

Overall Priority	Medium
------------------	--------

4.17.2 Stimulus/ Response Sequences

Use Case ID:	UC_PI_02		
Use Case Name:	Edit Pet Information		
Created By:	Sih Jia Qi	Last Updated By:	Sih Jia Qi
Date Created:	26th August 2024	Date Last Updated:	26th August 2024

Actor:	User
Description:	Allow users to update their pet information such as name, date of birth, breed, sex, weight, coat color, special markings and medic conditions.
Preconditions:	1. The user has a device with <i>PetCare</i> installed. 2. The user is logged in and authenticated. 3. The user has registered at least a pet.
Postconditions:	Save and display the user's updated pet information.
Priority:	Medium
Frequency of Use:	Low
Flow of Events:	1. The user taps on the image of one of their pets in the My Pets section. 2. The user taps on the "Edit" icon on the Pet Information modal. 3. The user is redirected to the EditPetForm page. 4. The user can update the details of their pet including: - Pet name - Date of birth - Breed - Sex - Weight - Coat color - Special markings - Medical conditions 5. The user taps on the "Done" button after editing the details. 6. The system redirects the user to the Home page with the updated pet information loaded when the user clicks on the pet image again.
Alternative Flows:	AF-S4: The user does not save the changes

	1. If the user clicks on the “Cancel” button on the editing page, all the edits will not be saved to the database. 2. The user will be redirected to the homepage with no pet information updated.
Exceptions:	EX4: Data update failure 1. If the system is unable to update the pet due to network issues or server issues, the system displays an error message and prompts the user to try again later. 2. The pet information will not change.
Includes:	None
Special Requirements:	None
Assumptions:	Database is connected.
Notes and Issues:	None

4.17.3 Functional Requirement

17. *PetCare* must allow the user to edit their pet’s information.

17.1 *PetCare* must display an “Edit” button on the Pet Information modal.

17.2 If the user taps on the “Edit” button:

17.2.1 *PetCare* shall allow editing of the Pet Name.

17.2.2 *PetCare* shall allow editing of the Date of Birth.

17.2.3 *PetCare* shall allow editing of the Breed.

17.2.4 *PetCare* shall allow editing of the Sex.

17.2.5 *PetCare* shall allow editing of the Weight.

17.2.6 *PetCare* shall allow editing of the Coat Color.

17.2.7 *PetCare* shall allow editing of the Special Markings.

17.2.8 *PetCare* shall allow editing of the Medical Conditions.

4.18 Add Pet

4.18.1 Description and Priority

Existing PetCare users can register their pets into the system including pet details and images.

Overall Priority	Medium
------------------	--------

4.18.2 Stimulus/ Response Sequences

Use Case ID:	UC_PI_03
--------------	----------

Use Case Name:	Add Pet		
Created By:	Sih Jia Qi	Last Updated By:	Sih Jia Qi
Date Created:	26th August 2024	Date Last Updated:	26th August 2024

Actor:	User
Description:	Allow users to add a new pet and its information such as name, date of birth, breed, sex, weight, coat color, special markings and medic conditions.
Preconditions:	1. The user has a device with <i>PetCare</i> installed. 2. The user is logged in and authenticated.
Postconditions:	The new registered pet is added successfully into the database and the system displays it in the My Pets section on the Home page
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. The user clicks on the '+' icon in the My Pets section on the Home page. 2. The system redirects the user to the AddPetForm page with the following input boxes: a. Pet name b. Date of birth c. Breed d. Sex e. Weight f. Coat color g. Special markings h. Medical conditions 3. The user taps on the "Done" button to submit the form after entering the pet details 4. The system redirects the user to the Home page.
Alternative Flows:	AF-S3: The user does not want to add a new pet 1. If the user taps on the "Cancel" button, the system will redirect the user back to the Home page with no new pet added.
Exceptions:	EX3: Form submission error 1. Form submission fails due to network issues or server errors, the system displays an error pop-up message asking the user to try again later.
Includes:	None
Special Requirements:	None

Assumptions:	Database is connected.
Notes and Issues:	None

4.18.3 Functional Requirement

18. *PetCare* must allow the user to add a new pet in the My Pets section on the Home page.

18.1 *PetCare* must display a '+' button on the My Pets section to add a new pet.

18.2 When the user clicks on the '+' button, *PetCare* must redirect the user to the add pet form screen.

18.1.1 *PetCare* shall display a text field for "Pet Name".

18.1.2 *PetCare* shall display a date field for "Date of Birth".

18.1.3 *PetCare* shall display a text field for "Breed".

18.1.4 *PetCare* shall display the 'M' and 'F' options for "Sex".

18.1.5 *PetCare* shall display a text field for "Weight".

18.1.6 *PetCare* shall display a text field for "Coat color".

18.1.7 *PetCare* shall display a multiline field for "Special markings".

18.1.8 *PetCare* shall display a multiline field for "Medic conditions".

18.3. *PetCare* shall display the "Save" and "Cancel" button on the Pet Information modal.

18.3.1. When the user taps on the "Save" button, *PetCare* shall save the data to the database after verifying the format of the inputs.

18.3.2. When the user taps on the "Cancel" button, *PetCare* shall return to the home page.

4.19 Delete Pet

4.19.1 Description and Priority

PetCare users can delete their registered pets from the system by simply clicking on the delta icon on the pet information modal. The pet will be removed from the database.

Overall Priority	Medium
------------------	--------

4.19.2 Stimulus/ Response Sequences

Use Case ID:	UC_PI_04
--------------	----------

Use Case Name:	Delete Pet		
Created By:	Sih Jia Qi	Last Updated By:	Sih Jia Qi
Date Created:	26th August 2024	Date Last Updated:	26th August 2024

Actor:	User
Description:	Allow users to delete their pet and its information.
Preconditions:	1. The user has a device with <i>PetCare</i> installed. 2. The user is logged in and authenticated. 3. The user has registered at least a pet.
Postconditions:	The deleted pet will be removed from the My Pets section.
Priority:	Medium
Frequency of Use:	Low
Flow of Events:	1. The user taps on the image or name of one of their pets in the My Pets section on the Home page. 2. The user taps on the “Delete” icon on the Pet Information popup. 3. The system displays a popup with the “Confirm” and “Cancel” buttons. 4. The user taps on the “Confirm” button. 5. The user is redirected to the Home page.
Alternative Flows:	AF-S3: Cancel delete option 1. If the user taps on the “Delete” button, the user will be redirected back to the Home page with no changes made to the pet information.
Exceptions:	EX4: Data deletion failure 1. If the system is unable to delete the pet due to network issues or server issues, the system displays an error message and prompts the user to try again later. 2. The pet is not deleted and will be displayed in the My Pets section.
Includes:	None
Special Requirements:	None
Assumptions:	Database is connected.
Notes and Issues:	None

4.19.3 Functional Requirement

19. *PetCare* must allow the user to delete an existing pet information.
 - 19.1 *PetCare* must display a “Delete” button on the Pet Information modal.
 - 19.2 When the user clicks on the “Delete” button, *PetCare* shall display a dialog with the “Confirm” and “Cancel” buttons.
 - 19.3 When the user taps the “Confirm” button, *PetCare* shall remove the pet and its details from the Home page.
 - 19.4 When the user clicks on the “Cancel” button, *PetCare* shall close the modal.

4.20 Display Active Missing Pets

4.20.1 Description and Priority

Existing users of *PetCare* can view a map of active missing pets reported by other users of *PetCare* application, displaying markers at each pets' last seen location. Users are also able to tap on any of the markers to display missing pet details and each sighting of the missing pet.

Overall Priority	High
------------------	------

4.20.2 Stimulus/ Response Sequences

Use Case ID:	UC_MP_1		
Use Case Name:	Display Active Missing Pet		
Created By:	Jasmine Tye	Last Updated By:	Jasmine Tye
Date Created:	25th August 2024	Date Last Updated:	19th October 2024

Actor:	User
Description:	Allows a user to view a list of active missing pets reported by other users.
Preconditions:	1. The user is logged in and authenticated to <i>PetCare</i> 2. The user has a GPS-enabled device. 3. The application has access to the user's current location.
Postconditions:	1. The system displays a map with markers showing the last seen location of missing pets. 2. The user can tap on a marker to view detailed information about the missing pet.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. The user navigates to the "Missing Pets" section in the app. 2. The app retrieves the user's current location using GPS. 3. The system sends queries to the database for active missing pet reports nearby to the user. 4. The system displays the missing pets' images as markers on the map at their first seen locations.

	5. The user taps on a marker to go to the selected missing pet's page to view more information about the selected missing pet: a. Name b. Breed c. Age d. Gender e. Photo f. Contact details of the owner g. Description posted by the owner 6. The system displays a new map interface displaying the record of multiple last seen details for the selected missing pet. 7. The system displays a GPS icon, allowing the user to quickly re-center the map on their current location with a single tap. 8. The new map interface includes: a. A timeline of sightings as markers on the map b. Date and time of each reported sighting c. Image posted by reporter d. Sighting description by reporter 9. When the user taps on the next button on the map which displays markers of sighting locations, the next sighting information is displayed.
Alternative Flows:	AF-S4: Incomplete Information 1. If some missing pets have incomplete information: a. Missing sighting image(s), system displays a default image b. Missing sighting description, system omits this information from the report.
Exceptions:	EX1: Location services unavailable 1. If GPS is disabled or the device is unable to determine the user's location, the system notifies the user and provides an option to manually input a location by postal code or keyword.
Includes:	None
Special Requirements:	Device permission for location access is enabled.
Assumptions:	The missing pet database is up-to-date and returns accurate information.
Notes and Issues:	None

4.20.3 Functional Requirement

20. *PetCare* must display a map interface that displays markers of active missing pets.

20.1 *PetCare* shall request permission from the user to access their location using a foreground permission request if it's not already granted.

20.1.1 If the user denies location access, the system shall display an error message indicating that location access is required to show their current location.

20.2 *PetCare* shall retrieve the user's current location using the device's GPS.

20.2.1 *PetCare* must display the user's current location on the map using a user marker that represents the user's position.

20.2.2 *PetCare* must display a GPS icon on the bottom right of the screen that centers the map on the user's current location, no matter where the user has moved the map to.

20.3 *PetCare* shall query the database and fetch all active missing pet reports.

20.4 *PetCare* must display the missing pets' images as markers on the map at their last seen location.

20.4.1 *PetCare* shall retrieve the longitude and latitude from the last seen location of each missing pet and display it accordingly on the map.

20.4.2 *PetCare* shall only display active missing pet cases.

20.5 *PetCare* shall allow users to tap on any of the active missing pet markers to redirect the user to the selected missing pet report page.

20.5.1 *PetCare* shall display the following information:

20.5.1.1 *PetCare* shall display the missing pet's name.

20.5.1.2 *PetCare* shall display the missing pet's breed.

20.5.1.3 *PetCare* shall display the missing pet's age.

20.5.1.4 *PetCare* shall display the missing pet's picture.

20.5.1.5 *PetCare* shall display the missing pet's gender.

20.5.1.6 *PetCare* shall display the owner's contact details.

20.5.1.7 *PetCare* shall display the description posted by the owner.

20.5.1.8 *PetCare* shall display the owner's name.

20.5.2 *PetCare* shall display a new map interface that displays the sighting markers for the selected missing pet.

20.5.2.1 The last seen details when the user first created the missing pet report is set as the first sighting and selected by default.

20.5.2.2 *PetCare* shall display the list of sightings as markers on the map.

20.5.2.3 *PetCare* shall display the sighting image.

20.5.2.4 *PetCare* shall display the sighting description.

20.5.2.5 *PetCare* shall display the sighting location in address format.

20.5.3 *PetCare* shall display a “->” icon on the bottom right of the sighting map view.

20.5.3.1 When the user taps on the “->” icon, the next sighting is selected and the sighting information in the sighting card is updated.

4.21 Report Pet as Missing

4.21.1 Description and Priority

Existing users of *PetCare* can report a missing pet by filling out a form that includes essential details like last seen location, date, time, an image, and a description. Basic pet information from the owner's registered pets is auto-filled for convenience. After

submitting, the report is verified, saved, and displayed as a marker on the map in the "Display Missing Pets" section, allowing other users to view the missing pet details.

Overall Priority	High
------------------	------

4.21.2 Stimulus/ Response Sequences

Use Case ID:	UC_MP_2		
Use Case Name:	Report Pet as Missing		
Created By:	Jasmine Tye	Last Updated By:	Jasmine Tye
Date Created:	25th August 2024	Date Last Updated:	25th August 2024

Actor:	Pet Owner
Description:	Allows the pet owner to report their missing pet by providing details about the pet and the circumstances under which it went missing.
Preconditions:	1. The pet owner is logged in and authenticated. 2. The pet owner has saved the profile of the pet that went missing.
Postconditions:	1. The system successfully saves and publishes the missing pet report made by the pet owner to the app. 2. Other users are able to view the missing pet report.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. The pet owner taps on the "Report Missing Pet" option on the Home page. 2. The system displays a list of the pet owner's registered pets from the "My Pets" section. 3. The pet owner selects the profile of the pet that went missing. 4. The system automatically fills in the report form with the selected pet's details: a. Name b. Breed c. Age d. Gender e. Photo 5. The pet owner inputs additional information in the form: a. Last seen location - postal codes only

	b. Last seen time c. Last seen date d. Last seen image e. Additional descriptions 6. The pet owner submits the report. 7. The system confirms the submission and displays the reported pet in the “Missing Pets” section. 8. The system displays the last seen information as the first sighting marker .
Alternative Flows:	AF-S2: No pets registered under “My Pets” 1. The system displays a prompt to add a pet profile before reporting it as missing so that the pet owner can get updates. 2. Once the user adds the missing pet’s profile, the pet owner can continue with step 2.
Exceptions:	EX3: Form submission error Form submission fails due to network issues or server errors, the system displays an error popup message asking the pet owner to try again at a later time. EX3: Invalid Last Seen Location 1. When the pet owner inputs an invalid postal code in last seen location, the system prompts the user to re-input a different valid postal code
Includes:	UC_PI_01-View Pet Information
Special Requirements:	None
Assumptions:	1. The pet owner’s registered pet information is accurate. 2. The pet owner consents to share the missing pet’s information with other users.
Notes and Issues:	None

4.21.3 Functional Requirement

21. Pet Owners must be able to report their pet as missing and display a marker in the “Display Missing Pet” page.

21.1 *PetCare* must display a ‘+’ icon on the bottom right of the “Display Missing Pets” page that allows the user to report a missing pet

21.2 *PetCare* shall display a form to add a missing pet report.

21.2.1 *PetCare* must display all the pet profile(s) registered under the user as a horizontal scroll view.

21.2.1.1 When the user taps on a pet profile, *PetCare* shall automatically populate the pet’s name in the text field.

21.2.1.2 When the user taps on a pet profile, *PetCare* shall automatically populate the pet’s age in the text field.

- 21.2.1.3 When the user taps on a pet profile, *PetCare* shall automatically highlight the pet's gender.
- 21.2.2 *PetCare* must display 5 input fields that are editable.
 - 21.2.2.1 *PetCare* shall display a text field for the pet's last seen location which accepts postal codes.
 - 21.2.2.2 *PetCare* shall display a text field for the pet's last seen date.
 - 21.2.2.3 *PetCare* shall display a text field for the pet's last seen time.
 - 21.2.2.4 *PetCare* shall display an image picker for the pet's last seen image sighting.
 - 21.2.2.5 *PetCare* shall display a text field for the description sighting of the pet.
- 21.3 *PetCare* must ensure that the user fills in all the fields before allowing users to mark their pet as missing.
 - 21.3.1 *PetCare* shall display an alert if all fields are not filled.
- 21.4 *PetCare* must verify all input information.
 - 21.4.1 *PetCare* must verify that the last seen location postal code exists.
 - 21.4.2 *PetCare* must verify that the pet is not already marked as missing.
 - 21.4.3 *PetCare* must provide an alert to the user to show feedback regarding the validation issues.
- 21.5 When a user submits the report, *PetCare* must add a missing pet report to the database.
 - 21.5.1 *PetCare* must display a success feedback to inform the user about a successful missing pet report.
 - 21.5.2 *PetCare* must display the new missing pet report as a marker on the map interface in the "Display Missing Pets" page.
- 21.6 *PetCare* must redirect the user to the newly created missing pet report page, where the missing pet details together with their respective details are displayed.

4.22 Report Sighting

4.22.1 Description and Priority

Existing users of *PetCare* can add sightings for active missing pet reports. Users can enter the sighting's date, time, location (postal code), image, and description. Upon submission, *PetCare* validates all information, updates the map with a new marker, and notifies the original pet owner. This feature helps the community collaborate by sharing real-time updates on missing pets' whereabouts, enhancing the chances of locating them.

Overall Priority	High
------------------	------

4.22.2 Stimulus/ Response Sequences

Use Case ID:	UC_MP_3
--------------	---------

Use Case Name:	Report Sighting		
Created By:	Jasmine Tye	Last Updated By:	Jasmine Tye
Date Created:	25th August 2024	Date Last Updated:	19th October 2024

Actor:	User
Description:	Allows other users to add a sighting for the selected missing pet.
Preconditions:	1. The original pet owner has made a missing report. 2. The selected missing pet marker is an active case in the system.
Postconditions:	1. The system updates the map with the new last seen location, time, and date for the missing pet. 2. All users of the app can see the updated information on the map. 3. The original reporting user receives a notification about the updated details.
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. The user navigates to the “Display Missing Pets” page in the app. 2. The system retrieves and displays the map with active missing pet markers. 3. The user selects an active missing pet marker on the map. 4. The system displays a pop-up of the details of the missing pet. 5. The user taps on the “Report Sighting” button. 6. The user enters the new time, date, location, sighting image and sighting description when the user last saw the pet. 7. The system validates the entered data. 8. The system updates the map with the new last seen location, time and date for the missing pet.
Alternative Flows:	AF-S2: User inputs incorrect data 1. If the user inputs an incorrect or incomplete address, time or date, the system displays an error message and prompts the user to change the input. AF-S7: User cancels form submission 1. The user can cancel the form submission by tapping on the “Back” button and return back to the “Display Active Missing Pets” page.
Exceptions:	EX3: Form submission error 1. Form submission fails due to network issues or server

	errors, the system displays an error popup message asking the user to try again at a later time.
Includes:	UC_MP_1 - Display Active Missing Pets
Special Requirements:	None
Assumptions:	Users provide accurate and honest updates about the missing pet.
Notes and Issues:	None

4.22.3 Functional Requirement

22. Users must be able to report a sighting for a missing pet through the “Report Sighting” button.

22.1 *PetCare* shall display a form for sighting details when the user taps on the “Report Sighting” button.

22.1.1 *PetCare* must include the following input fields in the form:

22.1.1.1 *PetCare* shall display an image picker for the last sighting image.

22.1.1.2 *PetCare* shall display a text field for the sighting postal code.

22.1.1.3 *PetCare* shall display a text field for the pet’s last seen date.

22.1.1.4 *PetCare* shall display a text field for the pet’s last seen time.

22.1.1.5 *PetCare* shall display a text field for the sighting description/ notes.

22.3 *PetCare* must ensure that the user fills in all the fields before allowing users to add a sighting.

22.3.1 *PetCare* shall display an alert if all fields are not filled.

22.4 *PetCare* must verify all input information.

22.4.1 *PetCare* must verify that the last seen location postal code exists.

22.4.3 *PetCare* must provide an alert to the user to show feedback regarding the validation issues.

22.5 When a user submits the report, *PetCare* must add a sighting to the selected missing pet report to the database.

22.5.1 *PetCare* must display a success feedback to inform the user about a successful sighting addition.

22.6 *PetCare* must redirect the user to the selected missing pet report page.

22.6.1 *PetCare* must display the new sighting on the timeline map interface for the selected missing pet.

22.6.2 *PetCare* shall add a marker with the updated sighting location, date, and time for the missing pet.

4.23 Mark Missing Pet as Found

4.23.1 Description and Priority

Existing users of *PetCare* can mark their missing pet as found and the pet will be removed from the missing pets list.

Overall Priority	Medium
------------------	--------

4.23.2 Stimulus/ Response Sequences

Use Case ID:	UC_MP_4		
Use Case Name:	Mark Missing Pet as Found		
Created By:	Jasmine Tye	Last Updated By:	Jasmine Tye
Date Created:	26th August 2024	Date Last Updated:	26th August 2024

Actor:	Pet Owner
Description:	Allows the pet owner to mark their missing pet report.
Preconditions:	1. The pet is currently in the missing pets database. 2. The pet owner has reported a pet as missing and it is currently active in the “Display Missing Pets” page.
Postconditions:	1. The system updates the pet’s status to inactive in the database. 2. The missing pet report is no longer being displayed on the “Display Missing Pets” page.
Priority:	Medium
Frequency of Use:	Low
Flow of Events:	1. The pet owner navigates to the “Display Missing Pets” page in the app. 2. The pet owner selects the marker of their missing pet on the map. 3. The details of the selected missing report is shown. 4. The pet owner taps on the “Found” toggle switch which is only accessible to the pet owner. 5. The toggle switch changes to “ON” to indicate the pet has been marked as found. 6. A pop-up message displays to confirm that the pet has been found. 7. The pet owner taps on the “OK” button. 8. The system updates the pet’s status to inactive in the database. 9. The pet listing is hidden from the “Display Missing Pets” map interface.

	10. The system shows a popup message to confirm that the report has been closed.
Alternative Flows:	AF-S3: Canceling the toggling of “Found” switch 1. The user taps on the “Cancel” button. 2. The pop-up messages closes without making any changes to the pet’s status. 3. The missing pet continues to be displayed in the “Display Missing Pets” map interface.
Exceptions:	EX2: Data update failure 1. If the system is unable to update the pet’s status due to network issues or server issues. 2. The system displays an error message and prompts the user to try again later. 3. The pet status has not changed.
Includes:	UC_MP_1 - Display Active Missing Pets
Special Requirements:	None
Assumptions:	The pet owner is authorized to update the status of the missing pet.
Notes and Issues:	None.

4.23.3 Functional Requirement

23. Pet owners must be able to mark their missing pet as found using the “Found” toggle switch.

23.1 *PetCare* shall display the “Found” toggle switch only to the pet owner when accessing their own missing pet report.

23.1.1 If the user accessing the report is not the pet owner, *PetCare* must hide the “Found” toggle switch.

23.2 When the pet owner taps on the “Found” toggle switch, *PetCare* must display a confirmation modal indicating that marking the pet as found will close the case and hide it from the “Display Missing Pets” page.

23.2.1 If the pet owner taps “No” on the modal, the action is canceled, and the pet’s status remains unchanged and the modal closes.

23.3 When confirmed, *PetCare* must update the pet’s status to inactive, remove it from the active missing pets on the map, and display a final confirmation of the closed report.

23.4 *PetCare* redirects the pet owner to the “Display Missing Pets” page to show feedback to the pet owner that the specific pet marker is now not being shown in the map interface.

4.24 Explore PetCare Amenities

4.24.1 Description and Priority

Existing *PetCare* users can view the list of nearby petcare amenities within 5km from the user's current location. The locations will be displayed as listings and map markers on the interactive map.

Overall Priority	High
------------------	------

4.24.2 Stimulus/ Response Sequences

Use Case ID:	UC_PA_1		
Use Case Name:	Explore Petcare Amenities		
Created By:	Sih Jia Qi	Last Updated By:	Sih Jia Qi
Date Created:	25th August 2024	Date Last Updated:	15th September 2024

Actor:	User
Description:	Allows a user to search for nearby petcare amenities including veterinarian clinics, groomers and pet adoption centers based on their current location using a GPS-enabled mobile device. The system will display the search results as landmarks on a map interface. It will also display a list of nearby amenities on the Petcare Amenities page, providing options to sort, filter and search for a specific amenity by keywords.
Preconditions:	1. The user has a device with <i>PetCare</i> installed. 2. The user is logged in and authenticated. 3. The user has a device with GPS capabilities. 4. The application has access to the user's current location. 5. The user's device has Internet access. 6. The GoogleMaps API is fully functionable and returns the latest pet amenities information.
Postconditions:	1. The system displays a map with nearby pet amenities marked. 2. The user can interact with the map to view the detailed information about a selected vet. 3. The user can search, filter and sort the list of nearby pet amenities.
Priority:	High

Frequency of Use:	High
Flow of Events:	1. The user clicks on the <i>PetCare</i> Amenities icon on the navigation bar. 2. The system prompts the user to grant location permissions. 3. The system retrieves the user's current location using GPS. 4. The system queries the GoogleMaps API to search for nearby <i>PetCare</i> amenities. 5. The system receives a list of nearby <i>PetCare</i> amenities. 6. The system displays the returned locations as markers on the map. 7. The system also displays them as listings with their information on the <i>PetCare</i> Amenities page. 8. The user can sort the amenities by rating in descending order or by distance in ascending order and the app will display the results in the order that user selects. 9. The user can filter the pet amenities by ratings and/ or whether the amenity is open now. 10. The system will display the results according to the filter applied. 11. The user can search for a specific pet amenity by name or a keyword using the search bar. 12. The user can tap on a marker in the map or a listing to view more detailed information about the amenity, including its storefront image, distance, rating, name, address, contact information, website, open now and operating hours.
Alternative Flows:	AF-S2: User declines location access 1. The system prompts the user to manually enter their location or search a specific area. 2. The system displays the nearby pet amenities based on the user input location. AF-S5: No nearby pet amenities found 1. No amenity listings will be displayed on the screen. AF-S7: Specific Information unavailable 1. Certain information of the selected amenity is unavailable: a. Storefront image - Display a default image provided by system b. Other info: Display “-” AF-S7: User wants to sort the amenities by rating 1. The system allows the user to sort the ratings of the <i>PetCare</i> amenities in descending order or distance in ascending order. 2. The system reorders the list of <i>PetCare</i> amenities based on the selected sorting options by the user.
Exceptions:	EX-S1: Location services unavailable 1. If the GPS is disabled or the system is unable to determine the user's location, the system notifies the user and provides an option to manually input a location.

	EX-S2: Google Maps API Error 1. If the Google Maps API fails or returns an error, the system displays an error message that suggests the user to try again later.
Includes:	UC_PA_2 - Filter <i>Petcare</i> Amenities UC_PA_3 - Search for <i>Petcare</i> Amenities
Special Requirements:	Location service is enabled on the user's device.
Assumptions:	Google Maps API provides accurate and up-to-date data for the pet amenities.
Notes and Issues:	None

4.24.3 Functional Requirement

24.1 *PetCare* must display a list of nearby petcare amenities on the Explore Petcare Amenities page.

24.2 *PetCare* must allow the user to view the details of each petcare amenity on the map or in the list of petcare amenities.

24.2.1 When the user taps on a marker on the map or a petcare amenity listing, *PetCare* must display the details of the selected amenity on a modal.

24.2.1.1 *PetCare* must display the photo of the selected amenity.

24.2.1.2 If no images were found for the amenity, *PetCare* shall display a default image provided by the system.

24.2.1.3 *PetCare* must display the name of the amenity.

24.2.1.4 *PetCare* must display the ratings of the amenity.

24.2.1.5 If no ratings were given for the amenity, *PetCare* shall display "No ratings yet" on the pet amenity listing.

24.2.1.6 *PetCare* must display the address of the amenity.

24.2.1.7 If no address details were found, *PetCare* shall display "-" on the pet amenity listing.

24.2.1.8 *PetCare* must display the contact number of the amenity .

24.2.1.9 If no contact number were found, *PetCare* shall display "-" on the pet amenity listing.

24.2.1.10 *PetCare* must display the website of the amenity.

24.2.1.11 If no website were found, *PetCare* shall display "-" on the pet amenity listing.

24.2.1.12 *PetCare* must display the operation hours of the vet.

24.2.1.13 If no operation hours were found, *PetCare* shall display "-" on the pet amenity listing.

24.2.1.14 *PetCare* must display the distance of the amenity from the user.

4.25 Filter PetCare Amenities

4.25.1 Description and Priority

Existing *PetCare* users can filter petcare amenities by selecting filtering options “rating” and/ or “opening hours”. *PetCare* will dynamically display the filtered result based on the selected filtering option.

Overall Priority	High
------------------	------

4.25.2 Stimulus/ Response Sequences

Use Case ID:	UC_PA_2		
Use Case Name:	Filter Petcare Amenities		
Created By:	Sih Jia Qi	Last Updated By:	Sih Jia Qi
Date Created:	26th August 2024	Date Last Updated:	15th September 2024

Actor:	User
Description:	Allows a user to sort the list of petcare amenities by rating in descending order or by distance in ascending order.
Preconditions:	1. The user has a device with <i>PetCare</i> installed. 2. The user is logged in and authenticated. 3. The user has a device with GPS capabilities. 4. The application has access to the user's current location. 5. The user's device has Internet access. 6. The GoogleMap API is fully functionable and returns the latest pet amenities information.
Postconditions:	1. The system displays the list of nearby petcare amenities sorted and/or filtered according to the options selected by the user. 2. The user is able to view the updated locations on the map and from the listings.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user browses the Petcare Amenities page. 2. The user taps on the “Filter” icon on the screen. 3. The system allows the following filtering conditions:)

	a. Rating (eg. Any, 2 stars and above, 3 stars and above, 4 stars and above, 5 stars) b. Open now (eg. Any, Open now, Open 24 hours) 4. The user selects one or more filtering options. 5. The system applies the filters and only displays amenities that fit the filtering conditions. 6. The user can interact and tap on any of the displayed amenities on the map to view its details.
Alternative Flows:	AF-S5: No results returned after filtering 1. If no amenities match the filtering options, no listings will be displayed on the screen. AF-S7: User wants to sort the amenities by rating 1. The system allows the user to sort the petcare amenity ratings in descending order or distance in ascending order. 2. The system reorders the list of petcare amenities based on the selected sorting options by the user.
Exceptions:	EX-S1: Location services unavailable 1. If the GPS is disabled or the system is unable to determine the user's location, the system notifies the user and provides an option to manually input a location
Includes:	None
Special Requirements:	The system must sort and filter the results without refreshing or reloading the page.
Assumptions:	The Petcare Amenities page must be fully functionable and display accurate results.
Notes and Issues:	None

4.25.3 Functional Requirement

25.1 *PetCare* must allow the user to filter the petcare amenities by ratings and/ or operating hours.

25.1.1 *PetCare* must display a filter icon on the Explore Petcare Amenities page.

25.1.2 When the user taps on the filter icon, *PetCare* must display a modal with filtering options such as "Ratings" and "Open now".

25.1.3 When the user selects a filtering option, *PetCare* must filter the pet amenities and display the results on the Explore Petcare Amenities page.

4.26 Search Petcare Amenities

4.26.1 Description and Priority

Existing *PetCare* users can search for nearby amenities that are within 50km of the user's current location. It will filter the locations by types such as "veterinary_care and pet_store". The locations will be displayed as listings and markers on the interactive map.

Overall Priority	Medium
------------------	--------

4.26.2 Stimulus/ Response Sequences

Use Case ID:	UC_PA_3		
Use Case Name:	Search Petcare Amenities		
Created By:	Sih Jia Qi	Last Updated By:	Sih Jia Qi
Date Created:	26th August 2024	Date Last Updated:	15th September 2024

Actor:	User
Description:	Allows a user to search for a specific pet amenity by its name or a keyword.
Preconditions:	1. The user has a device with <i>PetCare</i> installed. 2. The user is logged in and authenticated. 3. The user has a device with GPS capabilities. 4. The application has access to the user's current location. 5. The user's device has Internet access. 6. The GoogleMaps API is fully functionable and returns the latest pet amenities information.
Postconditions:	The system displays the petcare amenities that match the user's search query.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. The user browses the Petcare Amenities page. 2. The user inputs a location or keyword in the search bar and taps the "Search" icon. 3. The system uses Google Maps API to query for Petcare amenities that match the user input. 4. The map updates to only display the Petcare amenities that match the user input. 5. The user can interact and tap on any of the displayed Petcare amenities to view its details.
Alternative Flows:	AF-S4: Clear search bar 1. The user clears the input in the search bar. 2. The system displays all the nearby Petcare amenities on

	the map and shows the listings below. AF-S4: No results were found 1. If no Petcare amenity matches the search query (Google Maps API does not return any result), no amenity listings will be displayed on the screen.
Exceptions:	EX1: Location services unavailable 1. If the GPS is disabled or the system is unable to determine the user's location, the system notifies the user and provides an option to manually input a location
Includes:	None
Special Requirements:	None
Assumptions:	Google Maps API can query based on user input and return results based on the search query.
Notes and Issues:	None

4.26.3 Functional Requirement

26.1 *PetCare* must allow the user to search for a Petcare amenity.

26.1.1 *PetCare* must display a text field on the Explore Petcare Amenities page.

26.1.2 When the user input the name of an amenity or a keyword, *PetCare* shall display a list of amenities as the search result and update the markers on the interactive map.

26.1.3 If the amenity or keyword cannot be found through the Google Maps API, *PetCare* will not display any listings or map markers.

26.2 *PetCare* must display a sort icon on the Explore Petcare Amenities page.

26.2.1 When the user taps on the sort icon, *PetCare* will display the sorting options “rating” and “distance” for users to sort the amenities by rating in a descending order of distance in an ascending order.

26.2.2 *PetCare* must sort the pet amenities based on the selected option and display the result on the Explore Petcare Amenities page.

4.27 Add Adoption Pet

4.27.1 Description and Priority

The Adoption Center of *PetCare* can add listings of available pets for adoption in their shelter. Adoption can enter pet name, breed, age, color, gender, type, image, and description. Upon submission, *PetCare* validates for any missing information, update to the current listings. This feature provides a platform for adoption centers to show off their pets and streamline the adoption process.

Overall Priority	High
------------------	------

4.27.2 Stimulus/ Response Sequences

Use Case ID:	UC_ADPT_01		
Use Case Name:	Add Adoption Pet		
Created By:	Lim Chun Wen	Last Updated By:	Lim Chun Wen
Date Created:	26th August 2024	Date Last Updated:	10th October 2024

Actor:	Adoption Center
Description:	Allows the adoption center to upload pets that are available for adoption.
Preconditions:	1. The adoption center is logged in and authenticated in admin format 2. The adoption center has the pet and wants to publish an adoption.
Postconditions:	1. The system successfully saves and publishes the pets to the app. 2. Other users are able to view the available pets in listview format.
Priority:	High
Frequency of Use:	Low
Flow of Events:	1. The system displays a “Add” button at “Adoption” page for adoption center to post their listing on a particular pets to the “adoption center” 2. When adoption center selects the “Add Button”, the system will bring to another page to fill up the details on the pets 3. Adoption center will manually fills in the item form: a. Pets Name b. Description c. Age d. Breed e. Color f. Gender g. Upload Images through phone camera h. Type i. Adoption Center : Name

	j. Adoption Center : Phone Number k. Adoption Center : Email 4. If the adoption center selects the “Submit” button, the system will save this information into the database and display a prompt to inform the user that “pets have been successfully posted”.
Alternative Flows:	AF-S4: Missing or Incomplete form 1. The system displays a prompt to inform the user that there are missing fields / wrong data types that require attention. 2. Once everything is validated , the system will go to step 4.
Exceptions:	EX1: Form submission fails due to network issues or server errors, the system displays an error popup message asking the user to try again at a later time.
Includes:	None
Special Requirements:	System needs to validate input data.
Assumptions:	Pets are in good or proper condition for adoption.
Notes and Issues:	None

4.27.3 Functional Requirements

27. *PetCare* shall allow the adoption center to post pets that are available for adoption.

27.1 *PetCare* must display a “+” button at the “upload adoption” page.

27.2 When the user clicks on “+”.button, *PetCare* must display 8 input fields.

27.2.1 *PetCare* must display a text field for “Pets Name”

27.2.2 *PetCare* must display a text field for “Description”

27.2.3 *PetCare* must display a number field for “Age”

27.2.4 *PetCare* must display a text field for “Breed”

27.2.5 *PetCare* must display a text field for “Color”

27.2.6 *PetCare* must display a radio field for “Gender”

27.2.7 *PetCare* must display a upload images field to upload image of the pets

27.2.8 *PetCare* must display radio field for “Type”

27.3 *PetCare* must provide a “Submit” button to upload their listing to the adoption page .

27.4 *PetCare* must provide an Error message if system fail to upload

4.28 Edit Adoption Pet

4.28.1 Description and Priority

Existing Adoption Center of *PetCare* can make changes to their previously posted information about their pet . They are required to navigate to their own section of the app and select the edit icon , which opens a form pre-filled with the pets current details. Users can modify any fields which will then be reflected in the database.

Overall Priority	High
------------------	------

4.28.2 Stimulus/ Response Sequences

Use Case ID:	UC_ADPT_02		
Use Case Name:	Edit Adoption Pet		
Created By:	Lim Chun Wen	Last Updated By:	Lim Chun Wen
Date Created:	26th August 2024	Date Last Updated:	10th October 2024

Actor:	Adoption Center
Description:	Allows the users to edit the pet detail that have uploaded for listings
Preconditions:	1. The user is logged in and authenticated in admin state. 2. The user has uploaded or created the pet posting in the database
Postconditions:	1. The system successfully saves and updates the information in the app. 2. Other users are able to view the updated listing in listview format.
Priority:	High
Frequency of Use:	Low
Flow of Events:	1. The system displays a “Edit ” button at page under a particular pets 2. When user selects the “Edit Button”,User will be able to modify the information in the form: a. Pets Name b. Price c. Description d. Age e. Breed f. Color g. Gender h. Upload Images via phone storage

	i. Type j. Adoption Center : Name k. Adoption Center : Phone Number l. Adoption Center : Email 3. If the user selects the “Submit” button, the system will save this information into the database and display a prompt to inform the user that “Pets have been successfully edited”.
Alternative Flows:	AF-S3: Missing or Incomplete form 1. The system displays a prompt to inform users that there are missing fields / wrong data types that require attention. 2. Once everything is validated , the system will go to step 3
Exceptions:	EX1: Form submission fails due to network issues or server errors, the system displays an error popup message asking the user to try again at a later time.
Includes:	None
Special Requirements:	System needs to validate user input data
Assumptions:	User have uploaded information of the pets and the data are captured in the database.
Notes and Issues:	None.

4.28.3 Functional Requirements

28. *PetCare* must allow the adoption center to edit their listings.

28.1 *PetCare* must display an “edit” button at the “admin adoption” page.

28.2 When user click on “edit” button,

28.2.1 *PetCare* shall allow editing of “Pets Name”

28.2.2 *PetCare* shall allow editing of “Description”

28.2.3 *PetCare* shall allow editing of “Age”

28.2.4 *PetCare* shall allow editing of “Breed”

28.2.5 *PetCare* shall allow editing of “Color”

28.2.6 *PetCare* shall allow editing of “Gender”

28.2.7 *PetCare* shall allow editing of Images of Pet

28.2.8 *PetCare* shall allow editing of “Type”

28.3 *PetCare* must provide a “save” button to save the pets information.

28.4 *PetCare* must provide an Error message if the system fails to edit.

4.29 Delete Pet Adoption

4.29.1 Description and Priority

Existing Adoption Center of *PetCare* can delete their pet listing . They are required to navigate to their own section of the app and select the delete icon , which will be prompted for confirmation for the deletion.

Overall Priority	High
------------------	------

4.29.2 Stimulus/ Response Sequences

Use Case ID:	UC_ADPT_03		
Use Case Name:	Delete Pet Adoption		
Created By:	Lim Chun Wen	Last Updated By:	Lim Chun Wen
Date Created:	26th August 2024	Date Last Updated:	26th August 2024

Actor:	Adoption Center
Description:	Allows adoption center to delete the detail of the pets once it has been adopted or not for adoption anymore
Preconditions:	1. The user is logged in and authenticated in the adoption center account. 2. The user has created the pet posting in the database
Postconditions:	1. The system successfully deletes the pet's details and updates the information in the database. 2. Item is removed from the "my pets house" page.
Priority:	High
Frequency of Use:	Low
Flow of Events:	1. The system displays a "Delete" button at individual pet page under individual page(adoption user) 2. When the user selects the "Delete Button",the system will prompt the user for confirmation. 3. When confirmation is clicked ,the system will display a dialogue box informing the user that "Pet has been successfully deleted".
Alternative Flows:	AF-S2 : User cancel the deletion 1. System will go back to individual pet page
Exceptions:	None
Includes:	None

Special Requirements:	None
Assumptions:	User want to remove the pet from the adoption listings
Notes and Issues:	None

4.29.3 Functional Requirements

4.29. *PetCare* must allow the adoption center to delete their existing pets for adoption.

4.29.1 *PetCare* must allow the deletion of their pet listings to indicate the pets have been adopted / changes in adoption status.

4.30 Display adoption Pet (User)

4.30.1 Description and Priority

Existing users of *PetCare* can view the details of all the available pets for adoption from different adoption centers . Upon selecting a pet , more information will pop out and users will be able to make contact with the adoption center via Whatsapp. This feature serves as a platform for adoption centers to showcase their pets and streamline the adoption process. It helps adopters easily browse, find, and book appointments to meet pets they are interested in, ultimately facilitating the adoption experience for both parties.

Overall Priority	High
------------------	------

4.30.2 Stimulus/ Response Sequences

Use Case ID:	UC_ADPT_04		
Use Case Name:	Display Adoption Pet (User)		
Created By:	Lim Chun Wen	Last Updated By:	Lim Chun Wen
Date Created:	26th August 2024	Date Last Updated:	10th October 2024

Actor:	User
Description:	Allows the users to view the available pets uploaded by the adoption center and make contact with them.

Preconditions:	1. The user is logged in and authenticated. 2. The adoption center has the information of pets in the database.
Postconditions:	1. The user successfully makes contact with the adoption and makes an appointment.
Priority:	High
Frequency of Use:	Low
Flow of Events:	1. User navigates to the Adoption page. 2. System retrieves and display all the currently available pets for adoption 3. System will have a choice of filter by type Dog,Cats and Others 4. User click on one of the pet listing and a pop out modal will be shown to view the detail of the pets 5. User clicks on “Book appointment”,the system will open WhatsApp and make contact with the adoption center. 6. Buyer successfully booked an appointment with the adoption center.
Alternative Flows:	AF-S3 : User filter pet by type 1. System will filter out and only display the type of pet selected by the user at the filter section 2. Flow continue back to step 4
Exceptions:	EX1: The user does not have WhatsApp installed on their mobile device.
Includes:	None
Special Requirements:	None
Assumptions:	User do not misused the system
Notes and Issues:	None

4.30.3 Functional Requirements

4.30 *PetCare* must display all the available pets for adoption and allow the user to view each individual pet on the “listings” page.

4.30.1 *PetCare* shall display a list of pets available for adoption via a list interface.

4.30.2 For each pet, *PetCare* shall display pet information.

4.30.3 If the “Book an appointment” button is pressed, *PetCare* shall open WhatsApp and send a message to book a slot with the adoption center.

4.30.4 *PetCare* shall allow users to filter the type of pets (All, Cat, Dog, Others) with a button.

4.31 Display Adoption Pet (Adoption Center)

4.31.1 Description and Priority

PetCare's Adoption Center users can view the details of pets for adoption that have been uploaded from their own adoption centers . Upon selecting a pet , more information will pop out and the adoption center will be able to use various functions that belong to them .

Overall Priority	High
------------------	------

4.31.2 Stimulus/ Response Sequences

Use Case ID:	UC_ADPT_05		
Use Case Name:	Display Pet for Adoption (Adoption Center)		
Created By:	Lim Chun Wen	Last Updated By:	Lim Chun Wen
Date Created:	26th August 2024	Date Last Updated:	30th Oct 2024

Actor:	Adoption Center
Description:	Allows adoption center to view their own listings of pet and make any changes to the information
Preconditions:	1. The user is logged in and authenticated in adoption center role. 2. The adoption center page has the information of pets in the database.
Postconditions:	None
Priority:	High
Frequency of Use:	Low
Flow of Events:	1. The Adoption Center navigates to the Adoption page and clicks on their own name . 2. System retrieves and displays all the currently available pets belonging to them and all the relevant functions. 3. Adoption click on one of the pet listing to view more detail regarding the detail of the pets 4. Usecase continue to UC_ADPT_01 or UC_ADPT_02 OR UC_ADPT_03 according to the function Adoption Center click
Alternative Flows:	None

Exceptions:	None
Includes:	None
Special Requirements:	None
Assumptions:	User do not misused the system
Notes and Issues:	None

4.31.3 Functional Requirements

4.31 *PetCare* must display all the pets belongs to an individual adoption center and given permission to make changes to their own listings

4.31.1 *PetCare* shall display a list of pets belonging to the shelter via a list interface.

4.31.2 For each pet, *PetCare* shall display the pet information,edit and delete icon.

4.32 Interact with Chatbot

4.32.1 Description and Priority

Initially, the chatbot serves as the main interaction point, with GuideBot and PetBuddy functioning as extensions based on user selection.

Overall Priority	High
------------------	------

4.32.2 Stimulus/ Response Sequences

Use Case ID:	UC_CB_01		
Use Case Name:	Interact with Chatbot		
Created By:	Hng Cherng Khai	Last Updated By:	Hng Cherng Khai
Date Created:	26th August 2024	Date Last Updated:	30th Oct 2024

Actor:	User
--------	------

Description:	This use case serves as the main interaction point, with GuideBot and PetBuddy functioning as extensions based on user selection.
Preconditions:	1. The user has a device with PetCare installed. 2. The user is logged in and authenticated. 3. The user's device has Internet access.
Postconditions:	1. The chatbot has activated either GuideBot or PetBuddy based on the user's choice. 2. The chatbot is now prepared to respond to either app guidance or pet care-related queries, depending on the selected mode.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user taps on the chatbot icon. 2. The system navigates the user to the Chatbot page. 3. Upon navigating to the Chatbot page, the chatbot sends the initial response message: a. "Welcome! How can I assist you today? Please choose from the options below: 1 To get a guided tour of our app features and learn how to make the most out of it, input '1' to chat with GuideBot. 2 For pet care advice and research assistance, input '2' to chat with PetBuddy. Please enter your choice to proceed! (Note: This is case-sensitive.)" 4. The user inputs either "1" for GuideBot or "2" for PetBuddy. 5. The chatbot verifies the input and provides a response based on the user input. 6. If the user input is "1" following the initial response message of the chatbot, GuideBot is activated. 7. Upon the user selecting "1" at the initial prompt, GuideBot confirms activation with a message: a. "You are now chatting with GuideBot! How can I assist you with app navigation?" 8. Usecase continues to UC_CB_02.
Alternative Flows:	AF-S6: If user input is "2" following the initial response message of the chatbot 1. PetBuddy is activated. 2. Upon the user selecting "2" at the initial prompt, PetBuddy confirms activation with a message. a. The message is "You are now chatting with PetBuddy! How can I assist with pet care today?" 3. Usecase continues to UC_CB_03. AF-S6: If the user input is neither "1" nor "2" following the initial response message of the chatbot 1. the chatbot must display a message. a. The message is "Invalid choice. Please select 1 for GuideBot or 2 for PetBuddy."
Exceptions:	None
Includes:	UC_PA_02 - Respond to User's PetCare App Guidance Queries

	UC_PA_03 - Respond to User's Pet Care Related Queries
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

4.32.3 Functional Requirement

4.32. Users shall be able to activate either GuideBot or PetBuddy.

4.32.1. The user must be able to navigate to the Chatbot page.

4.32.1.1. The user shall be able to click on the chatbot icon to initiate interaction with the chatbot.

4.32.1.2. When the user clicks on the chatbot icon, *PetCare* shall navigate the user to the Chatbot page.

4.32.2 The Chatbot must display its initial response message.

4.32.2.1. When the user first navigates to the Chatbot page, the chatbot shall send an initial response message.

4.32.2.1.1. The initial response message shall be: "Welcome! How can I assist you today? Please choose from the options below: 1 To get a guided tour of our app features and learn how to make the most out of it, input "1" to chat with GuideBot. 2 For pet care advice and research assistance, input "2" to chat with PetBuddy. Please enter your choice to proceed! (Note: This is case-sensitive.)"

4.32.3. The chatbot must display a text prompt field.

4.32.3.1. The chatbot shall display a text field for the user to input text prompts.

4.32.3.2. The chatbot shall prevent the user from submitting a text prompt if it has not yet completed generating a response.

4.32.4. The chatbot shall display each user input prompt visibly within the chatbot interface.

4.32.4.1. The chatbot shall display its response message.

4.32.4.2. The chatbot shall display a response for each user input prompt after the user submits it.

4.32.5. The chatbot shall wait for the user to input a prompt before displaying any additional responses.

4.32.6. The user shall be able to input "1" to activate GuideBot.

4.32.6.1. Upon receiving the user input "1" following the initial response message of the chatbot, the chatbot shall display with a message.

4.32.6.1.1. The message is "You are now chatting with GuideBot!
How can I assist you with app navigation?"

4.32.7. The user shall be able to input "2" to activate PetBuddy.

4.32.7.1. Upon receiving the user input "2" following the initial response message of the chatbot, the chatbot shall respond with a message.

4.32.7.1.1. The message must be "You are now chatting with
PetBuddy! How can I assist with pet care today?"

4.32.8. If the user input is neither "1" nor "2" following the initial response message of the chatbot, the chatbot must display a message.

4.32.8.1. The message must be "Invalid choice. Please select 1 for GuideBot or 2 for PetBuddy."

4.33 Respond to User's PetCare App Guidance Queries

4.33.1 Description and Priority

GuideBot assists the user with app navigation by providing information from a PDF file containing all functionalities of the *PetCare* app, accessed via the Gemini API. This allows the user to learn about app features and receive navigation support.

Overall Priority	High
------------------	------

4.33.2 Stimulus/ Response Sequences

Use Case ID:	UC_CB_02		
Use Case Name:	Respond to User's PetCare App Guidance Queries		
Created By:	Hng Cherng Khai	Last Updated By:	Hng Cherng Khai
Date Created:	26th August 2024	Date Last Updated:	30th Oct 2024

Actor:	User
Description:	This use case assists the user with app navigation by providing information from a PetCare app PDF, accessed via the Gemini API. This allows the user to learn about app features and receive navigation support.
Preconditions:	1. The user has a device with PetCare installed.

	- The user is logged in and authenticated. - The user's device has Internet access. - The Gemini API is fully functional. - The user input is "1" following the initial response message of the chatbot.
Postconditions:	- The user is able to input a prompt that is related to app guidance to the chatbot. - The chatbot displays its response message to the user input prompt about app guidance.
Priority:	High
Frequency of Use:	High
Flow of Events:	- The user types a prompt in the text field. - The user submits the input prompt. - GuideBot generates its response message based on information extracted from a PDF file containing all functionalities of the PetCare app. - The response is generated by invoking the Gemini API. - GuideBot's response is displayed in the chatbot interface after processing the user's input prompt. - Step 1 to Step 4 repeats until the user clicks on the "back" button to close the chat.
Alternative Flows:	None
Exceptions:	EX1: APIs Unavailable: If Gemini API is unavailable, the chatbot must respond with a fallback message: "Error occurred. Please try again later." EX2: Server Unreachable: If the chatbot cannot connect to the backend server, it must respond with the following message: "Error occurred. Please try again later."
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

4.33.3 Functional Requirement

4.33. After activating GuideBot, the user shall be able to input additional prompts for app guidance.

4.33.1. The chatbot shall display a response to each user input prompt when GuideBot is active.

4.33.1.1. GuideBot's responses shall be generated based on information extracted from a PDF file containing all functionalities of the PetCare app.

4.33.1.2. GuideBot's responses shall be generated by invoking the Gemini API.

4.33.1.3. If the Gemini API is unavailable, the chatbot shall respond with a fallback message.

4.33.1.3.1. The message must be "Error occurred. Please try again later."

4.34 Respond to User's Pet Care Related Queries

4.34.1 Description and Priority

PetBuddy provides a response message to each user's pet care queries by invoking Tavily API and Gemini API.

Overall Priority	High
------------------	------

4.34.2 Stimulus/ Response Sequences

Actor:	User
Description:	This use case provides a response message to each user's pet care queries by invoking Tavily API and Gemini API.
Preconditions:	1. The user has a device with <i>PetCare</i> installed. 2. The user is logged in and authenticated. 3. The user's device has Internet access. 4. The Gemini API is fully functional. 5. The user input is "2" following the initial response message of the chatbot.
Postconditions:	1. The user is able to input a prompt that is related to pet care topics and does not involve map location search. 2. The chatbot displays its response message to the user input prompt that is related to pet care topics and does not involve map location search. 3. If a user query is unrelated to pet care or involves map location search, the chatbot has provided an informational message indicating the chatbot only assist with pet care-related input prompts that does not involve map location search.
Priority:	High
Frequency of Use:	High

Flow of Events:	1. The user types a prompt in the text field. 2. The user submits the input prompt. 3. PetBuddy verifies whether the user's input prompt is: a. Pet care-related and does not involve map location search. b. Not pet care-related. c. Involving a map location search. 4. If the user's input prompt is pet care-related and does not involve map location search, PetBuddy pulls information from online sources by invoking Tavily API. 5. If the user's input prompt is pet care-related and does not involve map location search, PetBuddy generates a response by invoking the Gemini API. a. The response shall be based on the user's input prompt. b. The response shall be based on the information pulled by invoking Tavily API from online sources. 6. PetBuddy's response is displayed in the chatbot interface after processing the user's input prompt. 7. Step 2 to Step 7 repeats until the user clicks on the "back" button to close the chat.
Alternative Flows:	AF-S6: If the user's input prompt is unrelated to pet care 1. The PetBuddy chatbot displays a limitation message clarifying its limitations: - "I'm sorry, I can only assist with pet care-related questions. Please ask me something about pets, their care, or related topics." 2. The system returns to step 4. AF-S6: If the user's input prompt involves map location search 1. The PetBuddy chatbot displays a limitation message clarifying its limitations: - "I'm sorry, I cannot assist in searching map locations. Please ask me something about pets, their care, or related topics that are not related to map location search." 2. The system returns to step 4.
Exceptions:	EX1: APIs Unavailable: If either the Tavily API or Gemini API is unavailable, the chatbot must respond with a fallback message: - "Error occurred. Please try again later." EX2: Server Unreachable: If the chatbot cannot connect to the backend server, it must respond with the following message: - "Error occurred. Please try again later."
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

4.34.3 Functional Requirement

4.34. The chatbot shall display a response for each user input prompt when PetBuddy is active.

4.34.1. PetBuddy shall verify each user's input prompt.

4.34.1.1. PetBuddy shall verify if a user input prompt is related to pet care.

4.34.1.2. PetBuddy shall verify if a user input prompt does not involve map location search.

4.34.2. If the prompt is pet care-related and does not involve map location search, PetBuddy's responses shall be generated based on the user input prompt. .

4.34.3. If the prompt is pet care-related and does not involve map location search, PetBuddy's responses shall be generated based on any information retrieved from the Tavily API.

4.34.3.1. The information retrieved from the Tavily API shall be from online sources.

4.34.4. If the user's input prompt is unrelated to pet care, the chatbot shall respond with a message.

4.34.4.1. The message shall be "I'm sorry, I can only assist with pet care-related questions. Please ask me something about pets, their care, or related topics."

4.34.5. If the user's input prompt involves a map location search, the chatbot shall respond with a message.

4.34.5.1. The message shall be "I'm sorry, I cannot assist in searching map locations to answer your question. Please ask me something about pets, their care, or related topics that are not related to map location search."

4.34.6. If the Tavily API is unavailable, the chatbot shall respond with a fallback message.

4.34.6.1. The message must be "Error occurred. Please try again later."

4.34.7. If the Gemini API is unavailable, the chatbot shall respond with a fallback message.

4.34.7.1. The message must be "Error occurred. Please try again later."

5. Other Nonfunctional Requirements

5.1 Performance Requirements

5.1.1 Page Load Time

The system must ensure that each page loads within 5 seconds in response to user actions to enhance user experience and responsiveness.

5.1.2 User-Interface Learnability

The app must have a user-friendly interface that can be learned within 10 minutes by a first-time user.

5.1.3 Readability

The text and visual elements must be clearly readable and accessible for users with color vision deficiencies.

5.1.4 Error handling and feedback

The system must display error messages that allow the user to know what went wrong when an error occurs. 80% of the users must be able to understand the error and not repeat the same flow of events.

5.1.5 Compatibility

Users must be able to see 100% of the content regardless of what mobile phones they are using.

5.2 Safety Requirements

5.2.1 Accuracy of Petcare Related Amenities

1. The system must be able to accurately reflect pet-care related amenities to ensure that the user can rely on the information provided.
 - 1.1 The system must ensure that only pet-care related amenities are displayed.
 - 1.2 The system must ensure that up-to-date information is displayed.

The system must implement regular updates to the database of pet-care amenities to reflect changes in availability, location, and contact information.
 - 1.3 The system must include user reviews scraped from Google Maps API.

5.2.2 Image Upload and Retrieval

2. The system must ensure secure and efficient storage and retrieval of user-uploaded images to protect user data and maintain the integrity of the images.
 - 2.1 The system must upload all images to Firebase Storage using secure HTTPS connections to prevent data interception during transfer.
 - 2.2 The system must generate signed URLs for image retrieval.
 - 2.2.1 The signed URLs should be time-limited and allow temporary access to the image, ensuring that images are only accessible to authorized users.

5.3 Security Requirements

5.3.1 User Authentication Requirement

1. The system must ensure that the user is logged in prior to accessing any functionalities provided within the app.
 - 1.1 The user must be logged in using their email and password.
 - 1.2 The system must verify the credentials are correct through Firebase Authenticator.
 - 1.3 The system must retain the login session if logout is initiated.
 - 1.4 The system must redirect users that are not logged in to the login page.
2. The system must not save sensitive information, specifically the user's password into the database.
 - 2.1 The password of the user should be kept confidential and not disclosed to both the developers and other users of the application.

5.3.2 Chatbot Requirement

1. Protect the user data during interaction with chatbot by limiting its domain only to pet related topic

5.4 Software Quality Attributes

5.4.1 Correctness

1. The system must ensure that the correct data is being retrieved from the database and returned to the user based on app functionality.
 - 1.1 When the user searches for a specific item by inputting a keyword in the search bar, the system must only display items that contain the specific keyword in the title or body text.
 - 1.2 When the user taps on the filter or sort button, the system displays only the results that fulfills the parameter set.
2. System must ensure that the user's account information is always accurately displayed.
 - 2.1 System must accurately display the username registered by the user.
 - 2.2 System must accurately display the email provided by the user.
 - 2.3 System must accurately display the profile picture provided by the user.
 - 2.3.1 If the user has not provided a profile picture, a default image is used.
 - 2.4 System must accurately display the address provided by the user.
 - 2.4.1 If the user has not provided their address, a placeholder asking user to input their address is displayed.

- 2.5 System must accurately display the contact number provided by the user.
 - 2.5.1 If the user has not provided their contact number, a placeholder asking user to input their contact information is displayed.
- 3. System must ensure that the user's registered pet information is always accurately displayed.
 - 3.1 System must accurately display the profile picture of the pets registered under the logged in user.
 - 3.2 System must accurately display the name of the pets registered under the logged in user.
 - 3.3 System must accurately display the breed of the pets registered under the logged in user.
 - 3.4 System must accurately display the sex of the pets registered under the logged in user.
 - 3.5 System must accurately display the date of birth of the pets registered under the logged in user.
 - 3.6 System must accurately display the weight of the pets registered under the logged in user.
 - 3.7 System must accurately display the coat color of the pets registered under the logged in user.
 - 3.8 System must accurately display the special markings of the pets registered under the logged in user.
 - 3.9 System must accurately display the medical conditions of the pets registered under the logged in user.
- 4. The system must ensure that information related to active missing pet reports is always accurately displayed.
 - 4.1 The system must accurately display the name of the missing pet.
 - 4.2 The system must accurately display the profile picture of the missing pet.
 - 4.3 The system must accurately display the breed of the missing pet.
 - 4.4 The system must accurately display the sex of the missing pet.
 - 4.5 The system must accurately display the age of the missing pet.
 - 4.6 The system must accurately display the relative time since the missing pet report was published.
 - 4.7 The system must accurately display the date the pet was reported missing.
 - 4.8 The system must accurately display the last seen location of the missing pet.
 - 4.9 The system must accurately display the last seen date of the missing pet.
 - 4.10 The system must accurately display the last seen time of the missing pet.
 - 4.11 The system must accurately display the last seen image of the missing pet.
 - 4.12 The system must accurately display the last seen description of the missing pet.
 - 4.13 The system must accurately display contact details for the person who reported the pet missing.
- 5. The system must ensure that all articles and related data are always accurately displayed.
 - 5.1 The system must accurately display the title of the article.
 - 5.2 The system must accurately display the author's name.
 - 5.3 The system must accurately display the author's profile picture.
 - 5.4 The system must accurately display the publication date of the article.

- 5.5 The system must accurately display the full content or body of the article.
- 5.6 The system must accurately display the thumbnail image of the article.
- 5.7 The system must accurately display the category tag associated with the article.
- 6. The system must ensure that adoption center details are always accurately displayed.
 - 6.1 The system must accurately display the name of the adoption center.
 - 6.2 The system must accurately display the profile picture of the adoption center.
 - 6.3 The system must accurately display the address of the adoption center.
 - 6.4 The system must accurately display the phone number of the adoption center.
 - 6.5 The system must accurately display a description of the adoption center, if available.
 - 6.6 The system must accurately display the list of pets available for adoption, with each pet's details.
 - 6.6.1 For each pet, the system must accurately display the profile picture, name, breed, age, and any other relevant information.

5.4.2 Extensibility

- 1. The system architecture should follow modular programming principles and adhere to the open-closed principle, ensuring that existing code remains unchanged while allowing for easy extension.
- 2. The system must maintain a consistent user interface and performance across both Android and iOS platforms, ensuring seamless portability without any degradation in user experience or system efficiency.

5.4.3 User-Friendliness

- 1. The system must provide a quick and seamless navigation between different pages.
- 2. The system must provide a loading indication to the user that a content is loading.
 - 2.1 A loading screen must appear when system is not done retrieving information from the back-end
 - 2.2 A loading screen must appear to the user when the user input is being processed.
- 3. The system must provide clear and suitable pop-up messages.
 - 3.1 The system must display detailed and specific validation messages when the user inputs incorrect or incomplete information.
 - 3.2 The system must display a successful pop-up after a user successfully completes an action that requires their input.

Appendix A: Data Dictionary

Term	Definition
Account	A registered user's personal profile associated with an

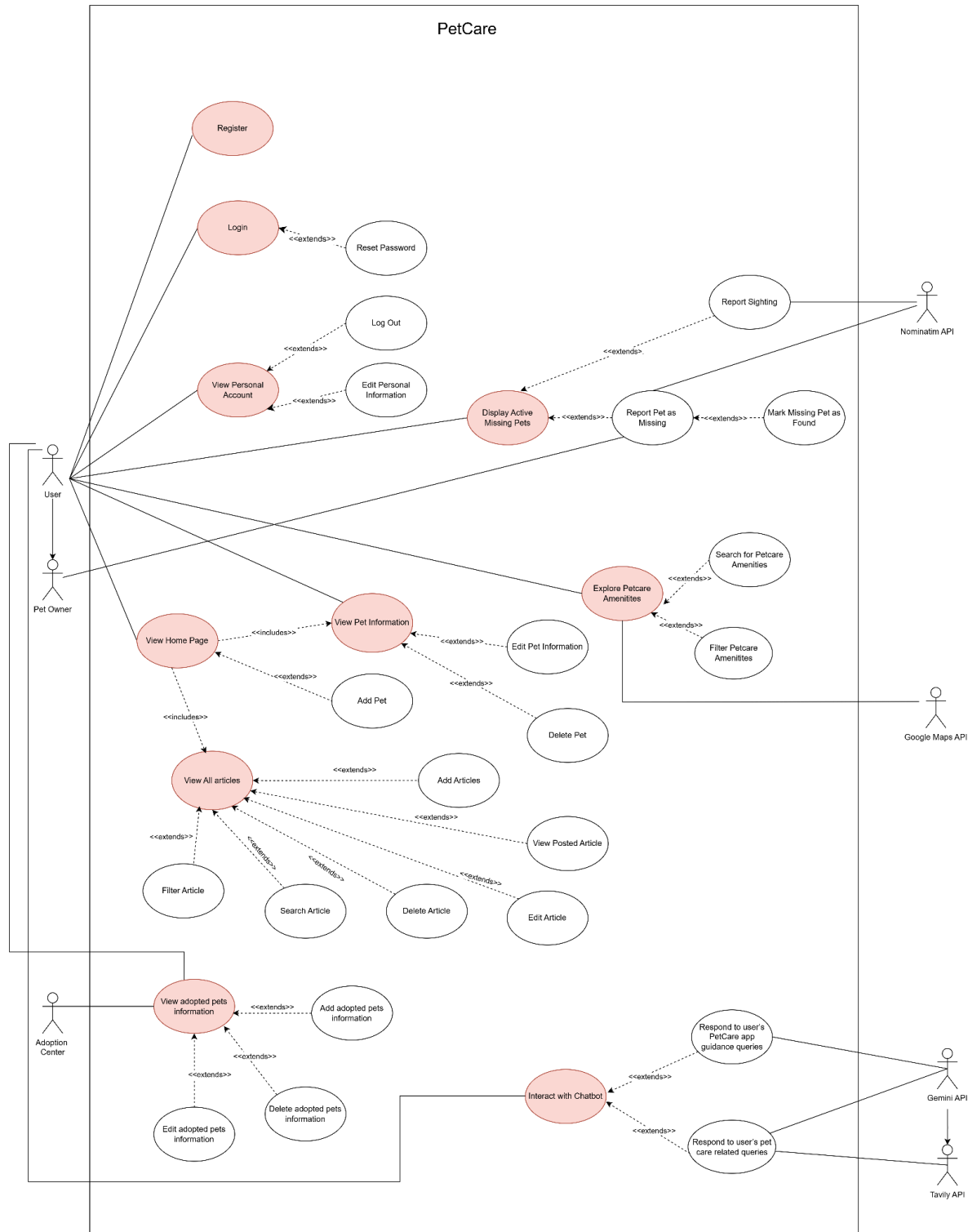
	application. It may include personal information, contact details, etc.
Application	A mobile application installed on users' smartphones that relies on an internet connection to fully utilize all of its intended features. The app may request certain system permissions to enable specific functionalities.
Adoption Center	Adoption center refers to a staffed facility where homeless animals are seized by authorities and are up for adoption.
API	API refers to mechanisms that enable two software components to communicate with each other using a set of definitions and protocols.
App Guidance	App guidance refers to providing users on how to use various <i>PetCare</i> application's functionalities.
Chatbot	Chatbot is an AI-powered virtual assistant integrated within the <i>PetCare</i> application, designed to enhance the user experience by offering real-time support for both app navigation and pet care-related inquiries.
Controller	The intermediary between the Presentation Layer (UI) and the Service Layer. It receives requests from the UI, processes those requests by invoking the appropriate services, and then sends the results back to the UI.
Database	Database refers to an organized collection of structured information, or data, typically stored electronically in a computer system
Dialog	A small area on a screen in which the user is prompted to provide information or select commands.
Display Article Page	A page that displays all the articles added by the user
Explore Petcare Amenities	A visual representation of the location of pet amenities, displayed using the GoogleMap API.
Fallback Message	A predefined message displayed to the user when the chatbot encounters issues, such as the unavailability of APIs or server connectivity issues. The fallback message notifies the user to try again later.
Gemini API	Gemini is a large language model used to generate conversational responses. In the <i>PetCare</i> app, the Gemini supports both the GuideBot and PetBuddy personas, allowing

	them to deliver accurate and context-specific responses.
GuideBot	GuideBot refers to a persona within the chatbot that provides responses to user input prompts based on information from a PDF file containing all functionalities of the PetCare app by leveraging Gemini API.
GPS	GPS, or the Global Positioning System, is a global navigation satellite system that provides location, velocity and time synchronization.
Home Page	Refers to the page that users see right after they log into their account. Users will be able to access all other pages from here.
Listings	A list or catalog.
Model	The Model Layer defines the structure of the data used across the application. It represents the objects that will be called by the Controller ,Service and Repositories layer which defines the rules and relationships for this data.
Modal	A modal is a dialog box or a pop-up window that is displayed on top of the current page. It is used to capture user input or display information without navigating away from the current page.
Map Location Search	A type of query involving location-based results or services. The <i>PetCare</i> chatbot, specifically PetBuddy, does not support map location searches and will notify users of this limitation.
Missing Pet Page	A visual representation of the location of a missing pet reported by the pet owner displayed using the map.
PetBuddy	PetBuddy refers to a persona within the chatbot that provides responses to user input prompts that are related to pet care and does not map location search by leveraging both Tavily API and Gemini API.
Pet Owner	Pet Owner refers to a person that own a pet and is responsible for its well being
Profile page	A page that displays information of the user.
Pop-up	A dialog that automatically appears (“pops up”) on the screen while the user performs a certain action.
Service	The Service Layer acts as the intermediary between the Presentation Layer and the business logic. It contains the core

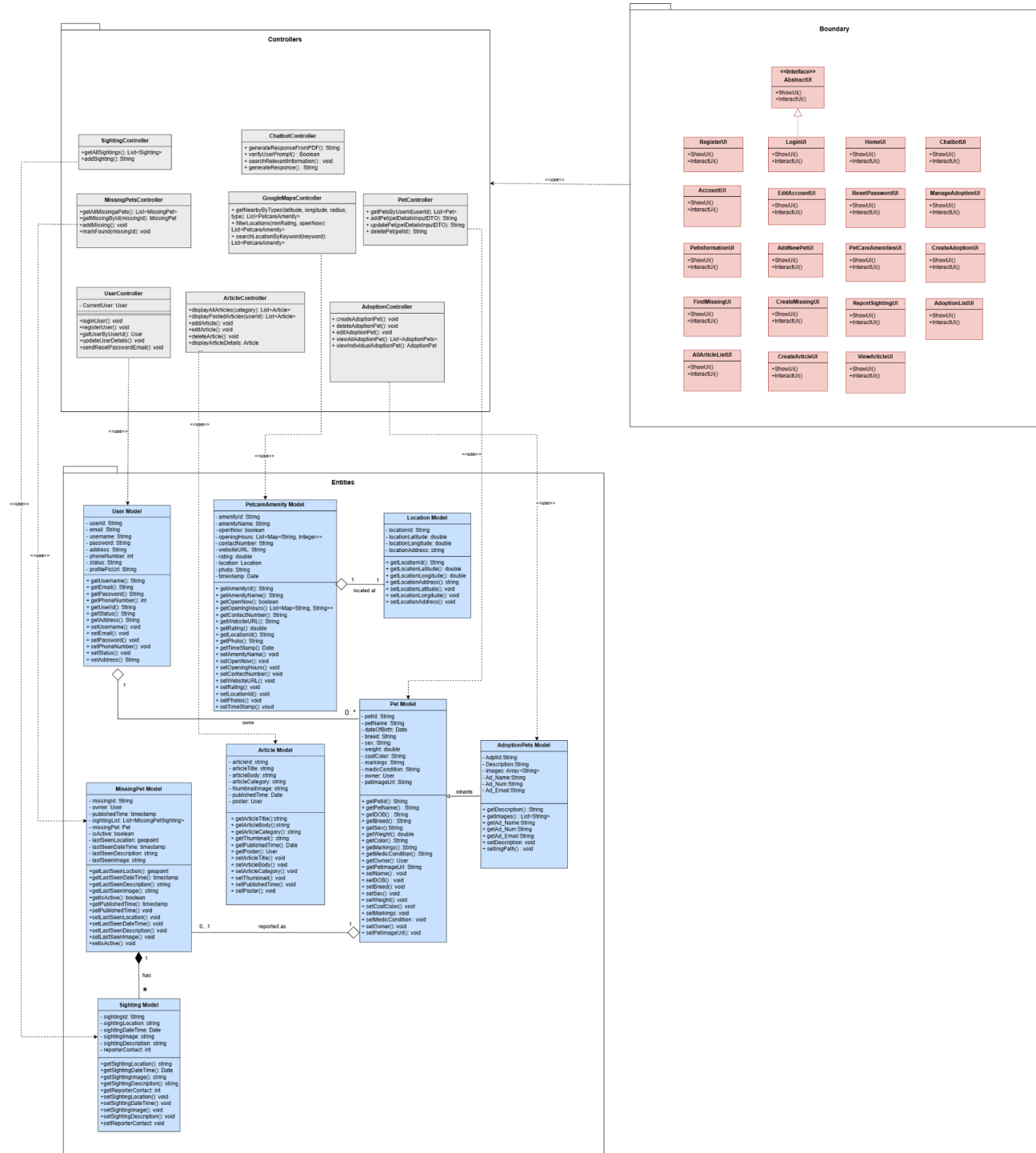
	functionality and application logic, handling the interactions between different components and implementing use cases.
System	<i>PetCare</i> application.
Tavily	Tavily is a search engine optimized for LLMs, optimized for a factual, efficient, and persistent search experience
User	The person using the <i>PetCare</i> mobile application after creating a <i>PetCare</i> account.
UI	User interface
User Input Prompt	A text entry submitted by the user in the chatbot interface.
View Pet Page	A page that displays all the information related to the pet.
View Adoption Center Pet Page	A page that displays all the available pets for adoption added by the adoption center for users to view.

Appendix B: Analysis Models

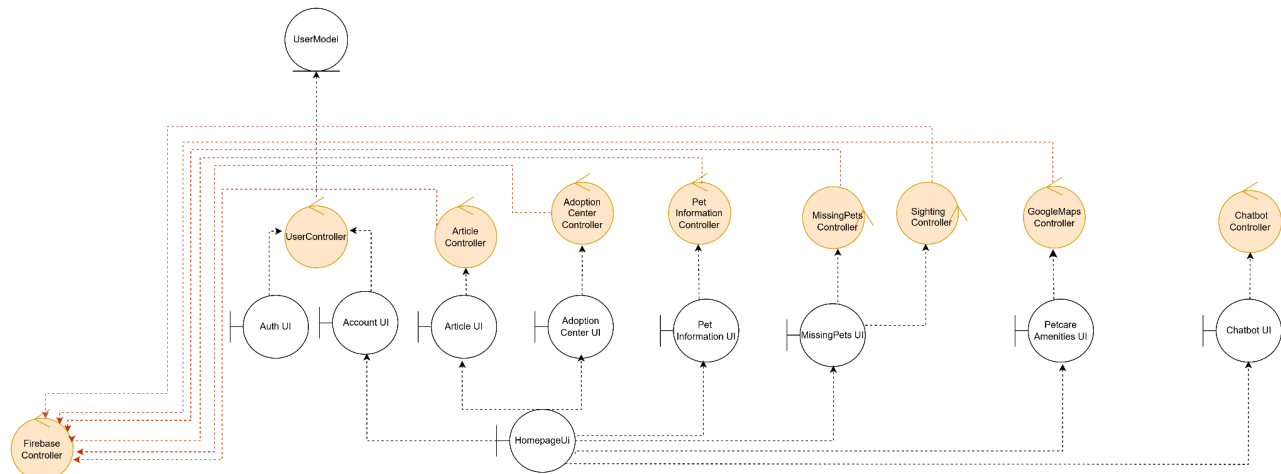
B.1 Use Case Model



B.2 Entity Class Diagram

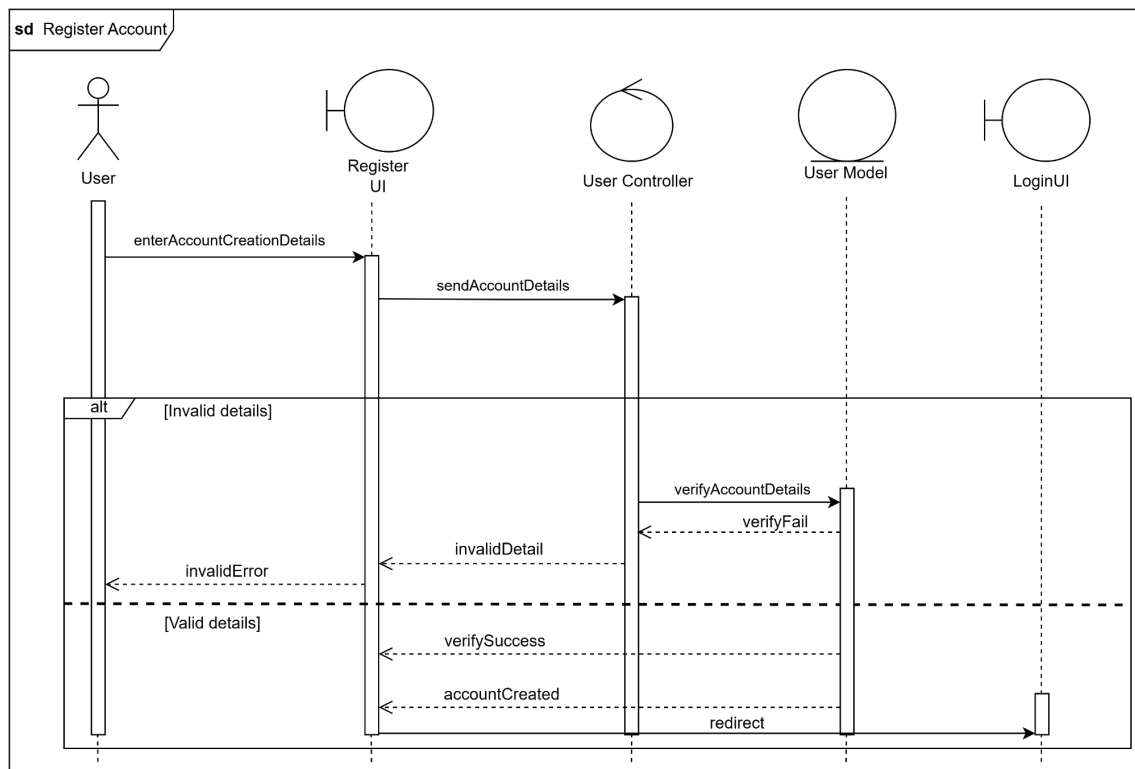


B.3 Control and Boundary Class Diagram

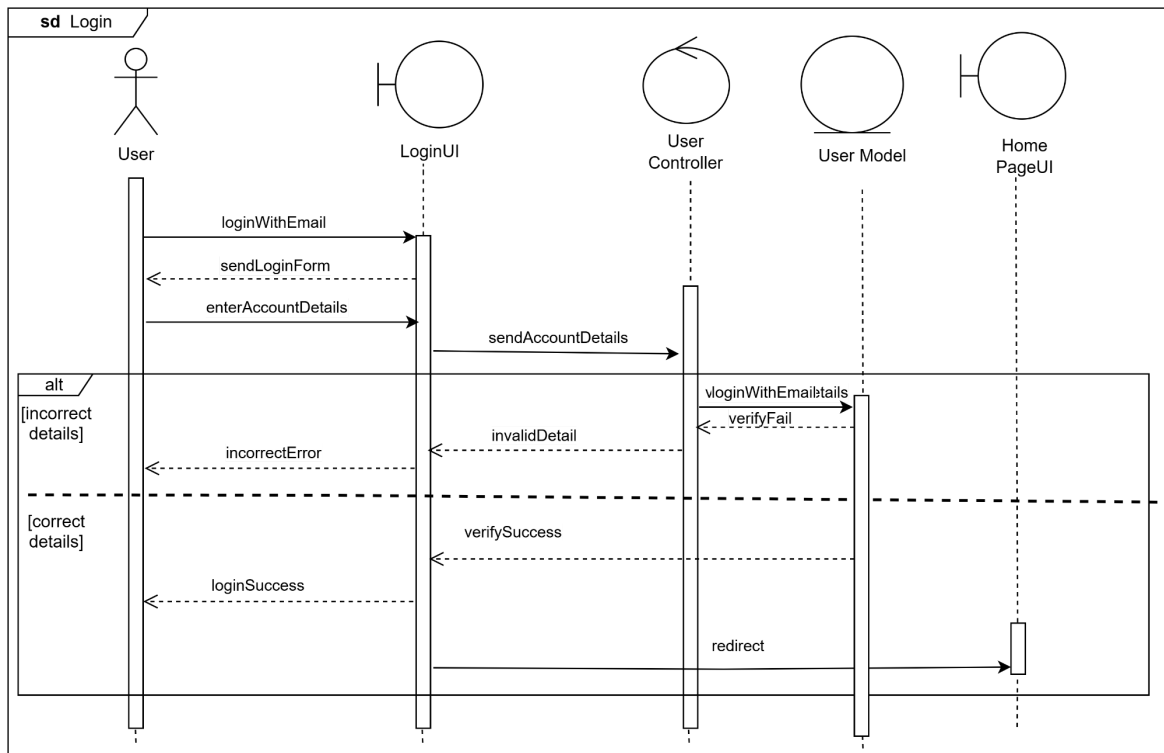


B.4 Sequence Diagram

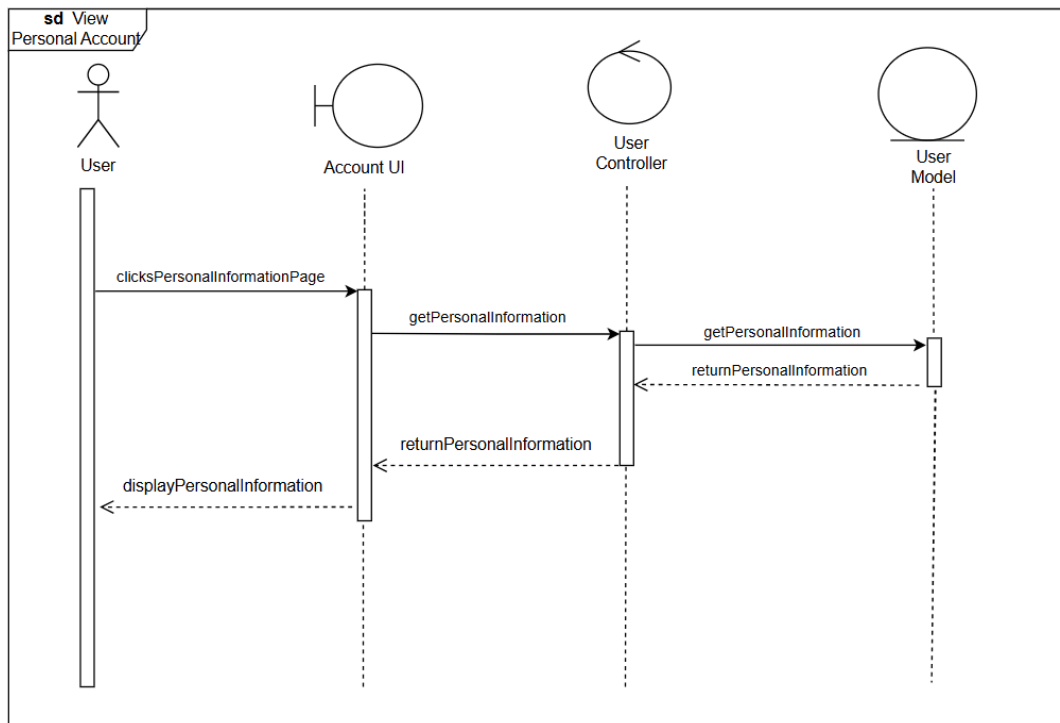
B.4.1 Register Account

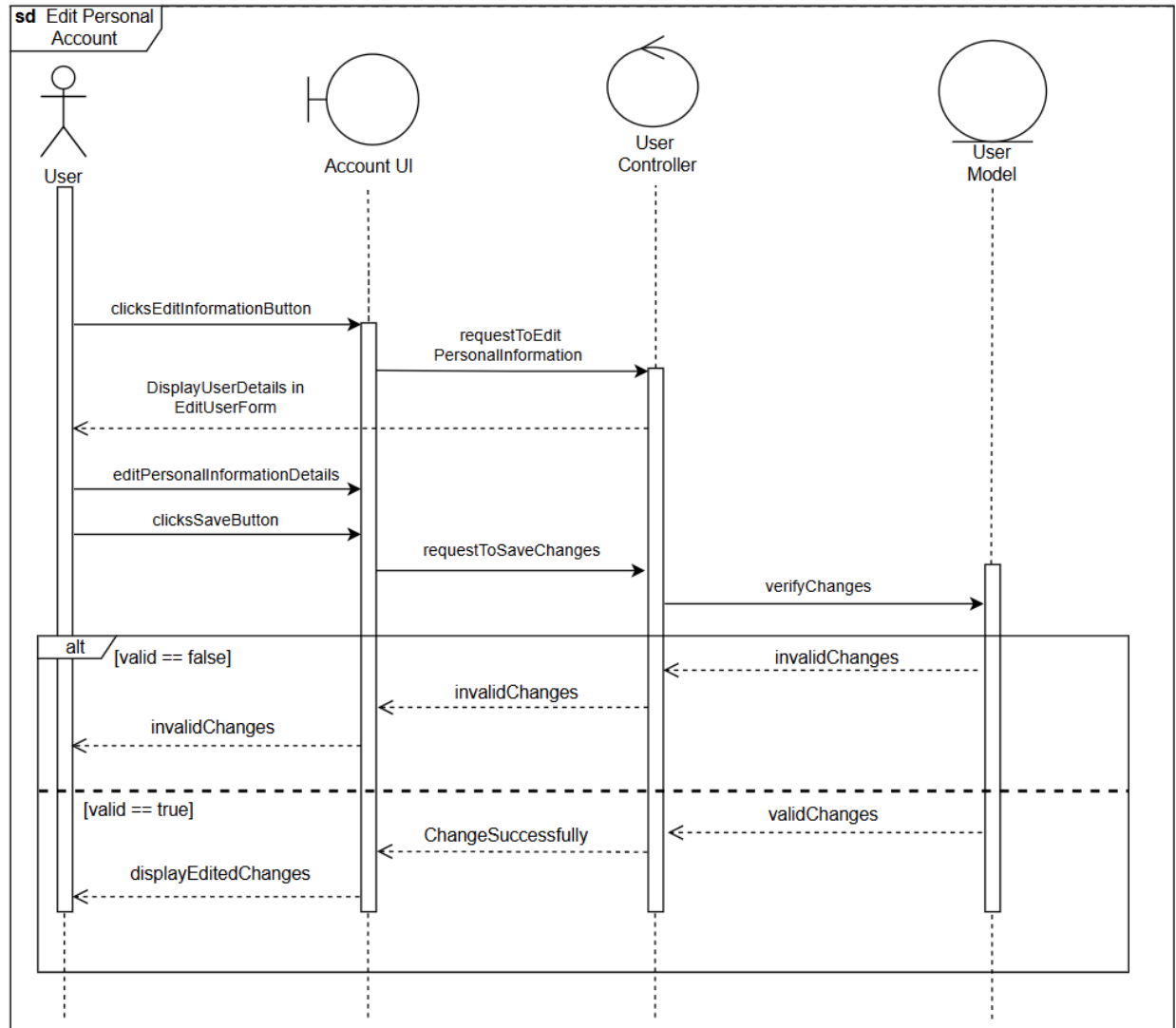


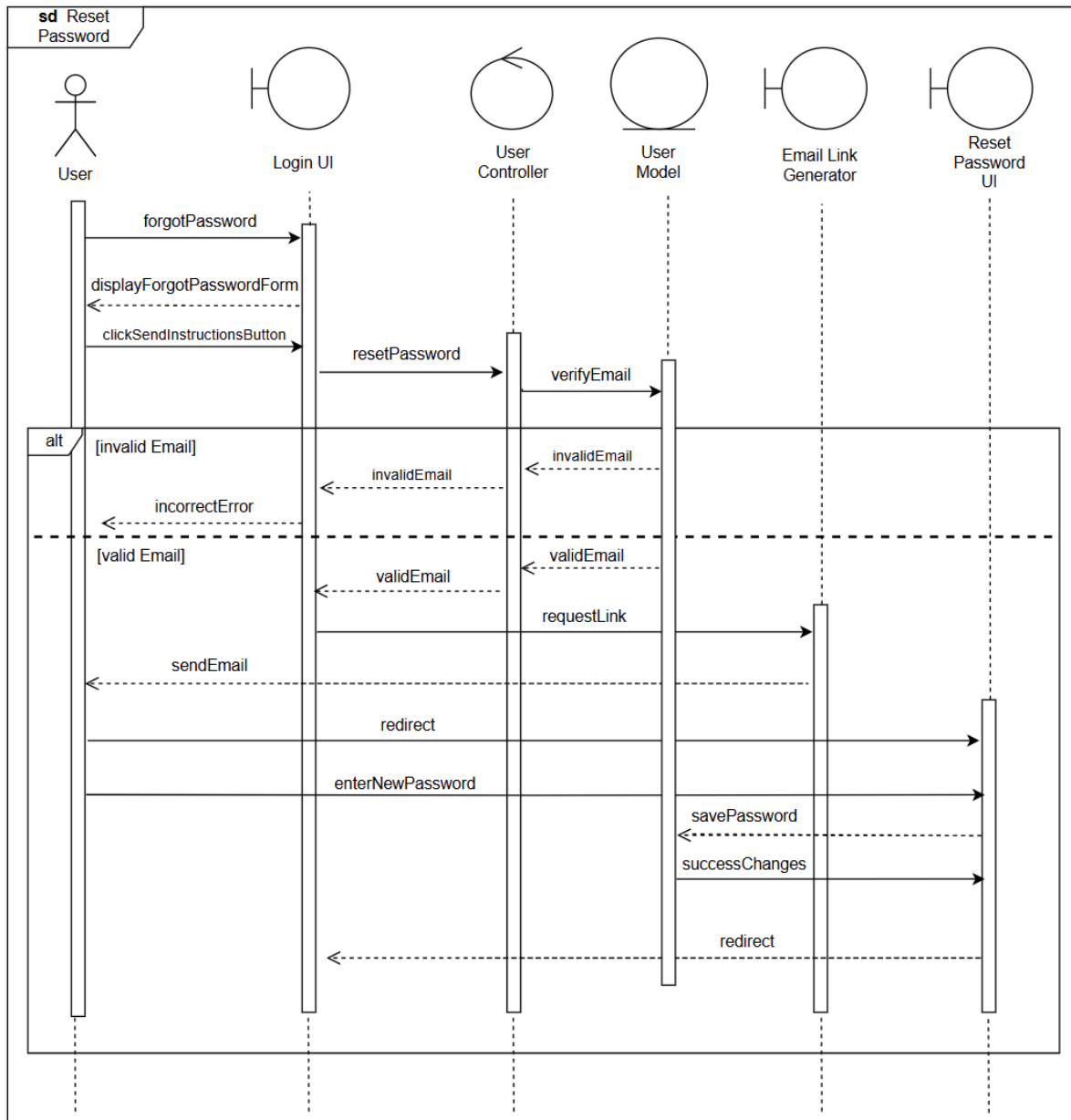
B.4.2 Login

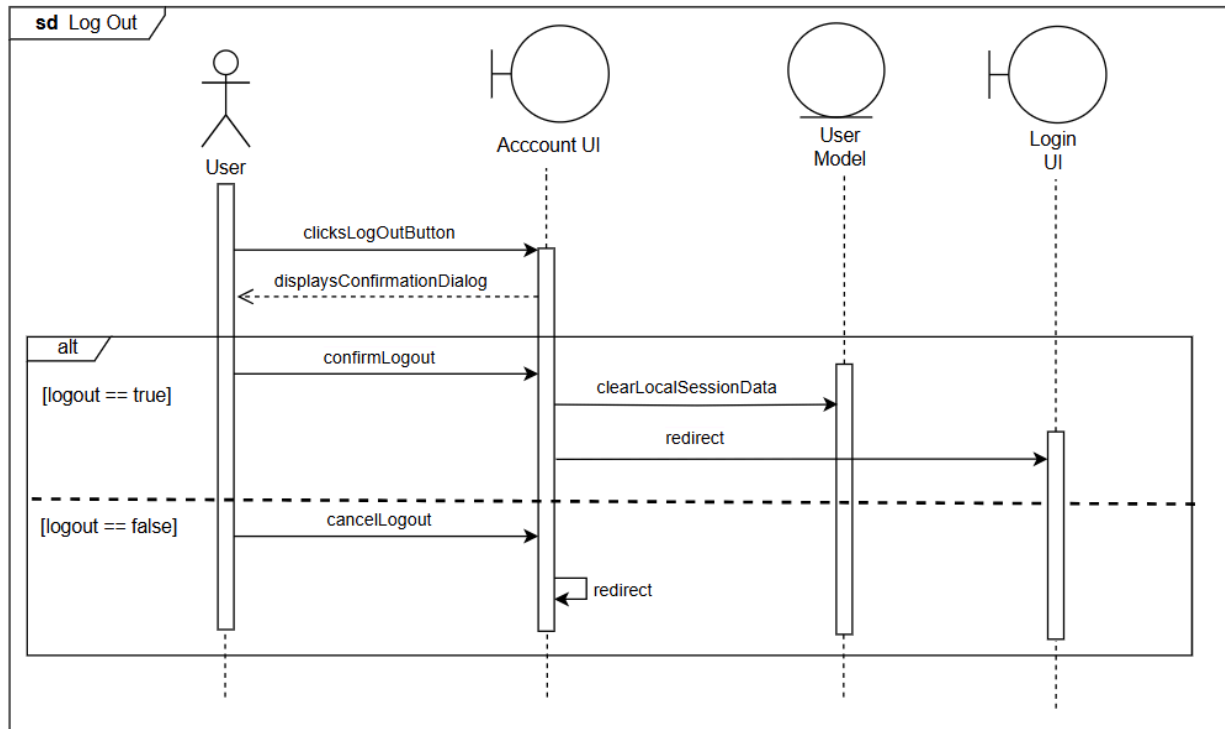


B.4.3 View Personal Account

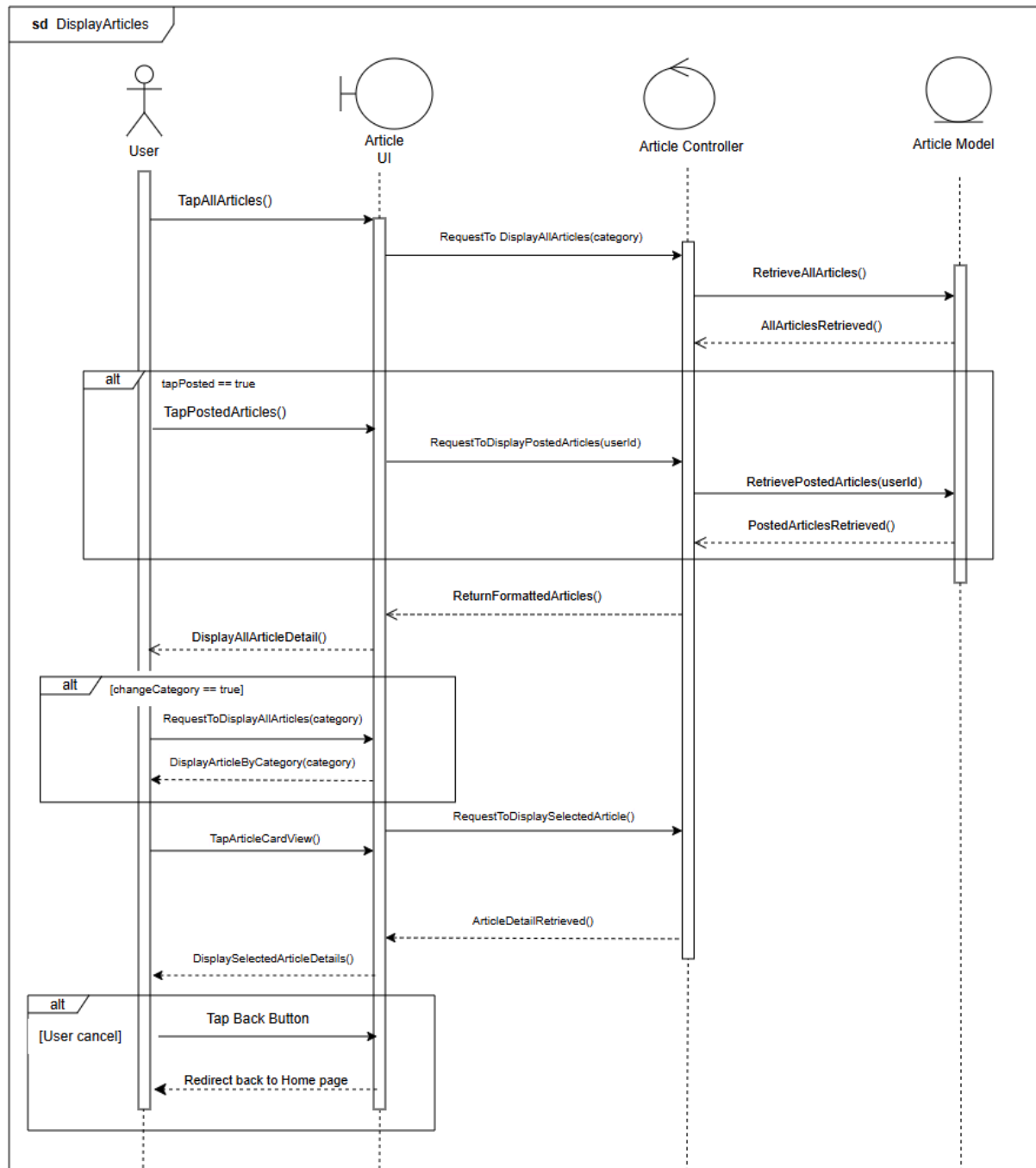


B.4.4 Edit Personal Account

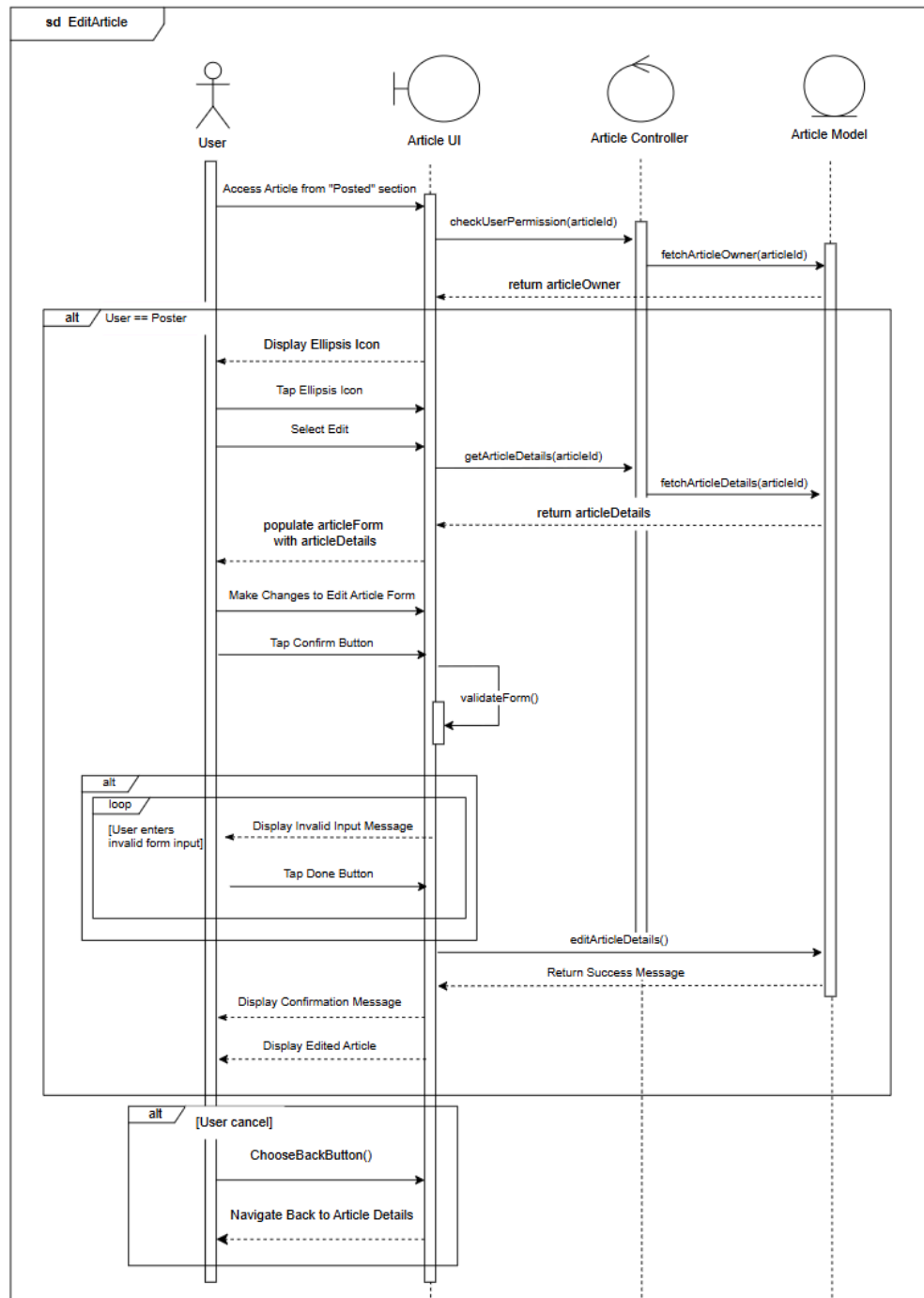
B.4.5 Reset Password

B.4.6 Log Out

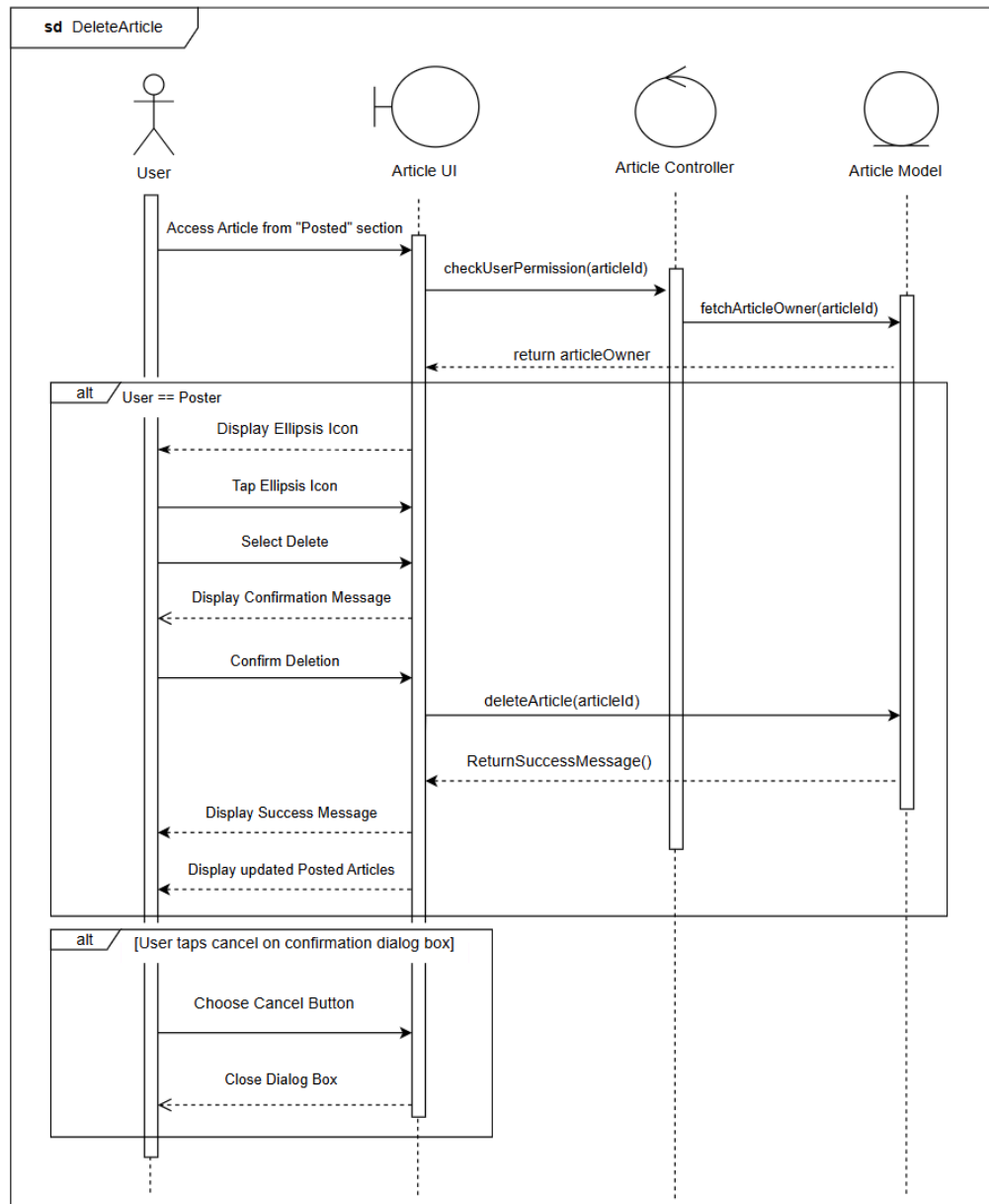
B.4.7 Display Articles



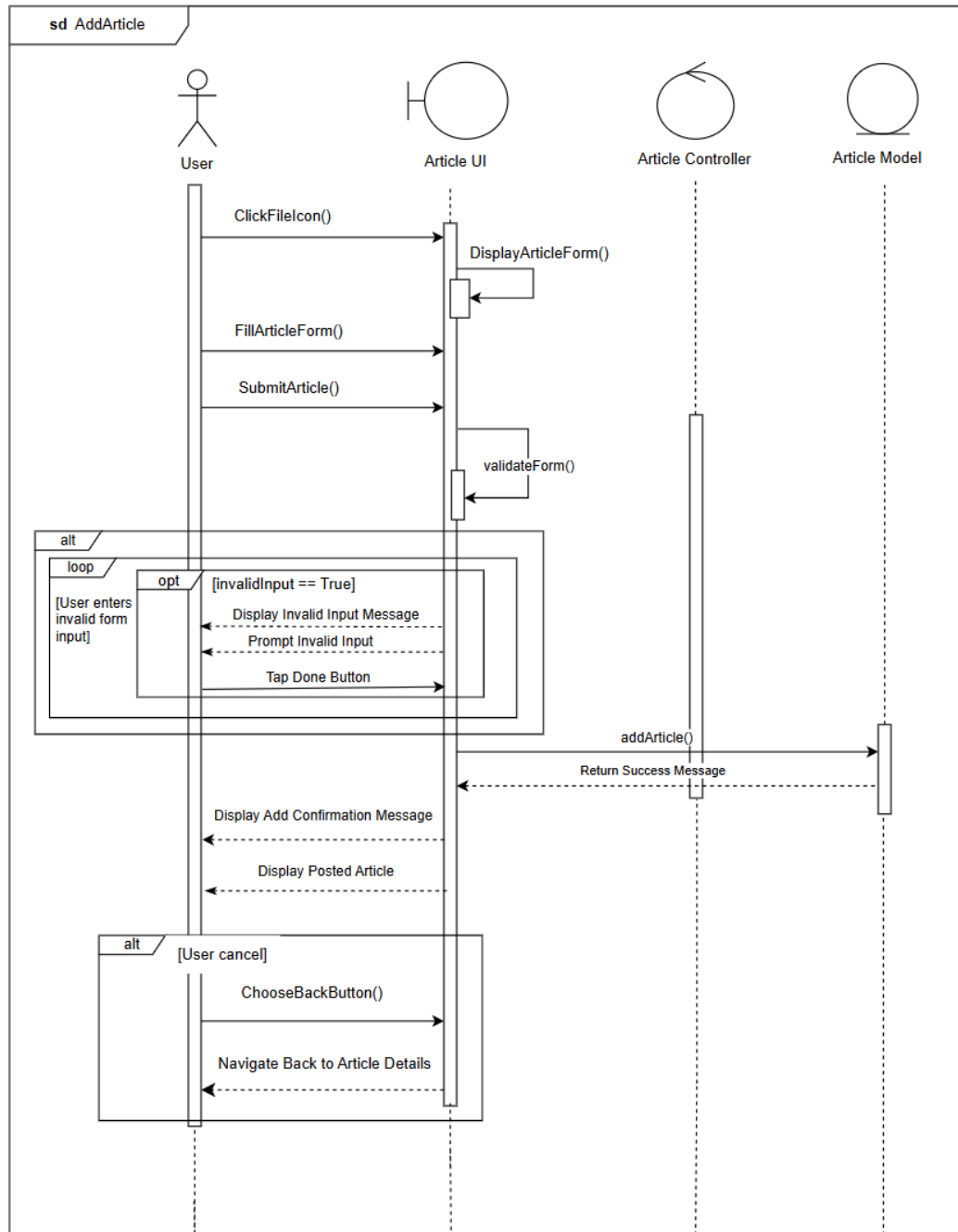
B.4.8 Edit Article



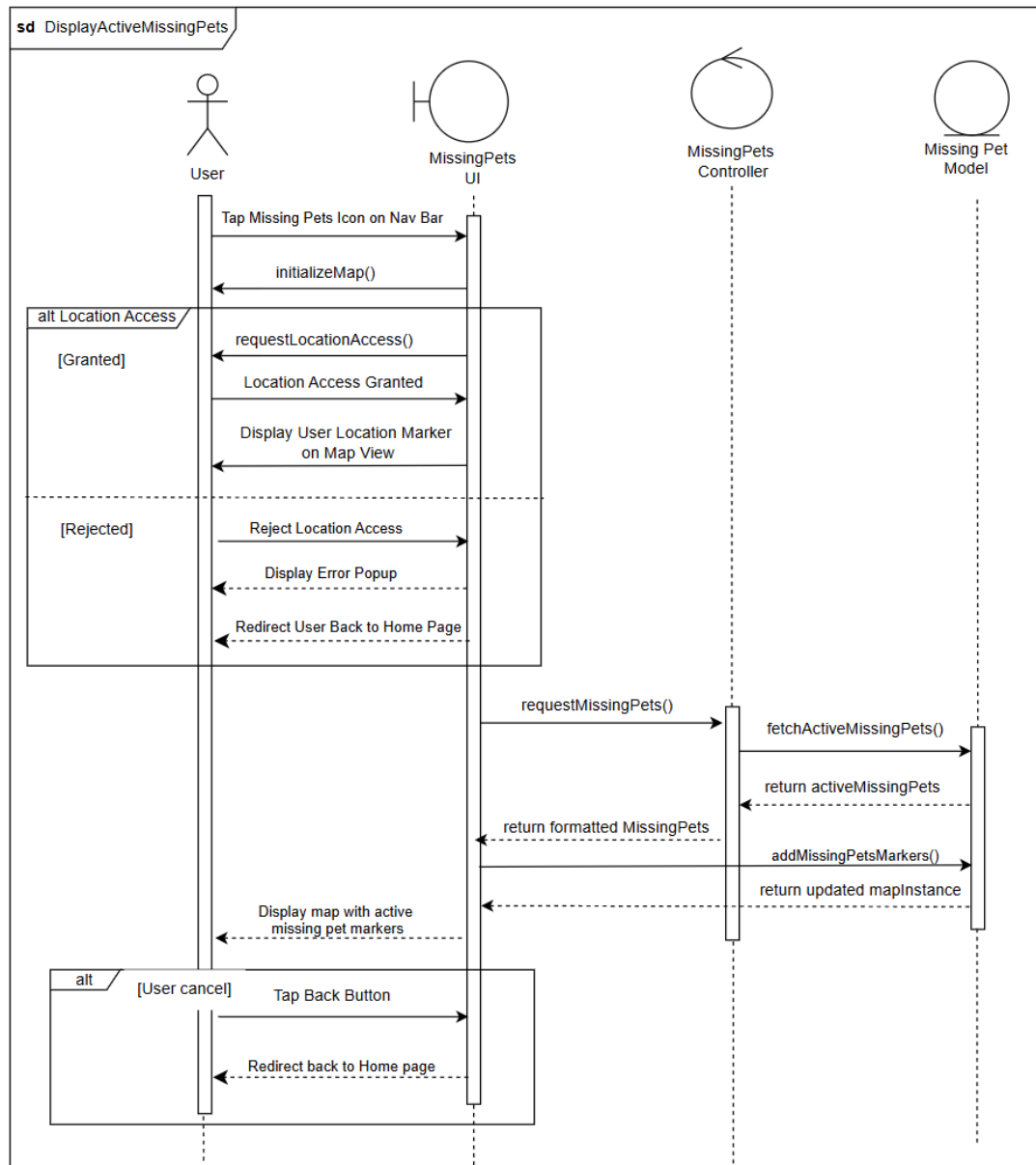
B.4.9 Delete Article

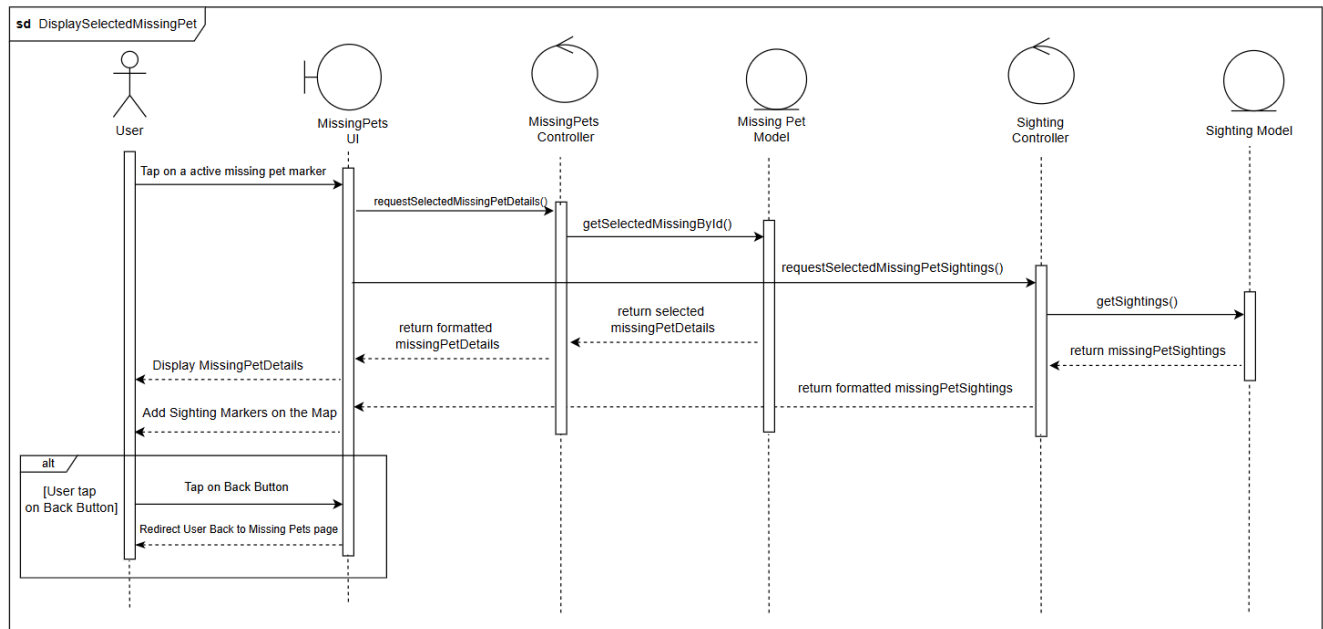


B.4.10 Add Article

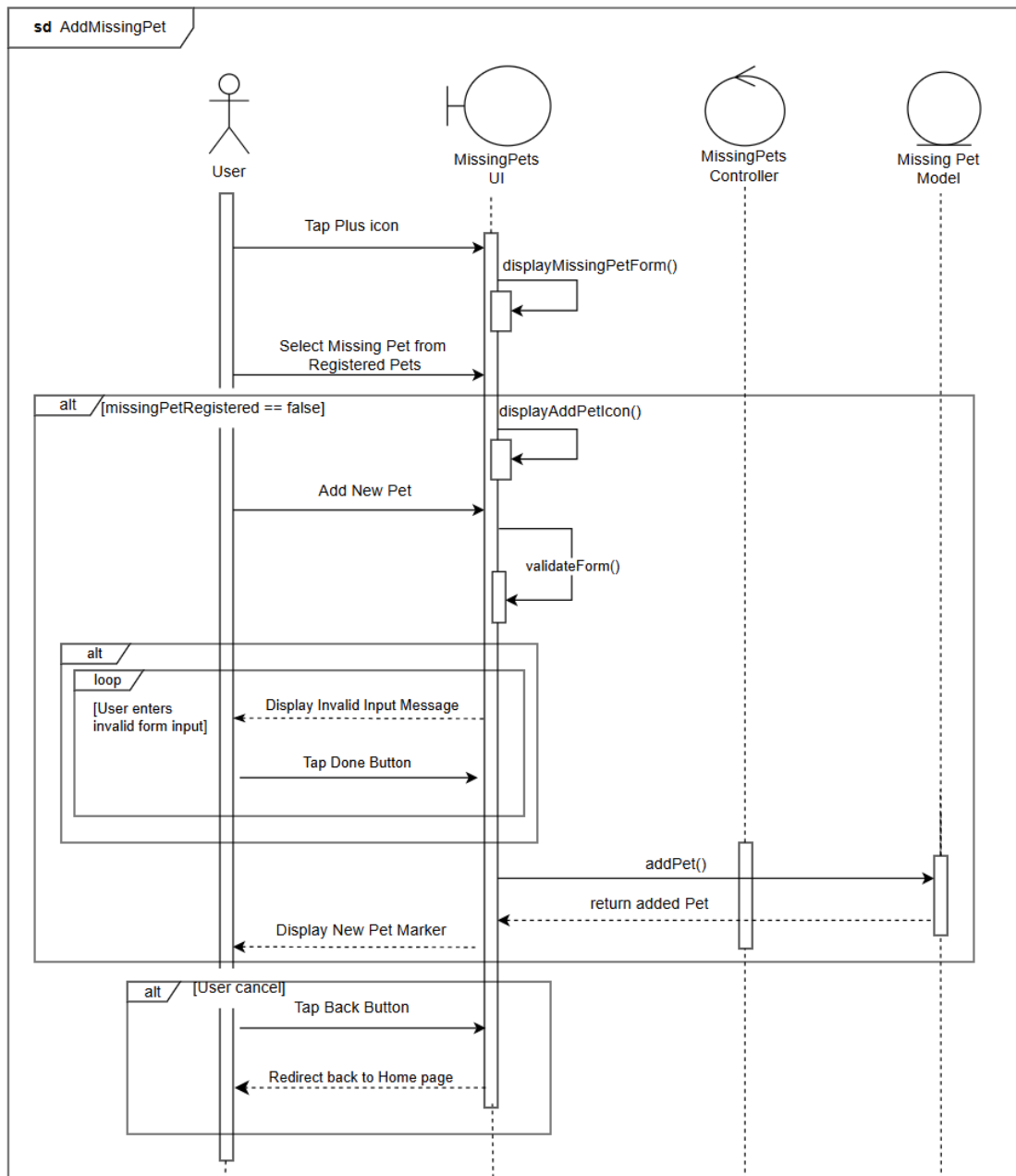


B.4.11 Display Active Missing Pets

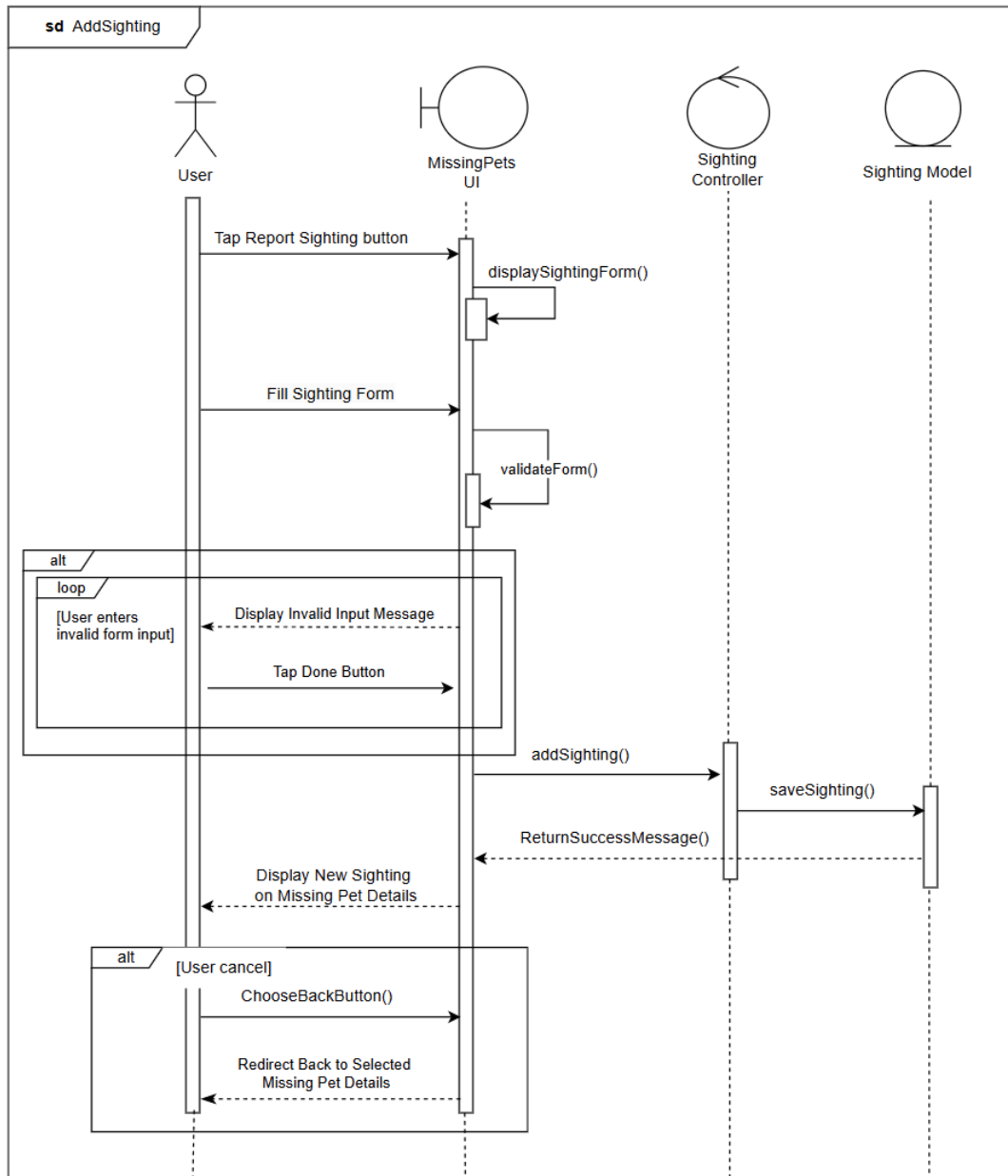


B.4.12 Display Selected Missing Pet

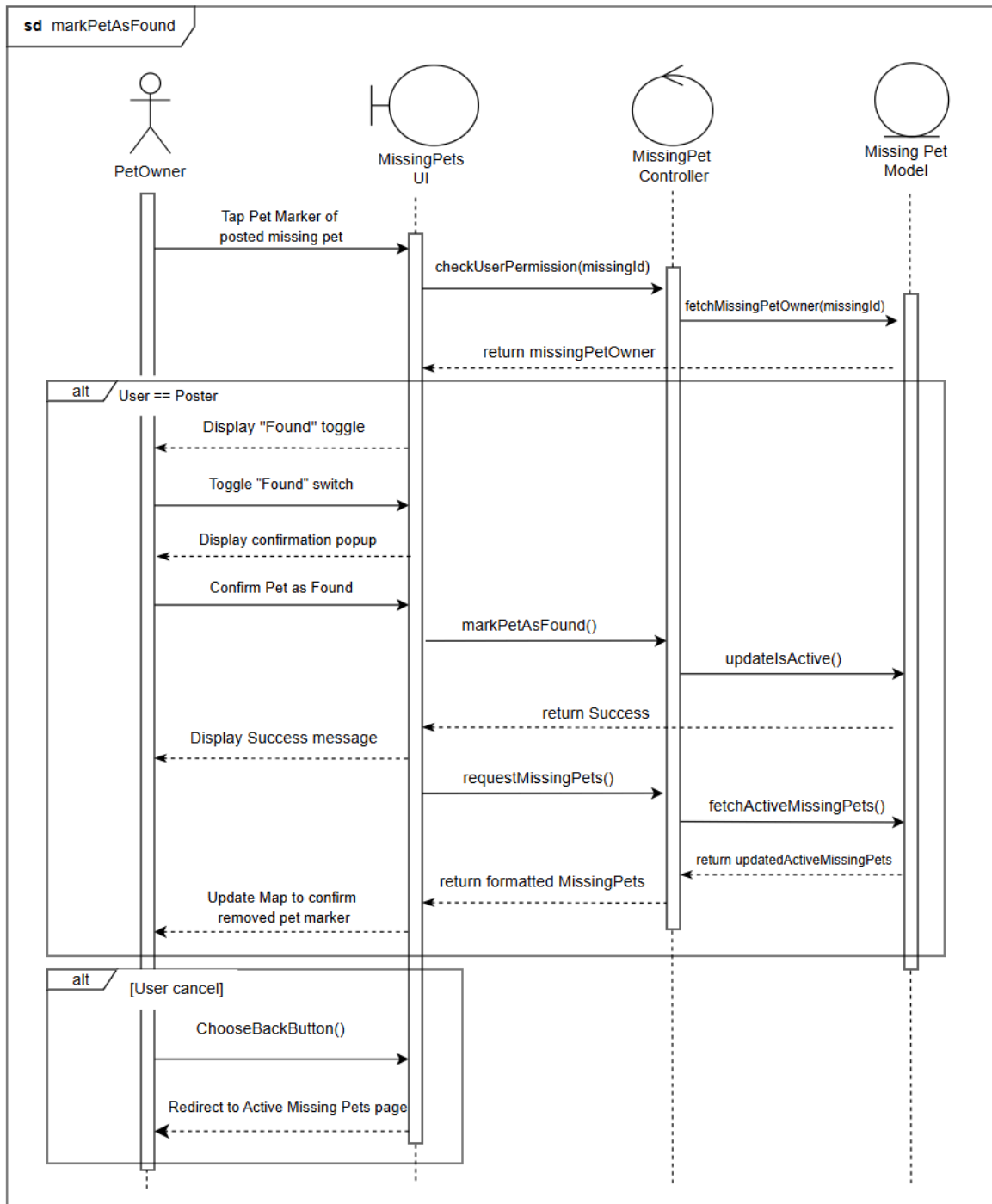
B.4.13 Add Missing Pet



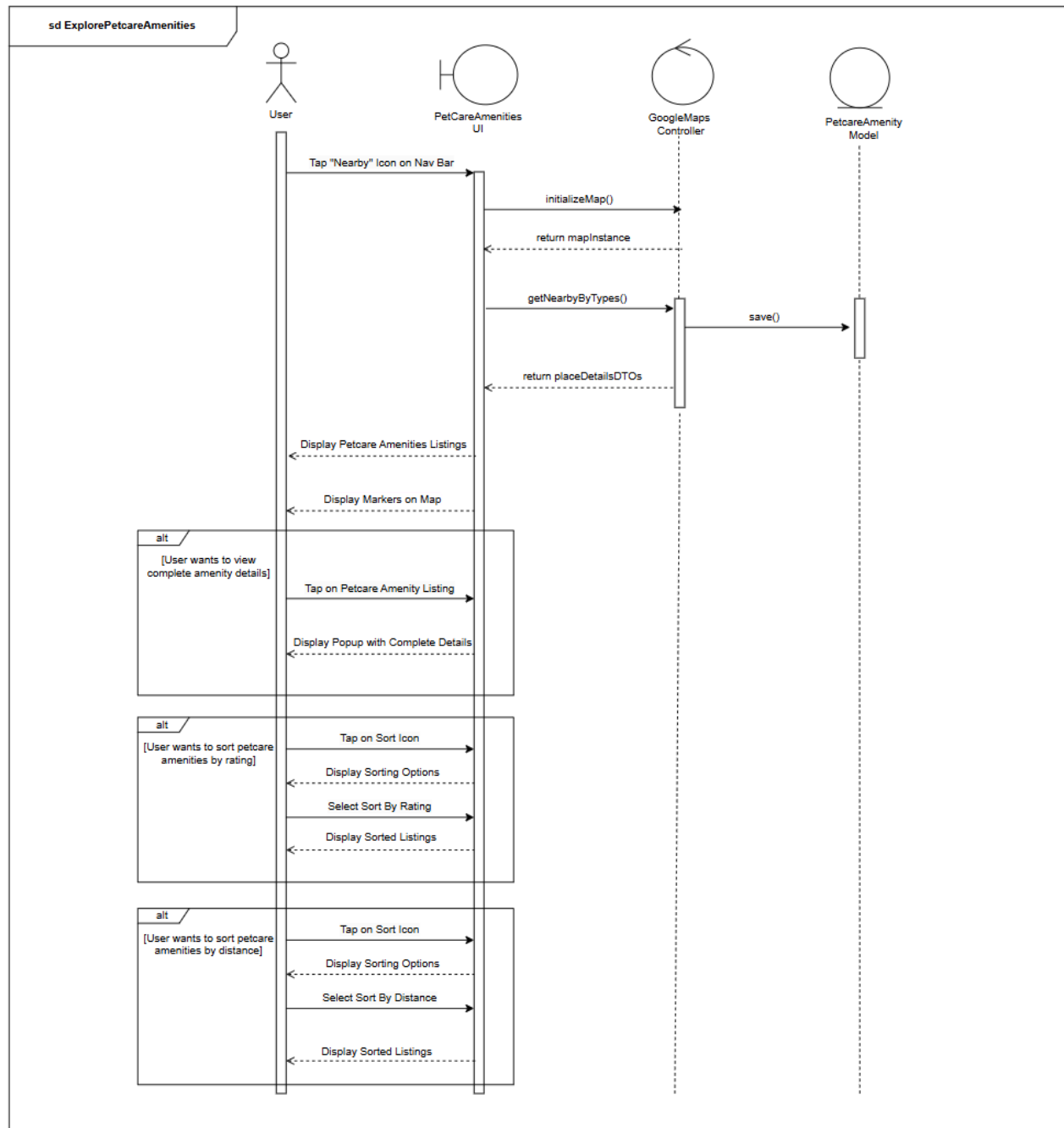
B.4.14 Add Sighting



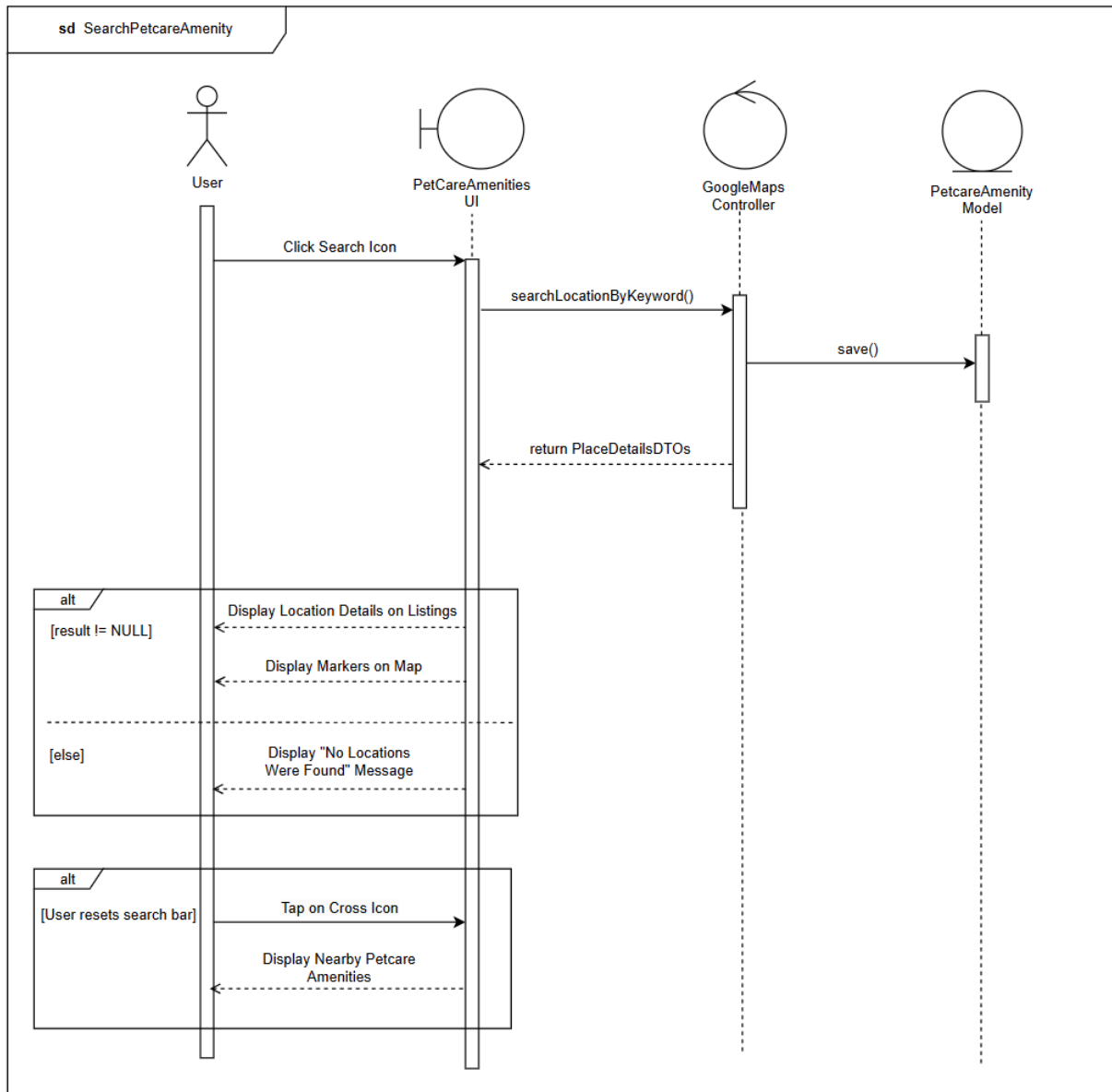
B.4.15 Mark Pet As Found



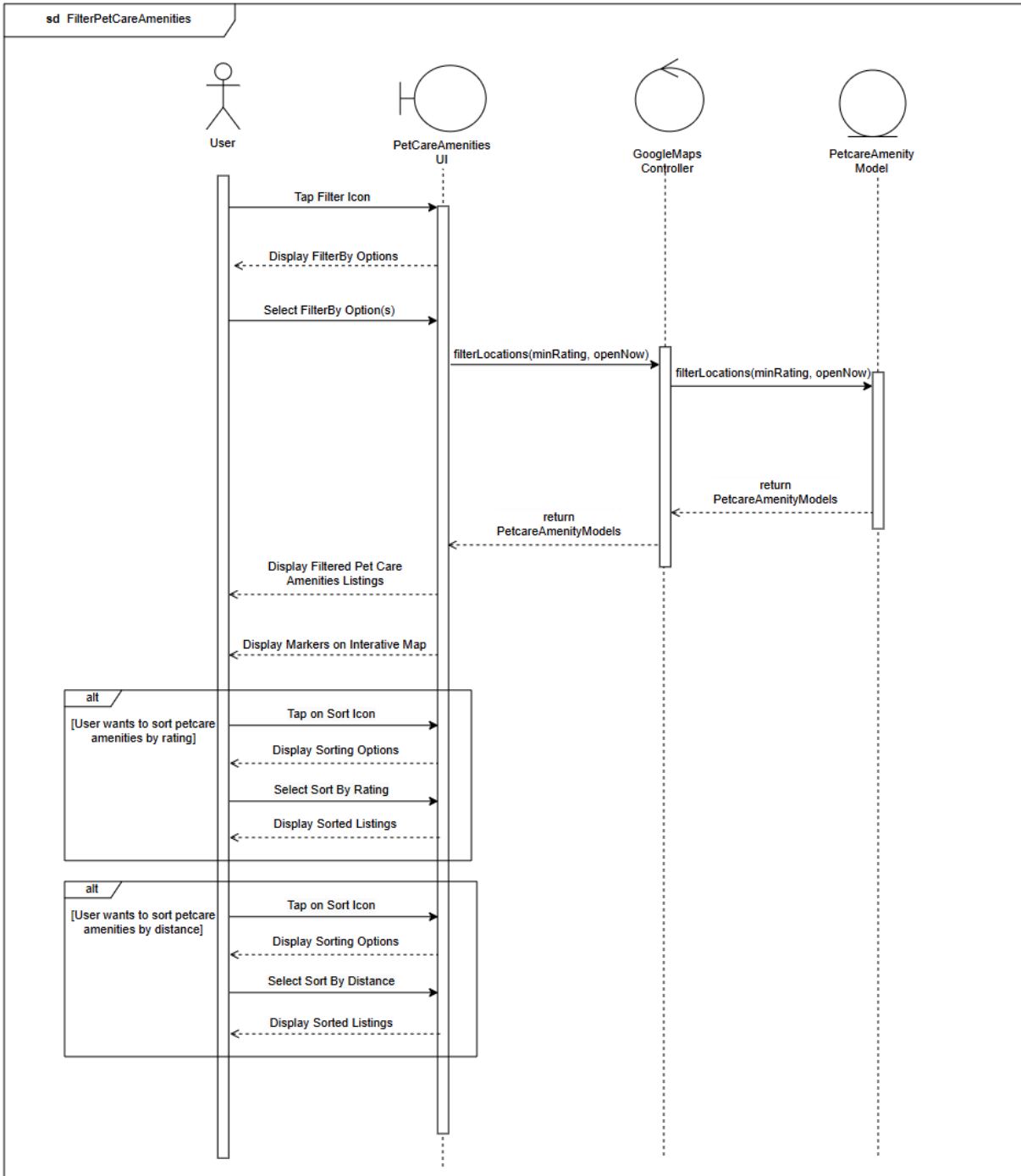
B.4.16 Explore PetCare Amenities

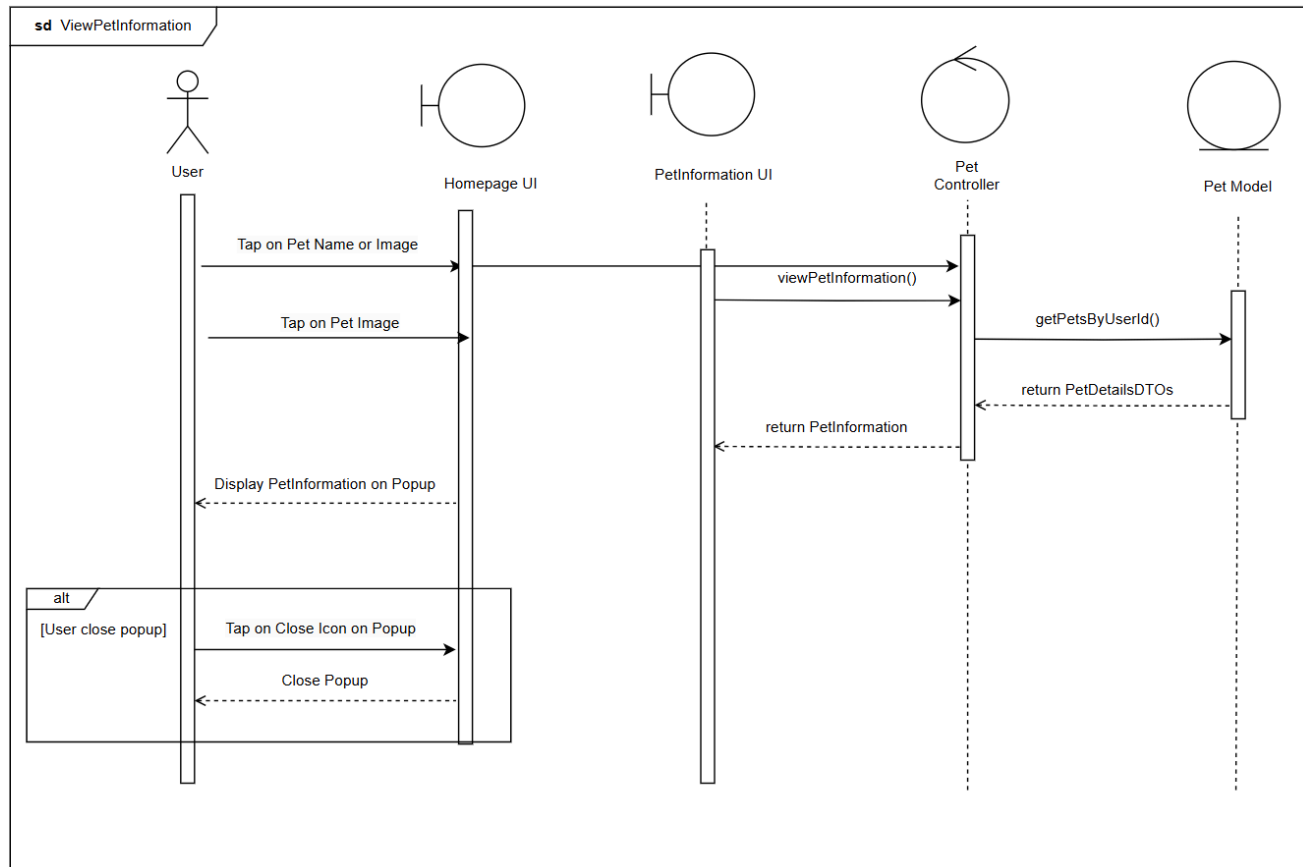


B.4.17 Search PetCare Amenity

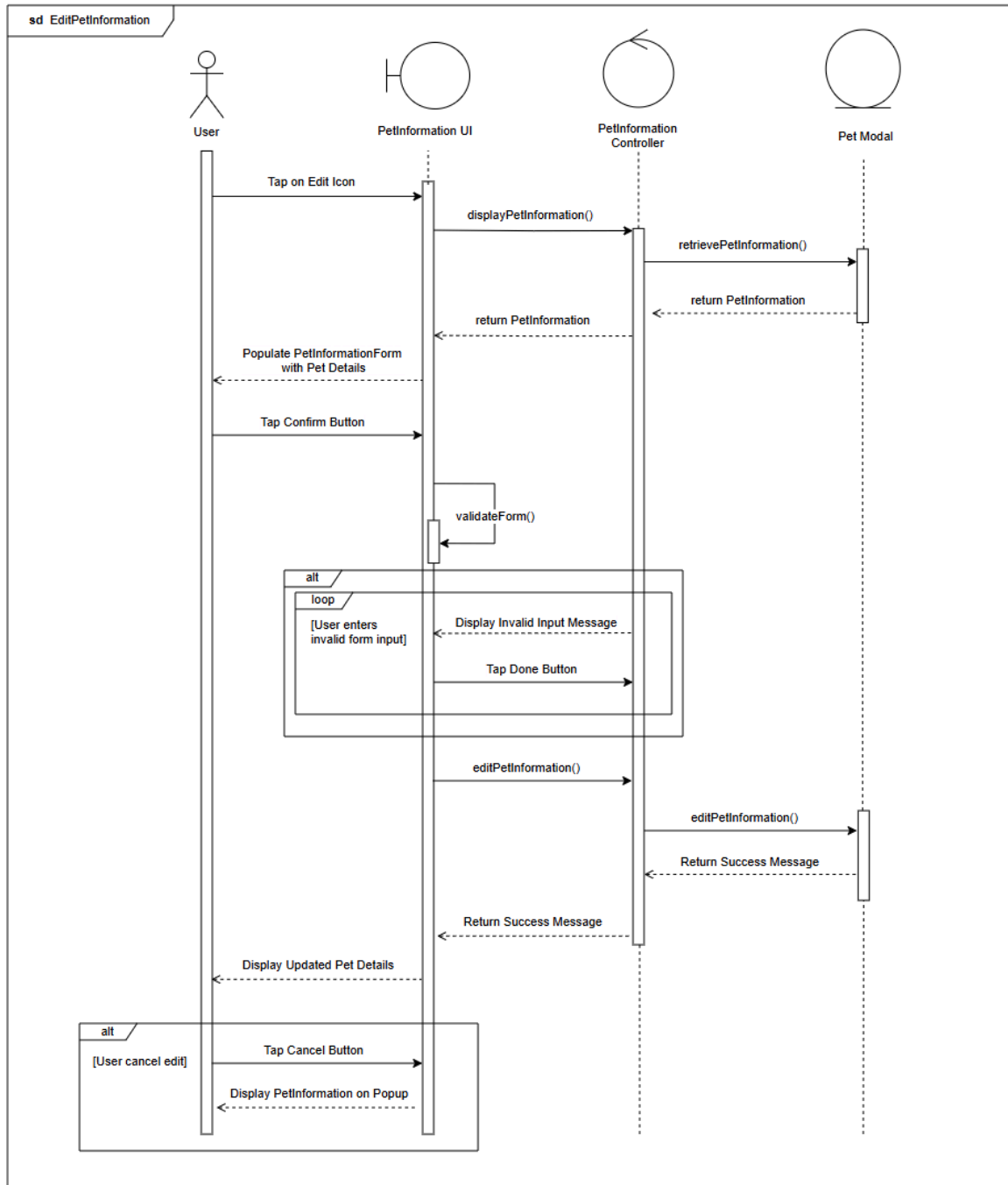


B.4.18 Filter PetCare Amenities

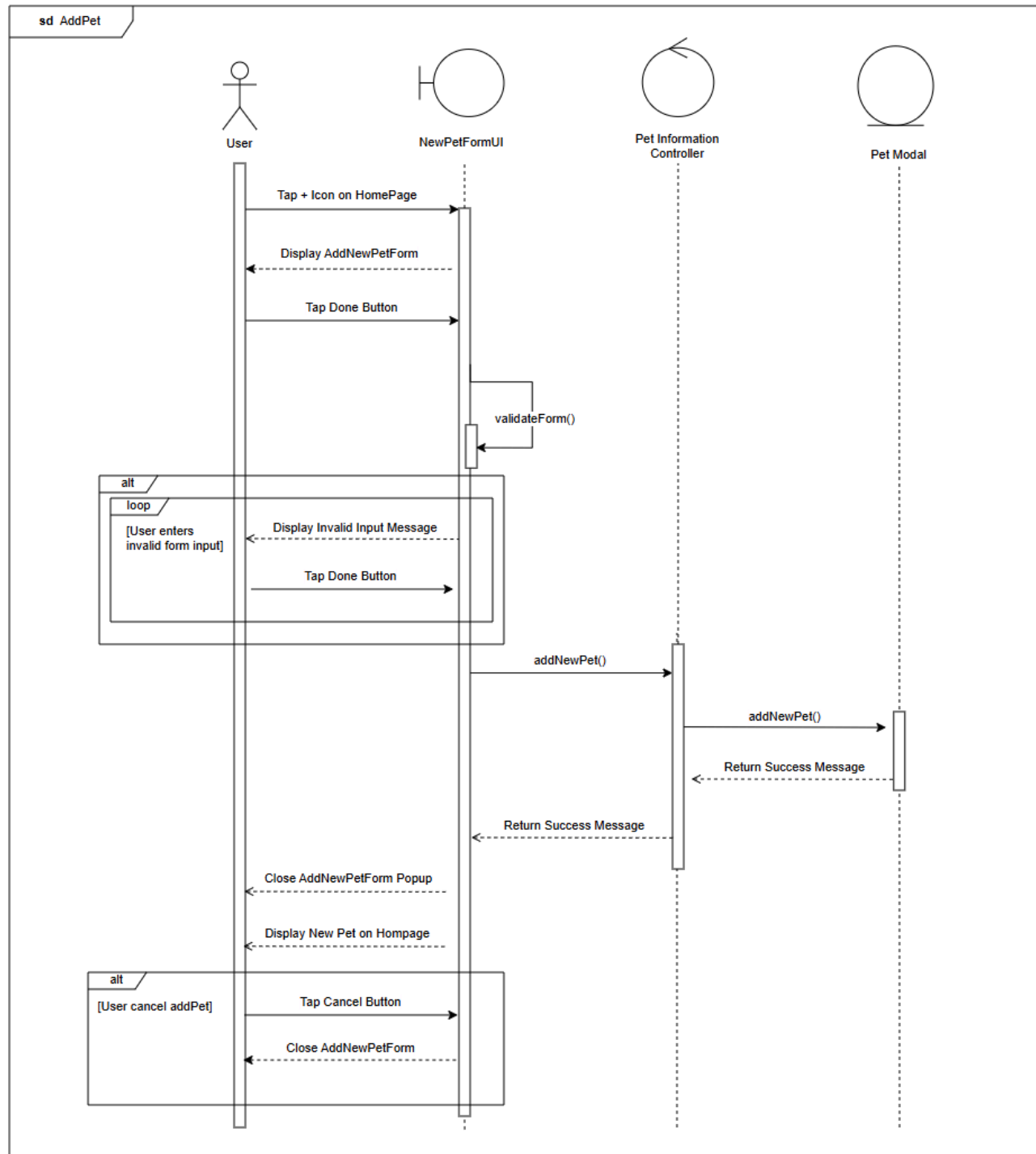


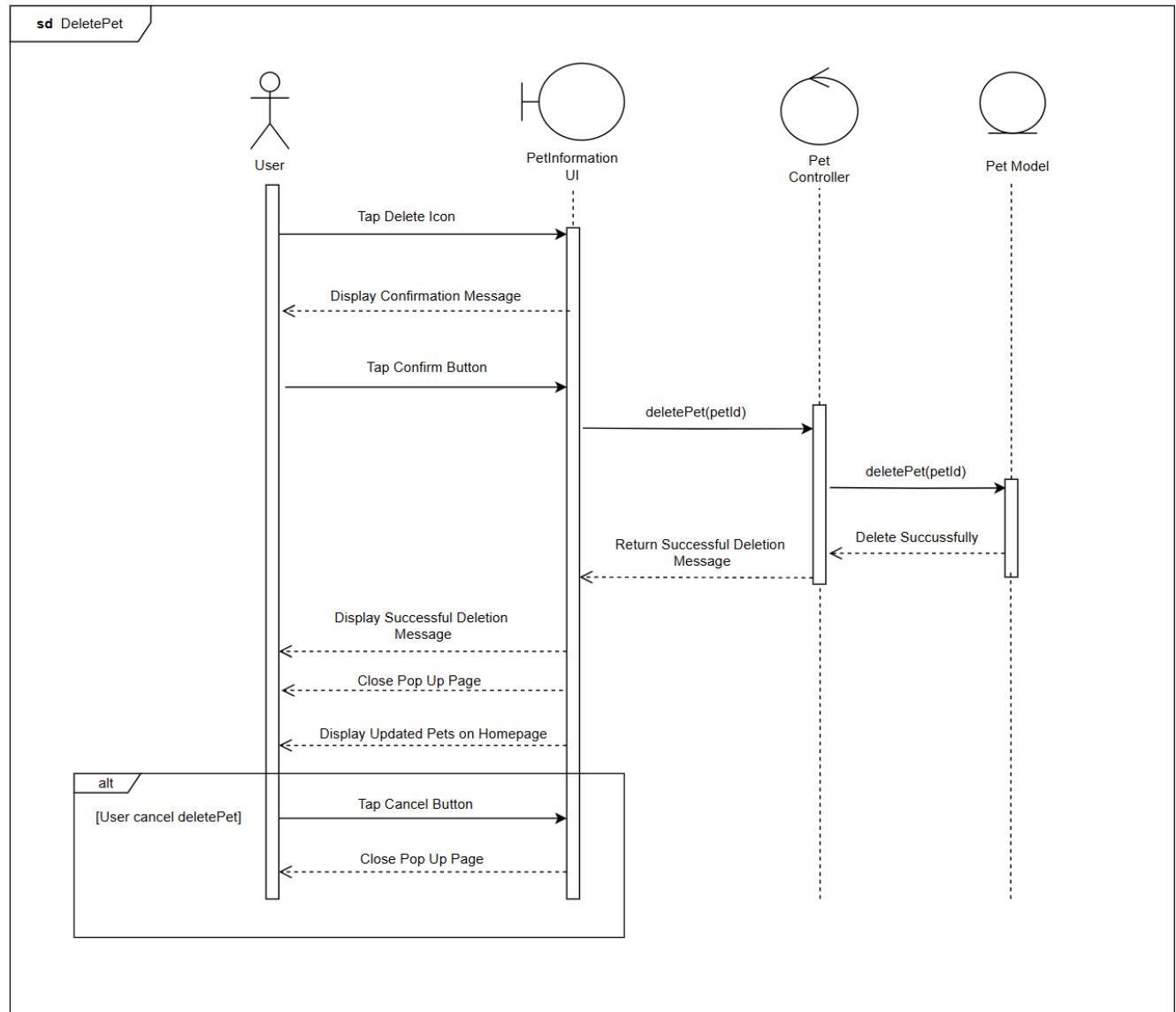
B.4.19 View Pet Information

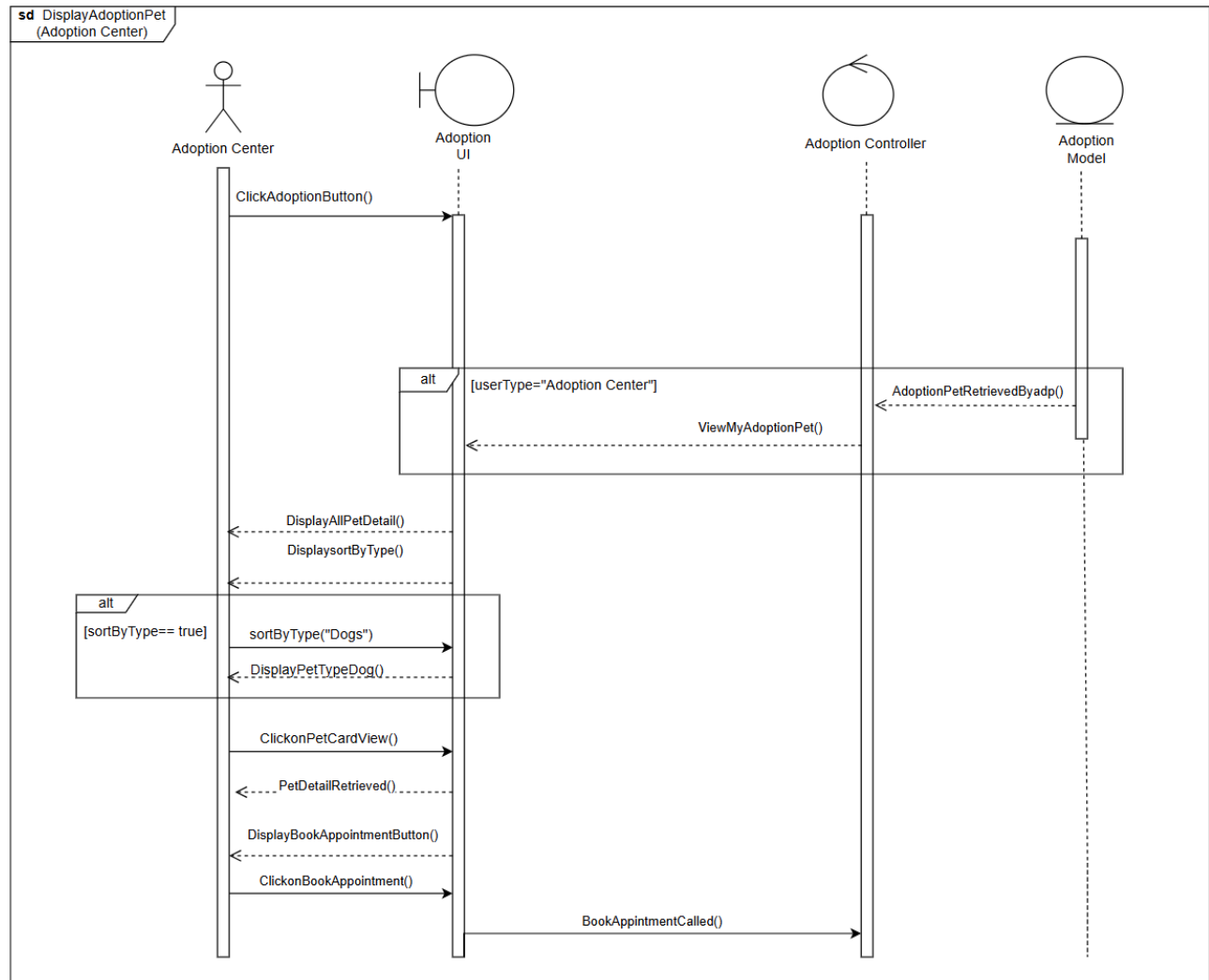
B.4.20 Edit Pet Information

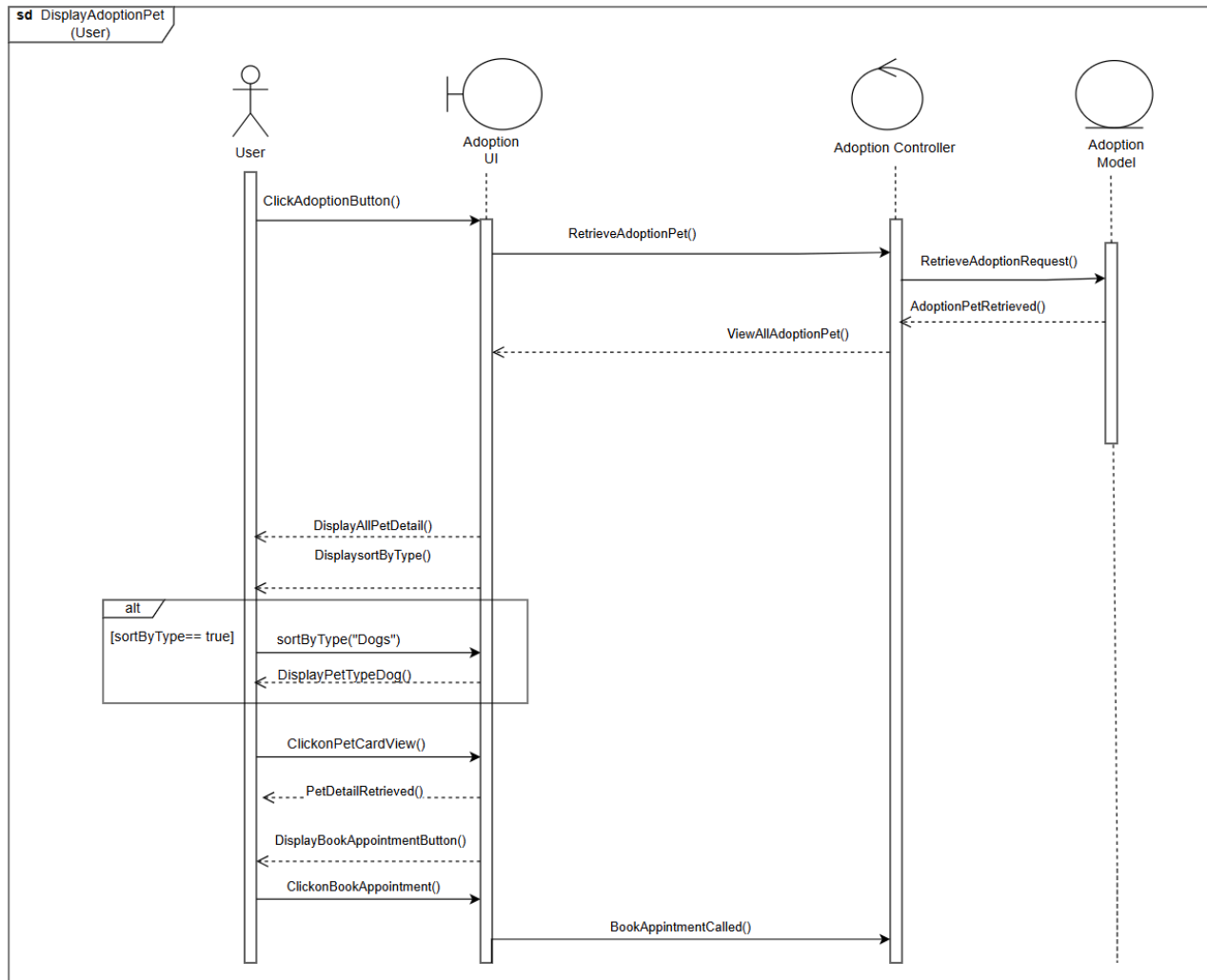


B.4.21 Add Pet

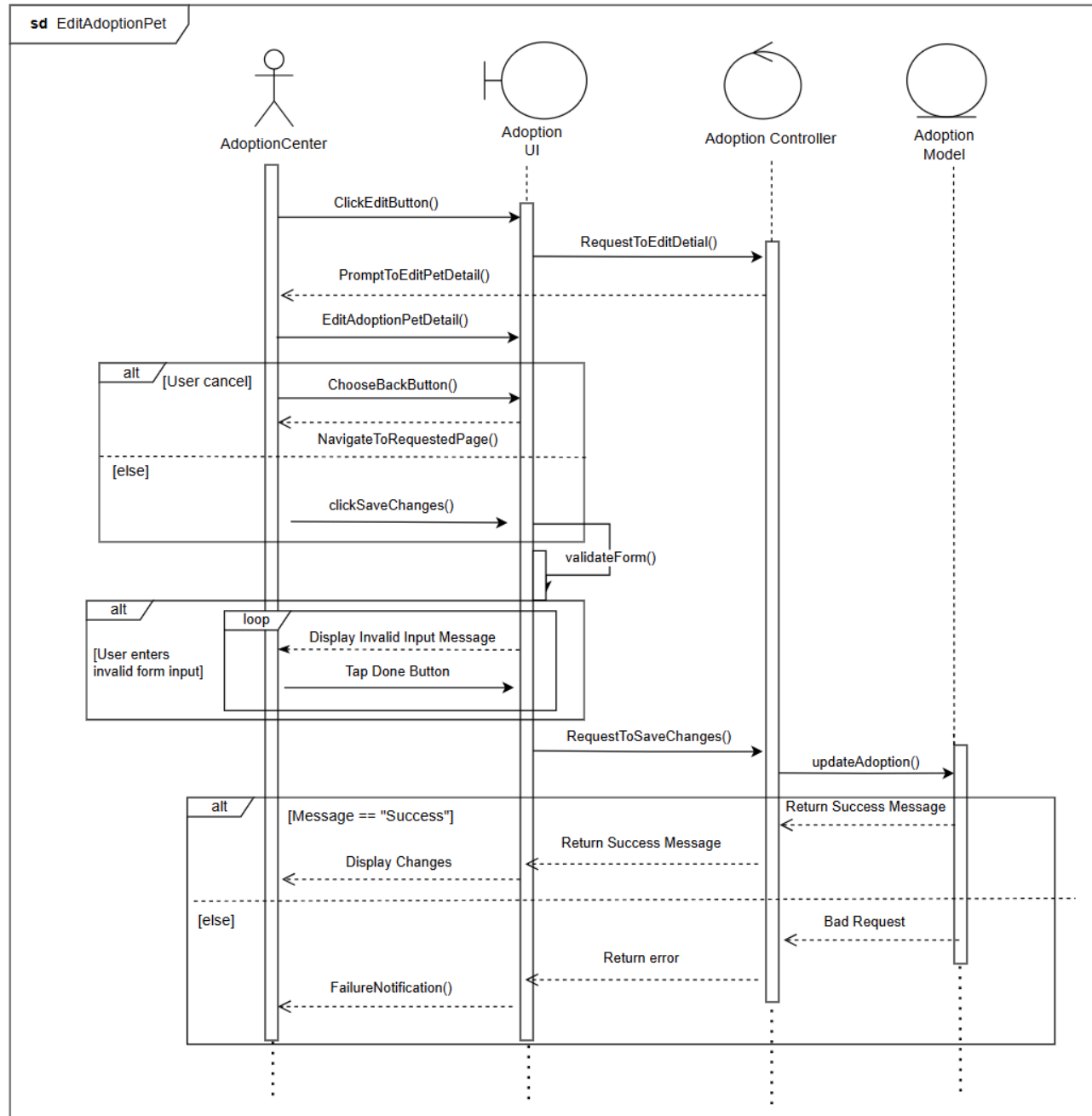


B.4.22 Delete Pet

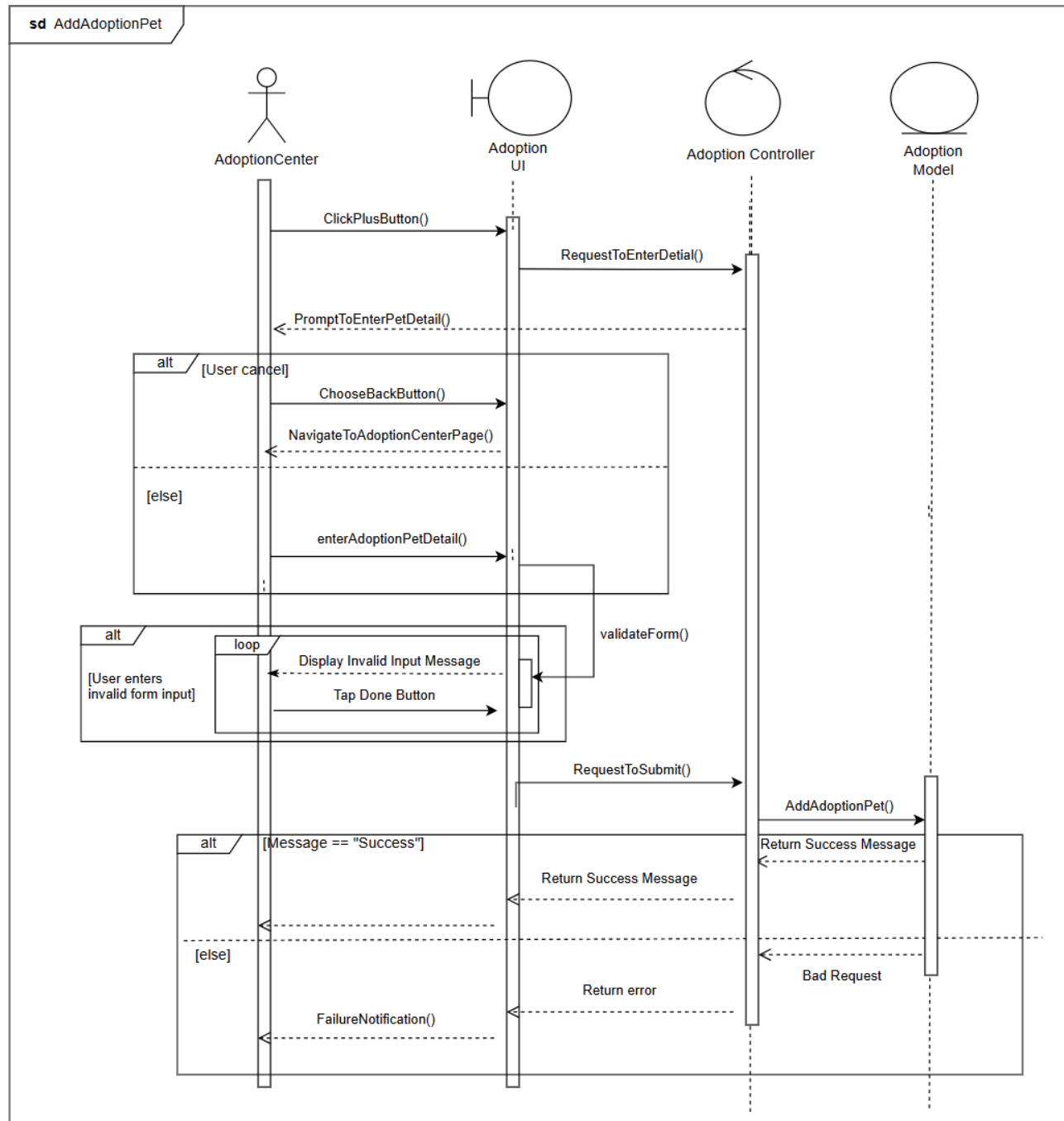
B.4.23 Display Adoption Pet (Adoption Center)

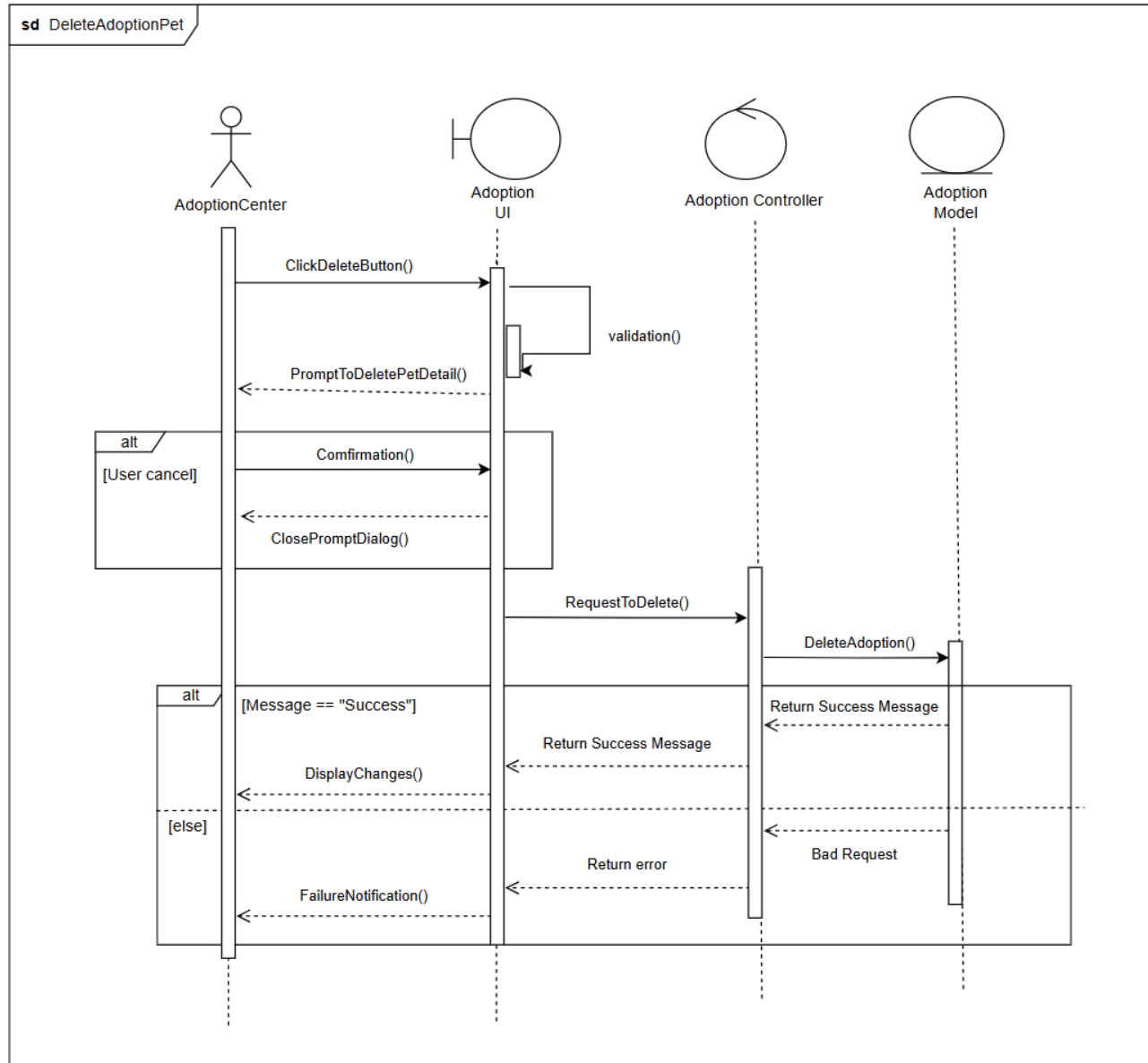
B.4.24 Display Adoption Pet (User)

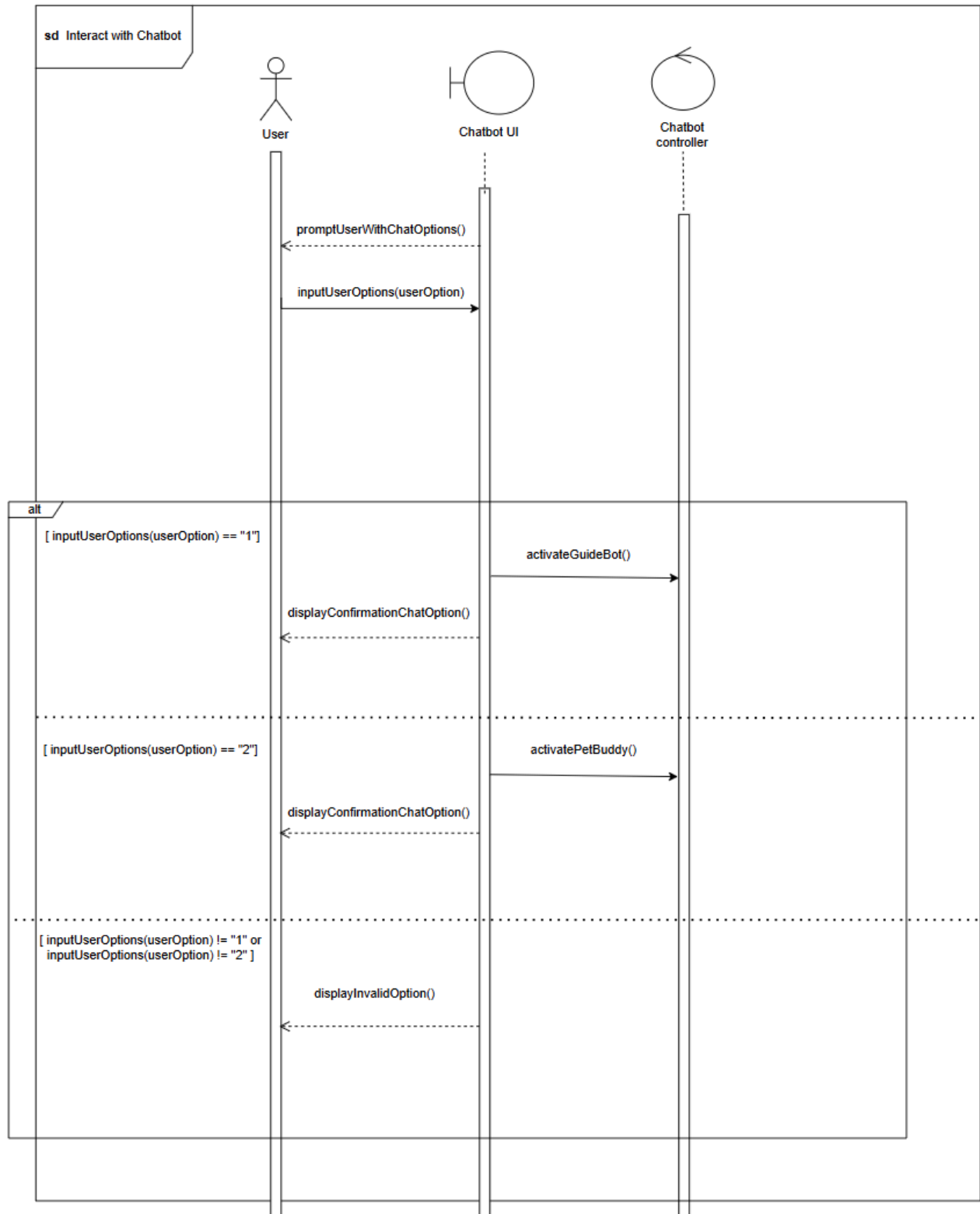
B.4.25 Edit Adoption Pet

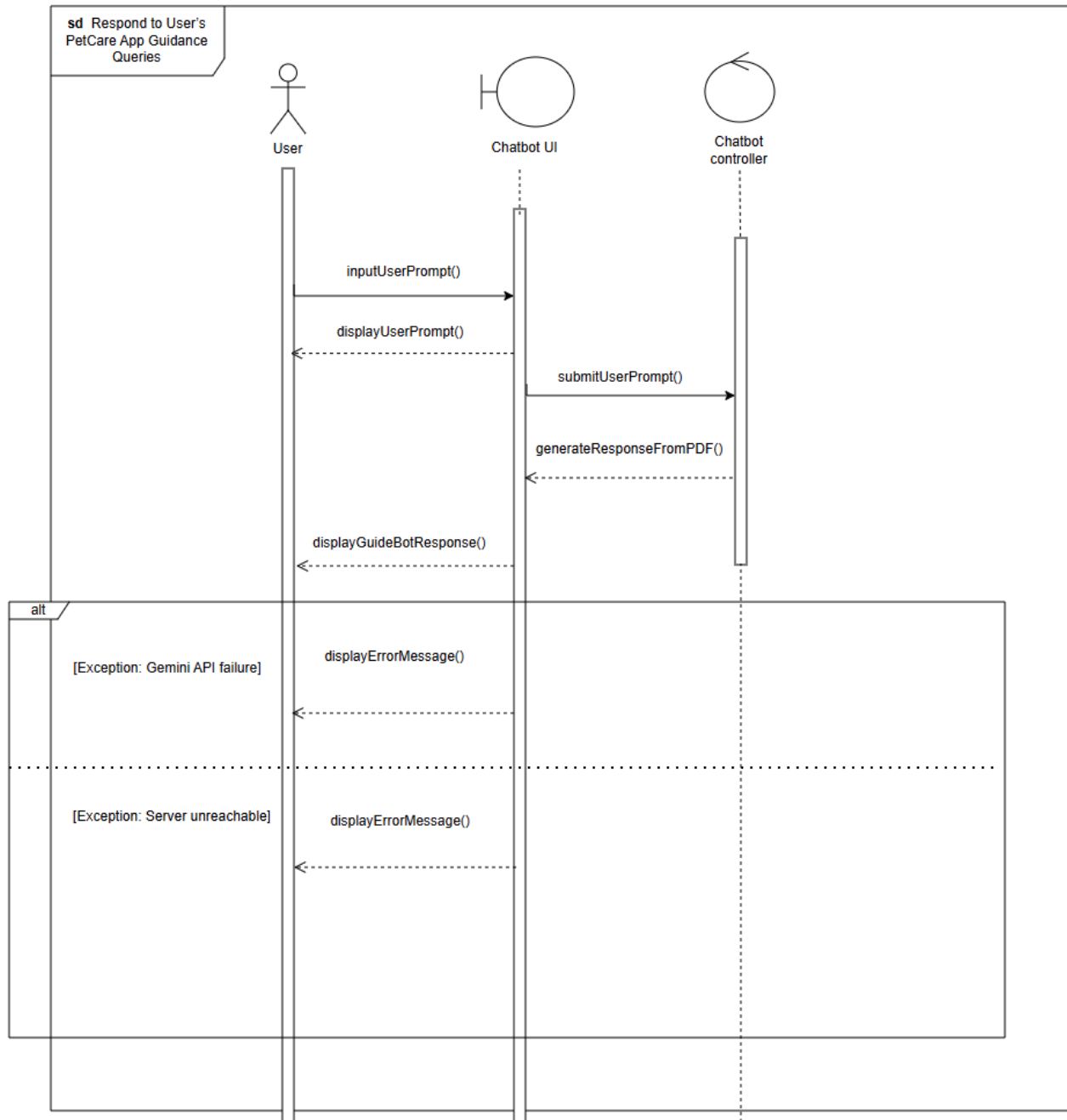


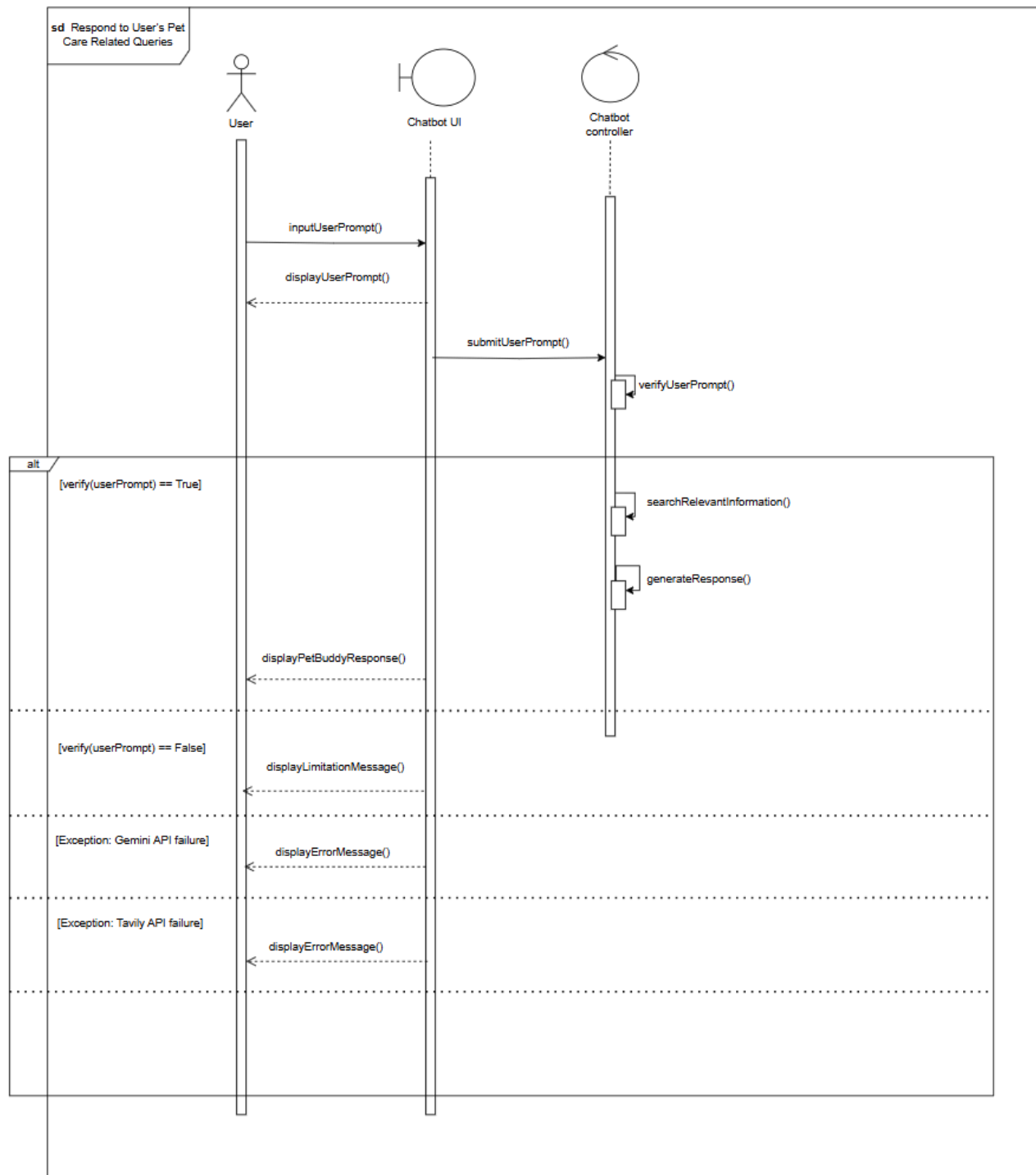
B.4.26 Add Adoption Pet



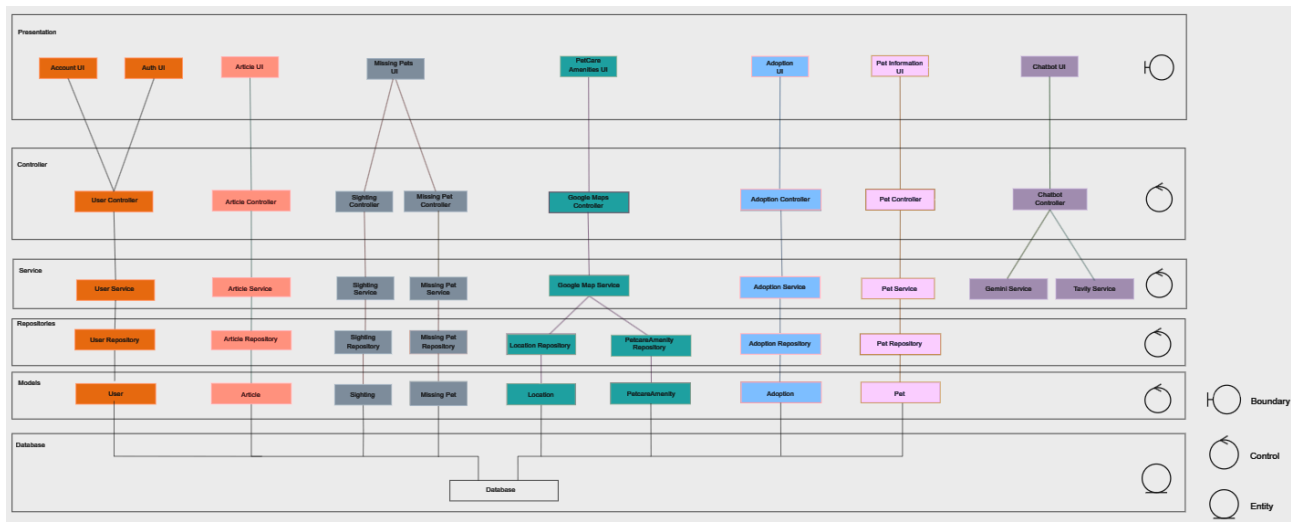
B.4.27 Delete Adoption Pet

B.4.28 Interact with chatbot

B.4.29 Responds to user's PetCare app guidance queries

B.4.30 Responds to user's PetCare Related queries

[illegible]



Appendix C: To Be Determined List

C.1 Postman API Documentation

[Postman API Test Cases](#)

The screenshot displays the Postman API documentation for the `registerUser` endpoint. On the left, the endpoint is identified as a **POST** request to `http://localhost:8080/api/auth/register`. The request body is in `formdata` format, with the following fields:

Field	Value
<code>userName</code>	John Lee
<code>email</code>	john@example.com
<code>password</code>	123456
<code>phoneNumber</code>	1234567890
<code>status</code>	normal
<code>profilePicUrl</code>	

On the right, the **Example Request** section shows the equivalent `curl` command:

```
curl --location 'http://localhost:8080/api/auth/register' \
--form 'userName="John Lee"' \
--form 'email="john@example.com"' \
--form 'password="123456"' \
--form 'phoneNumber="1234567890"' \
--form 'status="normal"' \
--form 'profilePicUrl=@p'.
```

The **Example Response** section shows the `Body` tab selected, indicating that no response body is returned for this request.

Screenshot of registerUser API documentation from Postman

